THE IMPACT OF AI DISCLOSURE ON CUSTOMER VALUE IN LLM-POWERED CUSTOMER SERVICE: EVIDENCE FROM A NATURAL EXPERIMENT AND FIELD EXPERIMENT

A thesis submitted in partial fulfillment of the requirements
for the degree of

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at

The University of Hong Kong

Zhen GUO

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Study Period: September 2023 – July 2024

Research Partner: Shulex Technology (data provider); Anker Innovations (field partner)

ABSTRACT

Research question. Does disclosing AI identity in inbound customer service reduce customer satisfaction, and can firms manage this effect through coupon compensation and hybrid service design? This question is commercially urgent — AI handles

billions of service interactions annually — and theoretically unresolved: prior algorithm aversion research relies on outbound sales experiments and laboratory studies that cannot speak to inbound service contexts where customers seek help rather than resist solicitation.

Methods. This dissertation reports two complementary empirical studies conducted in partnership with Shulex Technology and Anker Innovations. Study 1 analyzes 477,983 customer service tickets (September 2023 – May 2024) using high-dimensional fixed-effects regression with product category-by-intent interaction fixed effects, propensity score matching, and inverse probability weighting to address ticket routing endogeneity. Study 2 is a pre-registered 2 (disclosure) × 2 (coupon) × 2 (agent type) randomized controlled trial assigning 1,474 pilot tickets to hold service quality constant while varying only the information customers receive about their agent's identity.

Main results. Study 1 documents *efficiency-satisfaction decoupling*: AI reduces resolution rounds by 20–70% across all 29 category-intent cells while leaving NPS statistically unchanged in 25 of 29 cells (86%). Satisfaction penalties concentrate in purchase inquiry interactions (e.g., Edge Security: $\beta = -2.88$, $p < .001$), where customers seek relational trust rather than information retrieval. Study 2 pilot data confirm protocol viability and randomization balance; preliminary escalation patterns are directionally consistent with H1b (disclosed AI groups escalate approximately 2.1 percentage points more than non-disclosed groups); full outcome estimates await complete data collection.

Theoretical contributions. This dissertation proposes a *service value decomposition* framework for the AI era: transactional interactions deliver functional value that AI can match or exceed; relational interactions deliver trust value that AI disclosure disrupts. This reframes algorithm aversion theory from a unified psychological disposition into a boundary condition determined by which value dimension is primary for a given interaction type. The Bayesian persuasion logic of disclosure theory (Kamenica and Gentzkow, 2011) provides the economic foundation; social presence, attribution, fairness preference (Fehr and Schmidt, 1999), and expectation confirmation mechanisms elaborate the behavioral channels.

Policy implications. For firms facing mandatory AI transparency requirements under the EU AI Act, this research provides task-contingency guidance: disclosure is low-cost in transactional contexts (troubleshooting, product usage) and higher-cost in relational contexts (purchase advisory). In mandatory-disclosure, high-cost contexts, coupon compensation framed as cost-sharing — returning AI efficiency savings to the customer — is theoretically and experimentally motivated as the primary mitigation tool.

This dissertation addresses these questions through two complementary empirical studies with Shulex Technology and Anker Innovations, a major consumer electronics brand on the U.S. Amazon marketplace. Study 1 analyzes 477,983 customer service tickets (September 2023 – May 2024) using high-dimensional fixed-effects regression with product category-by-intent interaction fixed effects to control for endogenous ticket routing. The central finding is an *efficiency-satisfaction decoupling*: AI agents reduce resolution rounds by 0.34 to 2.91 per ticket across all 29 category-intent cells — a 20–70% efficiency gain — while leaving Net Promoter Score (NPS) statistically unchanged in 25 of 29 cells. Satisfaction penalties appear only in purchase inquiry interactions (e.g., Edge Security: $\beta = -2.88$, $p < .001$; TWS: $\beta = -1.70$, $p < .05$), where customers expect personalized relational judgment rather than information retrieval. This task-contingent pattern re-draws the boundary conditions of algorithm aversion theory: resistance is not a universal reaction to AI, but activates specifically when customers perceive that AI lacks the social and financial judgment their situation demands.

Study 2 is a randomized controlled trial (RCT) launched in July 2024, assigning 1,474 pilot-phase tickets across a 2 (AI disclosure) × 2 (coupon incentive) × 2 (agent type: pure AI vs. hybrid) factorial design. The RCT is the primary vehicle for causal identification: by holding service quality constant and varying only the information provided to customers about their agent's identity, it cleanly separates the *information effect* of disclosure from the *quality effect* of AI service that Study 1 captures. Randomization balance is confirmed across all treatment arms; outcome data collection through August 2024 is ongoing, and preliminary directional evidence is presented in Chapter 7. The causal estimates of disclosure, coupon, and agent-type effects await full-sample analysis; the RCT design and pre-registered analysis plan constitute a methodological contribution independent of the final estimates.

This dissertation makes four contributions. First, it provides the largest field dataset yet employed in the algorithm aversion literature — 477,983 tickets across 19 product categories — yielding statistical power for fine-grained heterogeneity analysis that laboratory studies structurally cannot achieve. Second, it documents *efficiency-satisfaction decoupling* as a systematic empirical phenomenon: AI-driven efficiency gains (20–70% reduction in resolution rounds) do not reduce satisfaction in the majority of service contexts, directly challenging the longstanding service quality assumption that faster resolution comes at a relational cost. Third, it delineates the boundary conditions of algorithm aversion at field scale — aversion activates in purchase inquiry tasks requiring social and financial judgment, not in technical troubleshooting — extending and empirically refining the theoretical framework of Castelo et al. (2019) and Huang and Rust (2018). Fourth, it introduces the first pre-registered experimental test of compensatory coupon incentives as a practical intervention for managing disclosure-induced dissatisfaction under mandatory AI transparency regulations such as the EU AI Act.

Keywords: AI disclosure, algorithm aversion, LLM customer service, efficiency-satisfaction decoupling, NPS, hybrid AI, field experiment, e-commerce

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I am equally indebted to my co-supervisor, Professor Li-An Zhou of Peking University. Professor Zhou's expertise in organizational economics, field experiments, and the Chinese technology industry provided critical perspective on the research design and the contextual interpretation of findings. His challenging questions consistently pushed me to think more carefully about mechanisms and alternative explanations, making this work substantially stronger.

This research would have been impossible without the extraordinary cooperation of the team at Shulex Technology, who generously provided access to their proprietary customer service platform data, assisted with data extraction and variable construction, and facilitated the field experiment deployment in July 2024. I am particularly grateful to the Shulex data and product teams for their patience in responding to countless data questions and for ensuring the integrity of the experimental randomization. I also thank Anker Innovations for agreeing to participate as the field partner and for permitting analysis of sensitive operational data under appropriate confidentiality agreements.

I wish to thank the HKU DBA program faculty, staff, and my cohort colleagues for creating an intellectually stimulating and supportive environment. The program's emphasis on rigorous business research with managerial relevance has profoundly influenced the direction and framing of this dissertation.

Finally, I owe an immeasurable debt to my family for their unwavering support and sacrifices during the years of this doctoral journey. Their patience and encouragement sustained me through the most demanding periods of this research.

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INTRODUCTION

Background and Motivation

LLM-powered customer service systems have moved from experiment to mainstream deployment in fewer than two years, yet the core question of whether customers accept AI service — or resist it — remains empirically unsettled. Earlier generations of chatbots operated on rigid decision trees that could handle only anticipated queries, forcing customers through frustrating menus and frequently failing to address their actual concerns. The latest generation of LLM-powered systems, built on architectures such as OpenAI's GPT-4, can engage in open-domain dialogue, understand nuanced complaints, generate personalized recommendations, and resolve a substantial proportion of queries entirely without human intervention. The speed of this transition has outpaced the empirical evidence firms need to make confident decisions about deployment strategy, disclosure policy, and human-AI system design.

The scale of this transformation is documented in market data. The global chatbot market was valued at approximately $5.1 billion in 2022 and is projected to exceed $27 billion by 2030, representing a compound annual growth rate of over 23 percent (Grand View Research, 2023). Investment in AI-powered customer service solutions has accelerated sharply since 2022, driven by the availability of capable foundation models through API access and the intensifying competitive pressure on firms to reduce operating costs while maintaining or improving service quality. For large-scale e-commerce operations in particular, customer service represents a substantial and growing cost center. Amazon alone processes hundreds of millions of customer inquiries annually across its marketplace, and third-party sellers operating on the platform face the challenge of managing post-sale support for geographically dispersed customers without the infrastructure advantages of large retailers.

Anker Innovations, the firm whose customer service data forms the empirical foundation of this dissertation, represents a paradigmatic case of this transformation. Founded in 2011 in Shenzhen, China, Anker has grown to become one of the largest third-party sellers on Amazon globally, with annual revenues of approximately two billion dollars and a product portfolio spanning consumer electronics across charging devices, audio equipment, home security cameras, and projection technology. Like most large Amazon sellers, Anker relies heavily on email-based customer service — the primary communication channel mandated by Amazon's seller communication policies — to handle post-sale inquiries ranging from simple usage questions to complex warranty claims and return logistics. The volume of these inquiries is enormous. Prior to the AI deployment studied here, Anker's customer service operation required a large team of human agents working around the clock to respond to inquiries across time zones, with all the attendant costs in labor, training, and quality control.

Shulex Technology, Anker's AI customer service vendor, deployed an LLM-powered response system in late 2023, integrating GPT-based language generation with Anker's proprietary product knowledge base and a ticket management infrastructure

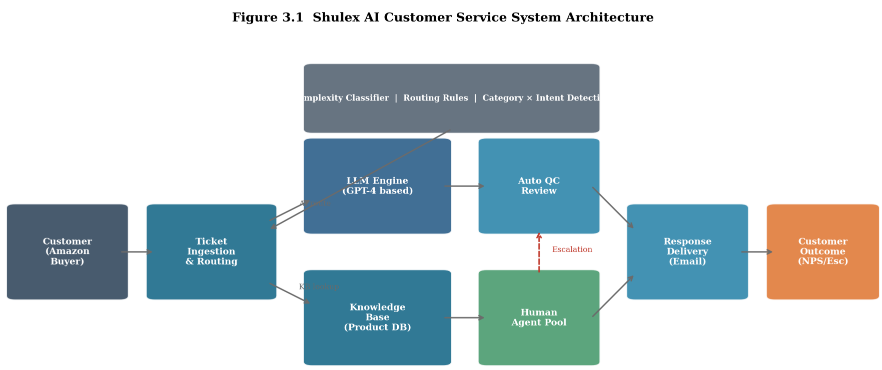

Figure 2.1: Shulex LLM Customer Service System Architecture

that routes, prioritizes, and tracks customer inquiries from initial contact through resolution. The adoption of AI within this system was rapid and striking. In September 2023, approximately 18.8 percent of tickets received at least one AI-generated response. By February 2024, that figure had risen to 37 percent of the total ticket volume.

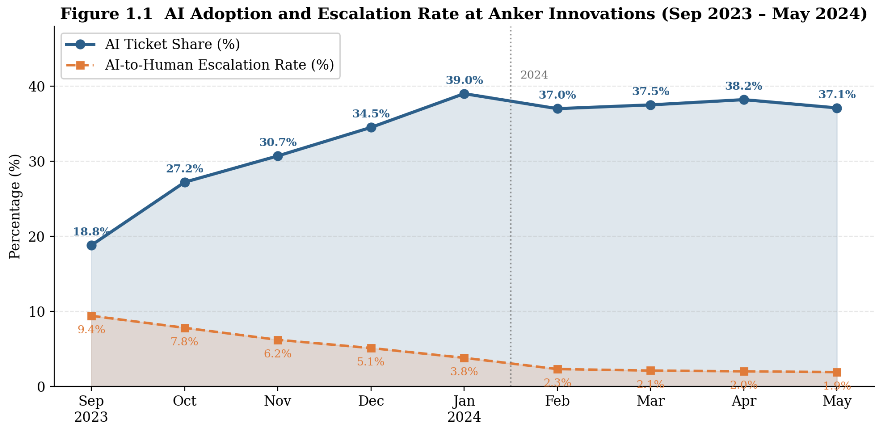

Figure 2.2: AI Adoption Trends in Customer Service (2019–2024)

This rapid adoption trajectory — driven by demonstrably lower per-ticket costs and consistent response quality for routine queries — captures the broader industry trend in compressed form. It also creates the empirical variation that makes rigorous study of AI effects on customer outcomes possible.

The commercial case for AI deployment is already compelling. AI agents can respond instantly around the clock, handle unlimited concurrent tickets, and at Anker Innovations specifically, resolve routine queries in 1.0 to 1.3 message exchanges on average — compared to 3.6 to 4.2 exchanges for hybrid AI-human systems and 1.7 to 2.1 exchanges for pure human agents. For large-scale e-commerce operations handling hundreds of thousands of tickets monthly, this efficiency differential translates directly into millions of dollars in annual labor savings. Firms are consequently deploying AI at speed, often before they have evidence on whether customers accept it.

On the other side, a growing body of academic literature in behavioral science and management documents what researchers have termed ”algorithm aversion” — a systematic and often irrational tendency for people to distrust and avoid algorithmic or AI-based systems, even in cases where those systems demonstrably outperform human judgment. The seminal work by Dietvorst, Logg, and Loewenstein (2015) demonstrated in a series of controlled laboratory experiments that after observing an algorithm make an error, individuals consistently preferred the less-accurate judgment of a human expert. Crucially, this aversion was not eliminated even when participants were explicitly informed that the algorithm was more accurate on average

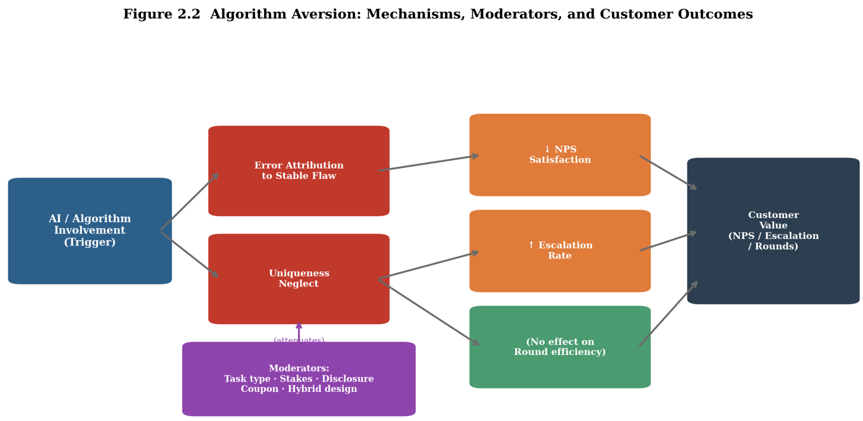

Figure 2.3: Algorithm Aversion Mechanisms and Boundary Conditions

: the emotional and psychological response to observing AI errors was powerful enough to override purely rational calculation.

The practical implications of algorithm aversion for customer service AI are potentially severe. If customers who discover they are being served by an AI agent become less satisfied, more likely to escalate to human agents, or less likely to make future purchases, then the efficiency gains from AI deployment may be partially or entirely offset by losses in customer lifetime value. The question of whether, when, and to what degree AI use in customer service triggers algorithm aversion therefore has direct implications for how firms should design, deploy, and communicate about their AI systems.

The most directly relevant prior empirical study is Luo, Tong, Fang, and Qu (2019), which examined the effects of AI chatbot identity disclosure in an outbound sales context. When a chatbot introduced itself as an AI agent before beginning a sales conversation, conversion rates dropped by 79.7 percent relative to baseline. This dramatic finding suggests that AI disclosure can be commercially catastrophic — at least in sales contexts where the AI is initiating contact and asking something of the customer. However, several critical features of that setting may limit the generalizability of those findings to the inbound customer service context. In outbound sales, the customer is being contacted unsolicited; the immediate reaction may be a combination of AI aversion proper and resistance to the unsolicited commercial contact itself. In inbound customer service, by contrast, the customer has chosen to initiate contact specifically because they need assistance. Their motivation is to have a problem resolved, not to purchase something. It is entirely possible that in this context, an AI's ability to resolve the problem quickly and accurately would dominate any AI aversion effect, particularly for routine technical queries.

This dissertation was motivated precisely by this gap. The critical questions that Luo et al. (2019) and subsequent studies leave unanswered are: Does AI aversion manifest in inbound customer service, where customers seek help rather than resist solicitation? Is AI aversion uniform across all types of customer service tasks, or does it concentrate in particular contexts? Can firms use compensatory interventions — such as providing coupons to customers who are told they are being served by AI — to mitigate any negative effects of disclosure? And does the design of the AI system itself, specifically the presence or absence of a human backup within a hybrid architecture, affect customers' experience of AI service? These are the questions this dissertation addresses.

The timing of this research could not be more consequential. As regulators worldwide move toward mandating transparency about AI use in consumer interactions — the European Union's AI Act explicitly requires disclosure when AI systems interact with humans, and the U.S. Federal Trade Commission has issued guidelines on AI transparency — firms face growing pressure to disclose AI involvement regardless of their preferences. Understanding the customer consequences of such disclosure is therefore not merely an academic exercise but a matter of practical urgency for every firm operating AI-powered customer service systems at scale.

Research Questions and Objectives

This dissertation is organized around three primary research questions, progressing from description to causal identification to intervention design.

RQ1: Does AI involvement in LLM-powered customer service affect customer outcomes? Specifically, does it alter Net Promoter Score (NPS), escalation rates, and resolution efficiency (response rounds), and are these effects uniform across all service task types? This foundational question must be answered before asking whether *disclosure* of AI involvement matters — because any disclosure effect is superimposed on the underlying effect of AI service quality itself. The observational study (Study 1) addresses RQ1 using natural variation in AI adoption across product categories and time, controlled through high-dimensional fixed effects.

RQ2: Does disclosing AI identity affect customer satisfaction and escalation behavior? This question moves from the observational question of what AI involvement does to the causal question of what *knowing* about AI involvement does. The field experiment (Study 2) provides clean identification by randomly assigning customers to disclosed versus non-disclosed conditions while holding actual service quality constant. This distinction matters because the most influential prior field study (Luo et al., 2019) found a 79.7% collapse in purchase rates after AI disclosure in *outbound* sales — a result that may not generalize to inbound service, where customers seek help rather than resist solicitation.

RQ3: Can coupon incentives mitigate disclosure-induced dissatisfaction, and does agent type (pure AI versus hybrid) moderate outcomes? This question addresses the intervention dimension of the research. For firms facing mandatory AI transparency regulations — the EU AI Act requires disclosure when AI systems interact with humans — the practical question is not whether to disclose but how to disclose while minimizing satisfaction costs. Coupons represent a commercially realistic compensatory mechanism; hybrid architecture represents a design-level intervention. The RCT tests both.

Together, these three questions define a sequential, internally coherent research agenda. Figure 1.1 illustrates the logic of progression: Study 1 (large-scale observational panel) answers RQ1 and identifies the *heterogeneous* pattern of AI effects across task types — necessary because any disclosure intervention will be superimposed on this underlying AI quality effect. Study 2 (RCT) answers RQ2 and RQ3 by holding AI service quality constant and varying only the information customers receive, providing clean causal identification of the disclosure and coupon mechanisms that Study 1 cannot disentangle. The two studies are therefore not parallel investigations but an evidence chain: Study 1 establishes the phenomenon and motivates the experimental design; Study 2 provides causal estimates of the interventions most relevant for managerial practice.

Significance and Contributions

This dissertation makes four academic contributions and three managerial contributions.

Academic Contribution 1: Scale and ecological validity. The overwhelming majority of algorithm aversion research relies on laboratory experiments or small-scale field studies. Dietvorst et al.'s (2015) foundational study used N = 346 participants in a forecasting task; Luo et al.'s (2019) field study examined outbound telemarketing calls. The present dissertation analyzes 477,983 tickets across 19 product categories over eight months — to the author's knowledge, the largest field dataset yet employed in this literature. This scale provides the statistical power to detect heterogeneous effects that are systematically underpowered in smaller studies.

Academic Contribution 2: The efficiency-satisfaction decoupling. Study 1 documents a previously unrecognized empirical phenomenon: AI simultaneously achieves major efficiency gains — reducing resolution rounds by 20–70% — while maintaining customer satisfaction at statistically indistinguishable levels from human service in 25 of 29 category-intent cells. This finding challenges the service quality literature's longstanding assumption that efficiency and quality exist in inherent tension. The SERVQUAL framework (Parasuraman et al., 1988) implicitly treats empathy and responsiveness as competing under resource constraints; the present evidence shows that AI can enhance responsiveness without systematically eroding satisfaction, in contexts where information retrieval — not emotional judgment — is the customer's primary need.

Academic Contribution 3: Task-contingent AI aversion at field scale. Castelo et al. (2019) and Longoni et al. (2019) demonstrated task-dependence of algorithm aversion in laboratory settings. This dissertation documents the same phenomenon in a live deployment: satisfaction penalties concentrate in purchase inquiry interactions (where customers expect social and financial judgment) and are absent in routine troubleshooting (where customers primarily need information). This field-scale confirmation specifies the commercial boundary conditions of AI aversion theory with a precision that laboratory studies cannot achieve.

Academic Contribution 4: Novel experimental identification of disclosure effects. The crossed factorial design of disclosure × coupon × agent type, implemented in a live commercial system, provides identification advantages unavailable to prior studies. By holding service quality constant and varying only the information customers receive about their agent, Study 2 isolates the pure informational effect of AI disclosure from the quality effect of AI service. The coupon condition — to the author's knowledge, the first experimental test of compensatory incentives as a disclosure moderator — introduces a novel behavioral intervention with both theoretical and regulatory relevance.

On the managerial side, firms deploying or considering AI customer service systems currently make deployment and disclosure decisions without evidence from inbound service settings. This research offers three evidence-based inputs. First, it provides a task-contingency framework for AI deployment: pure AI is justified for routine troubleshooting tasks (no detectable satisfaction cost; large efficiency gain) but should be complemented by human capacity for purchase inquiry tasks (measurable satisfaction penalty). Second, it quantifies the efficiency differential in commercially interpretable terms — response rounds — enabling firms to construct data-grounded business cases for AI investment. Third, it offers the first experimental evidence on compensatory coupon incentives as a disclosure management tool, directly relevant to firms navigating mandatory transparency requirements under regulations such as the EU AI Act.

Overview of Research Design

This dissertation employs a two-study sequential design, combining the large-scale observational power of a high-dimensional fixed-effects panel study with the experimental rigor of a randomized controlled trial. The two studies are complementary: Study 1 establishes the factual baseline of AI effects on service process and customer satisfaction in the observational data, providing the context and motivation for the experimental interventions tested in Study 2.

Study 1 is an observational study using 477,983 customer service tickets processed through the Shulex platform on behalf of Anker Innovations between September 2023 and May 2024. The natural variation in AI adoption over this period — driven by Shulex's progressive deployment rollout and Anker's product-category-level routing policies — creates the empirical variation needed to identify AI effects. Not all customers received AI service, and the variation in who did is partly exogenous to customer characteristics (it was driven by the platform's deployment schedule and by ticket characteristics such as complexity and product category routing rules). The primary analytical approach is high-dimensional fixed-effects (HDFE) regression, which controls for the systematic differences in ticket types routed to different agent types by including product category-by-intent interaction fixed effects (absorbing category-intent combinations as a single high-dimensional control) along with ticket complexity as an additional covariate. This approach provides quasi-experimental evidence on the causal effect of AI involvement on response rounds and NPS while acknowledging that full randomization is not available in the observational data.

The key outcomes analyzed in Study 1 are: (1) the number of response rounds (N_responses), as a measure of service efficiency and process; (2) NPS score on a 1-10 scale, as a measure of customer satisfaction; and (3) escalation rate, as a measure of customer-revealed preference for human service. The key independent variable is agent type: pure AI, AI-plus-human hybrid, or pure human. The analysis is conducted both at the aggregate level and with interaction terms to test for heterogeneous effects by product category, ticket intent, and complexity level.

Study 2 is a randomized controlled trial conducted in July 2024, with ticket assignments beginning from that date. A total of 1,474 tickets were assigned to one of six experimental conditions defined by a 2 (AI disclosure: disclosed vs. not disclosed) × 2 (coupon: provided vs. not provided) × 2 (agent type: pure AI vs. hybrid) factorial design, with one condition removed to ensure adequate power per cell. The random assignment of disclosure conditions holds AI service quality constant while varying only the information provided to customers about the nature of their service agent, enabling clean causal identification of disclosure effects. The coupon condition tests whether providing a small monetary incentive (a discount coupon on future Anker purchases) alongside AI disclosure mitigates any negative effect of disclosure on NPS. The agent type dimension provides experimental evidence on whether hybrid versus pure AI architecture affects customer satisfaction after controlling for ticket complexity. Analysis of Study 2 will employ intention-to-treat (ITT) analysis with randomization-based inference, supplemented by regression adjustment for any residual imbalance in pre-treatment covariates. Results of Study 2 will be reported in Chapter 8.

Organization of the Dissertation

The remainder of this dissertation is organized as follows. Chapter 2 provides a comprehensive review of the relevant literatures, encompassing the evolution of AI in customer service, the theoretical and empirical foundations of algorithm aversion and AI resistance, the literature on AI disclosure and its effects on consumer behavior, the behavioral economics of incentive-based compensation, and the emerging literature on hybrid AI-human systems. Chapter 2 concludes by synthesizing the three primary gaps in the existing literature that this dissertation addresses.

Chapter 3 describes the research setting and data in detail. It introduces Shulex and Anker Innovations, describes the AI customer service system architecture, presents the data sources and their properties, formally defines all key variables, and provides descriptive statistics that characterize the sample and document the key patterns that motivate the hypotheses.

Chapter 4 develops the theoretical framework and formally states the research hypotheses. It presents the conceptual framework integrating social presence theory, attribution theory, expectation confirmation theory, and service recovery theory, and derives the five hypotheses (H1a, H1b, H1c, H2, H3) that structure the empirical analysis.

Chapters 5 and 6 present the methods and results of Study 1 (the observational study). Chapter 5 describes the identification strategy, empirical specification, and planned robustness checks. Chapter 6 reports the main findings on response rounds, NPS, and task-contingent effects, along with sensitivity analyses.

Chapters 7 and 8 present the methods and results of Study 2 (the field experiment). Chapter 7 describes the experimental design, randomization protocol, balance checks, outcome variables, and analytical strategy. Chapter 8 reports the treatment effects of disclosure, the coupon moderation, and the agent type moderation.

Chapter 9 provides the discussion and conclusion, synthesizing findings across both studies, articulating theoretical contributions, deriving managerial implications, acknowledging limitations, and suggesting directions for future research. The dissertation concludes with a complete reference list and appendices containing the NPS survey instrument, experimental protocols, AI disclosure message templates, additional regression output, and the QC error type classification scheme.

LITERATURE REVIEW

AI in Customer Service: Evolution and Current State

The use of automated systems to handle customer inquiries is not new. What is new — and what distinguishes the current generation of AI customer service from its predecessors — is the fundamental shift in capability enabled by large language models. To appreciate the significance of this shift, it is useful to trace the developmental trajectory from the earliest automated service systems through to the present moment.

The history of automated customer service can be divided into at least three distinct technological eras. The first era, spanning roughly from the 1970s through the late 1990s, was characterized by interactive voice response (IVR) systems and early rule-based text interfaces. These systems operated on predefined decision trees: the customer selected from a menu of options, and the system routed the inquiry accordingly. The defining limitation of this generation was brittleness — the system could handle only anticipated queries, and any departure from the expected path led to failure or transfer to a human agent. Customer frustration with IVR systems became so widespread that ”press zero to speak to a representative” became a culturally recognized shorthand for the failure of automated service. Despite their limitations, however, IVR systems demonstrated that automation could handle a meaningful fraction of routine inquiries and reduce labor costs, establishing the commercial rationale for continued investment in service automation.

The second era, spanning roughly the 2000s through the mid-2010s, introduced machine learning-based dialogue systems. Advances in natural language processing (NLP) — particularly in intent classification and entity extraction — enabled chatbots that could understand a broader range of customer inputs and respond with more contextually appropriate replies. Companies such as LivePerson and Nuance developed commercial platforms that could handle defined domains of customer inquiry with reasonable accuracy. Retrieval-augmented systems, which matched customer queries against databases of pre-written responses, became common in e-commerce and telecom customer service. Academic research on these systems documented their limitations: they struggled with ambiguous or multi-intent queries, required extensive domain-specific training data, and failed gracefully only if the customer understood and accepted the system's constraints (Woebot et al., 2017; McTear, 2022). Importantly, this generation of systems did not generate novel language — they retrieved or assembled responses from predefined templates — which placed hard limits on their ability to handle the diversity of real customer inquiries.

The third and current era, beginning with the commercial deployment of large language model-based systems in 2023, represents a qualitative leap that makes the technology genuinely transformative rather than merely incremental. LLMs such as GPT-3.5 and GPT-4, trained on vast corpora of human-generated text, can generate fluent, contextually appropriate, and highly personalized responses to virtually any natural language input. They are not retrieval systems: they generate text dynamically, which means they can handle novel query types, explain technical concepts in accessible language, adapt tone to customer emotional state, and maintain coherent multi-turn conversations. Crucially for customer service applications, LLMs can be fine-tuned or prompted with domain-specific knowledge — product specifications, return policies, troubleshooting guides — enabling them to provide accurate, on-brand responses to brand-specific inquiries without requiring the extensive annotation and training data that earlier NLP systems demanded.

The implications for the economics of customer service are profound. A well-configured LLM-based system can handle, at marginal cost approaching zero, the same query that would require five to ten minutes of a skilled human agent's time. For large-scale e-commerce operations, where customer service volumes number in the hundreds of thousands of tickets monthly, the potential labor cost savings are measured in millions of dollars annually. Wirtz, Patterson, Kunz, Gruber, Lu, Paluch, and Martins (2018) provide an early theoretical treatment of AI service robots entering the service frontline, identifying both the potential value creation and the managerial challenges of deployment. Their framework anticipates many of the issues this dissertation addresses empirically, including the risk of negative customer reactions to AI service agents and the importance of the human-AI handoff in hybrid systems.

Empirical research on the effects of LLM-based customer service systems on customer outcomes is, as yet, limited. This is unsurprising given the recency of deployments: most firms only began moving from rule-based to LLM-based customer service systems in 2023 and 2024, and academic research on deployments at the scale required for statistical inference requires access to proprietary firm data that is rarely available. The present dissertation is therefore contributing to a nascent but rapidly growing area of empirical inquiry.

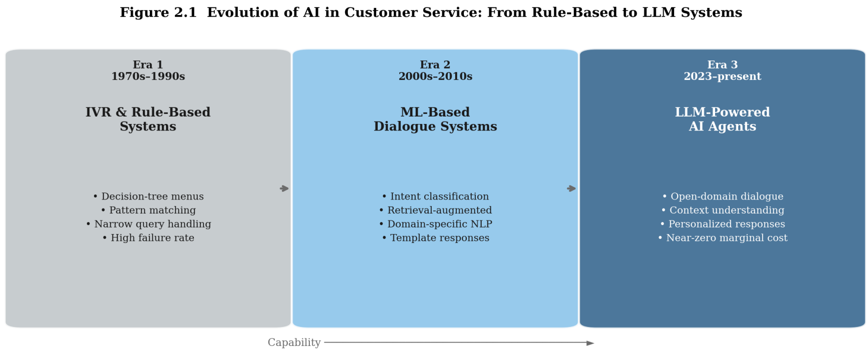

Figure 3.1: Evolution of AI in Customer Service: Three Eras

What does exist is a set of studies examining earlier AI-based customer service systems that provides partial evidence on customer reactions to automation. Crolic, Thomaz, Hadi, and Stephen (2022) found that customers were more likely to blame chatbots for service failures than human agents, even when the actual quality of service was identical — a phenomenon they termed the ”blame the bot” effect. This finding is directly relevant to the present dissertation because it suggests that the agent identity of the service provider affects attribution processes in ways that can damage satisfaction even when service quality is controlled. If customers attribute AI errors to stable technological limitations while attributing human errors to correctable individual failures, the subjective experience of equivalent objective service quality will differ systematically across agent types. Crolic et al.’s (2022) finding thus provides a mechanism through which AI involvement may reduce satisfaction independent of actual service quality differences.

Huang and Rust (2018) developed a theoretical framework for understanding AI service quality that distinguishes mechanical intelligence (performing standardized tasks), thinking intelligence (analysis and problem-solving), feeling intelligence (empathy and interpersonal sensitivity), and social intelligence (managing relationships and social norms). Their framework predicts that AI will outperform humans on mechanical tasks, approach or equal human performance on thinking tasks, but remain inferior on feeling and social intelligence tasks for the foreseeable future. This theoretical prediction maps directly onto the task-contingency hypothesis developed in Chapter 4 of this dissertation: if AI aversion is driven by perceived deficiencies in feeling and social intelligence, then aversion should concentrate in service tasks that require emotional sensitivity or relational judgment, such as handling purchase inquiries or resolving complaints from distressed customers.

The evidence on hybrid AI-human systems in customer service is more recent and less settled. A small number of studies have examined the effectiveness of systems in which AI handles first-line responses and humans intervene for complex or escalated cases. Brynjolfsson and McAfee (2014) argued theoretically for the complementarity of human and AI capabilities, predicting that hybrid systems would outperform either pure AI or pure human systems across most knowledge-work contexts. In customer service specifically, hybrid systems offer the theoretical advantage of combining AI's speed and consistency for routine queries with human empathy and judgment for complex or emotionally charged interactions. The present dissertation contributes empirical field evidence on this theoretical prediction in a large-scale e-commerce context.

Within Anker's deployment specifically, the AI system integration timeline followed a rapid trajectory that is representative of the industry more broadly. Shulex's GPT-based platform, deployed in late 2023, processed its first AI responses in September 2023, with AI involvement at 18.8 percent of all tickets that month. By November 2023, AI involvement had reached 30.7 percent, and by January 2024, 39.0 percent of tickets received at least one AI-generated response. This growth trajectory was driven not by any change in customer behavior — customers did not choose or know about their service agent type — but by a progressive rollout of AI routing across product categories and ticket complexity levels. This exogenous source of variation forms the basis of the identification strategy in Study 1.

Algorithm Aversion and AI Resistance: Theory and Evidence

Algorithm aversion refers to a systematic human tendency to prefer the judgment of human experts over that of algorithmic systems, even in situations where objective evidence indicates that the algorithm makes better predictions or decisions. The concept was formally introduced and rigorously documented by Dietvorst, Logg, and Loewenstein (2015) in a landmark paper that remains the cornerstone of this literature. Understanding the theoretical foundations, documented evidence, and boundary conditions of algorithm aversion is essential context for interpreting the effects of AI disclosure in customer service settings.

Dietvorst et al. (2015) conducted a series of six experiments involving N = 346 participants in a human forecasting task (predicting graduate school performance from quantitative profiles of applicants). In the key experimental manipulation, participants were shown the predictions of both a human forecaster and an algorithmic system, along with feedback on forecast accuracy over multiple rounds. When participants observed the algorithm making errors, they reliably preferred the human forecaster's predictions in subsequent rounds — even when the algorithm was, on average, more accurate than the human. Crucially, the aversion effect persisted even when participants were explicitly told the algorithm was more accurate: observing the error triggered an emotional-evaluative response that override the rational updating that Bayesian decision theory would predict. This finding was striking because it demonstrated not merely an information deficit (people not knowing the algorithm was better) but a motivated avoidance driven by the psychological discomfort of having trusted an algorithm that made a visible mistake.

The psychological mechanisms underlying algorithm aversion have been theorized along several dimensions. Dietvorst et al. (2015) proposed that the experience of watching an algorithm err created a sense of discomfort or betrayal that was qualitatively different from the reaction to human error — humans can explain and contextualize their mistakes in ways that feel meaningful, whereas algorithms cannot. Subsequent researchers have elaborated this basic insight into a richer mechanistic account. Three mechanisms deserve particular attention for the present dissertation.

The first mechanism is opacity: algorithmic processes are often opaque to users, who cannot follow the reasoning chain from inputs to outputs. This opacity, sometimes described as the "black box" problem, creates unease because it prevents the user from evaluating whether the algorithm is using sensible criteria or will continue to behave appropriately in the future. When a human advisor gives bad advice, the recipient can often understand why the mistake was made and adjust their trust calibration accordingly. When an algorithm gives a poor output, the opacity of the process makes it difficult to diagnose the source of error or predict when similar errors will recur, leading to a more generalized distrust (Jussupow, Spohrer, Heinzl, and Gawlitza, 2020). In customer service, this opacity mechanism suggests that AI service failures may produce more lasting satisfaction damage than equivalent human failures because customers cannot understand why the AI got it wrong.

The second mechanism concerns accountability and control. Human agents are embedded in accountability systems — they can be complained about, supervised, disciplined, or replaced — that give customers recourse when service fails. Algorithmic systems, particularly those not publicly understood, lack these accountability structures in ways that customers perceive. Dietvorst, Logg, and Loewenstein (2018) demonstrated that the key psychological driver of algorithm aversion was not the algorithm's error per se but the lack of control over the algorithmic process: when participants were allowed to modify the algorithm's predictions slightly (even by a trivially small amount), their willingness to use the algorithm increased dramatically, even though the modifications reduced average accuracy. The control finding is particularly important for hybrid AI-human systems: by preserving a human in the loop, hybrid systems may offer customers an implicit sense of accountability and oversight that reduces the psychological threat associated with AI, even if the human's role in any given ticket is minimal.

The third mechanism involves what Longoni, Bonezzi, and Morewedge (2019) termed "uniqueness neglect": the perception that AI, because it processes cases through standardized algorithms, cannot account for the unique individual characteristics of the specific person it is serving. In medical contexts, Longoni et al. (2019) showed that resistance to AI medical advice was driven primarily by the belief that AI could not understand or account for the individual patient's specific circumstances, even when the AI was described as highly accurate at the population level. This mechanism is likely to be particularly relevant in customer service contexts where the customer's specific situation is important — for example, a purchase inquiry involving the customer's specific budget, use case, and prior product history — and less relevant in contexts where the query is essentially generic, such as a troubleshooting question about a product malfunction that has a standard solution.

Castelo, Bos, and Lehmann (2019) formalized the task-dependence of algorithm aversion in a series of laboratory experiments, distinguishing between "objective" tasks (those with factual answers determinable from input data alone, such as numerical predictions) and "subjective" tasks (those requiring social or personal judgment, such as evaluating the appropriateness of a behavior or the attractiveness of a partner). They found that participants preferred algorithmic recommendations for objective tasks but strongly preferred human recommendations for subjective tasks, and that this preference was mediated by perceptions of the task's objectivity rather than beliefs about comparative accuracy. This finding provides the theoretical foundation for the task-contingency hypothesis (H1c) developed in Chapter 4 of this dissertation: purchase inquiry tasks, which require understanding and responding to the customer's individual preferences, budget constraints, and social context, are more analogous to Castelo et al.'s subjective tasks than to objective ones, suggesting that AI aversion should be stronger in purchase inquiry than in technical troubleshooting.

Beyond these laboratory demonstrations, field evidence on algorithm aversion in naturalistic contexts has been mixed. Liu, He, and Tang (2023), studying algorithmic recommendations for ridesharing drivers in a large-scale observational dataset, found that drivers were significantly less likely to follow the platform's route recommendations after experiencing a recommendation that led to a poor outcome (long wait, no pickup), even when following recommendations on average improved earnings. This finding, extending the Dietvorst et al. (2015) laboratory results to a high-stakes real-world context, provides important evidence that algorithm aversion is not merely a laboratory artifact. However, the ridesharing context involves repeated interactions and clear outcome feedback, which may amplify aversion dynamics relative to a single-encounter customer service context.

Zhang and Gosline (2023) offer a conceptual reframing that is valuable for the present study. Rather than framing the phenomenon primarily as ”AI aversion,” they argue that the more accurate description is ”human favoritism”: humans have a positive preference for human-produced content that is activated in ambiguous or comparable situations. This reframing shifts the locus of the effect from a negative reaction to AI to a positive preference for human contact, which may have different implications for disclosure design. If the mechanism is human favoritism rather than AI aversion, then disclosure strategies that emphasize what AI does well (speed, accuracy, availability) may partially overcome the effect, whereas strategies that merely identify AI without providing this positive framing may leave human favoritism unchecked.

Burton, Stein, and Jensen (2020) conducted a systematic review of the algorithm aversion literature and reached the measured conclusion that aversion is not universal: its occurrence depends heavily on contextual factors including the nature of the task, the stakes involved, whether the user has experience with the algorithm, and the framing of the algorithmic recommendation. This finding is consistent with the present dissertation’s emphasis on heterogeneous effects and task contingency, and motivates the differentiated analytical approach pursued in Study 1. Jussupow et al. (2020) reached a similar conclusion in a separate comprehensive literature review, identifying antecedents of aversion including task uncertainty, algorithm transparency, user expertise, and outcome feedback availability.

Two additional findings from the algorithm aversion literature deserve mention as context for the present study. Dietvorst et al. (2018) demonstrated the ”algorithm modification” effect: giving people even minimal ability to modify algorithmic outputs dramatically increased their willingness to use the algorithm. This finding suggests an intervention design principle — building user control or customization features into AI customer service systems might reduce algorithm aversion — that is related to but distinct from the hybrid architecture design explored in H3. Logg, Minson, and Moore (2019) documented ”algorithm appreciation” — a countervailing tendency for some individuals, particularly those with less confidence in their own judgment, to prefer algorithmic recommendations over human ones — suggesting that the population of customers responding to AI customer service is likely to be heterogeneous, with some customers actively welcoming AI service and others resisting it.

The cumulative evidence from the algorithm aversion literature thus suggests a nuanced picture: aversion exists, is psychologically real, is driven by identifiable mechanisms (opacity, control deprivation, uniqueness neglect, human favoritism), but is not universal and depends heavily on task and context. This dissertation applies this understanding to the specific context of inbound e-commerce customer service, where the stakes are different from those in medical, financial, or forecasting contexts, and where the specific task dimensions (routine troubleshooting versus relational purchase inquiry) may determine whether and to what degree aversion is activated.

Effects of AI Disclosure on Consumer Behavior

The question of how disclosure of AI identity affects consumer behavior is distinct from, though related to, the algorithm aversion question. Algorithm aversion can in principle operate even without explicit disclosure — if customers infer that they are being served by AI from contextual cues such as response speed, writing style, or conversational patterns — but the regulatory and ethical context increasingly mandates explicit disclosure, making the behavioral effects of such disclosure a matter of practical urgency. This section reviews what is known about the effects of AI disclosure across consumer contexts.

The general presumption in the transparency and disclosure literature in marketing and consumer behavior is that disclosure of relevant information to consumers is welfare-enhancing and typically increases trust, assuming the disclosed information is positive or neutral. Extensive research on persuasion knowledge (Friestad and Wright, 1994) and disclosure of advertising intent (Campbell and Kirmani, 2000) has established that consumers are sophisticated information processors who use disclosed information to calibrate their responses. In the context of human service agents, disclosures about agent expertise, credentials, or role have generally positive effects on customer confidence and satisfaction (Kelley et al., 1990). However, the literature on AI identity disclosure suggests that the dynamics are meaningfully different when the disclosed information is that the service provider is not human.

The most important and most cited study on AI disclosure in a consumer context is Luo, Tong, Fang, and Qu (2019), who examined the effects of disclosing chatbot identity in outbound sales calls for an insurance product in China. The experimental design varied whether and when the chatbot disclosed its AI identity: in some conditions, the chatbot identified itself as AI before beginning the sales conversation; in other conditions, it disclosed AI identity only after the interaction had already generated some engagement; and in control conditions, no disclosure was made. The headline finding — that pre-interaction AI disclosure reduced purchase rates by 79.7 percent relative to the no-disclosure control — attracted enormous attention from both academics and practitioners when published and has been widely cited as evidence that AI disclosure is commercially catastrophic.

However, several features of the Luo et al. (2019) design limit its generalizability. Most importantly, the outbound sales context is fundamentally different from the inbound service context studied in this dissertation. In outbound sales, the customer is being contacted unsolicited and must decide whether to engage with an unexpected solicitation. The reaction to AI disclosure in this context may be compounded by resistance to the solicitation itself — customers may use the AI identity as a socially convenient reason to terminate an interaction they were already predisposed to decline. Relatedly, the purchase decision being solicited (insurance) is a high-involvement financial decision with significant stakes, where the perceived trustworthiness of the advice-giver is paramount. AI disclosure in this context may trigger concerns about algorithmic advice in high-stakes decisions (related to the Longoni et al. uniqueness neglect mechanism) that are quite different from the dynamics in routine post-sale customer service.

A second important feature of the Luo et al. (2019) finding is the timing effect: disclosure after some initial engagement had much smaller negative effects than pre-interaction disclosure. This timing sensitivity suggests that the mechanism is not purely informational — simply knowing that the agent is an AI — but is partly about expectation setting and conversational momentum. When customers had already engaged with the chatbot before learning its identity, they had accumulated positive experience that partly offset the negative reaction to disclosure. When they knew before engaging that the agent was AI, the expectation of a qualitatively inferior interaction may have become self-fulfilling through motivated information processing. This timing finding has direct design implications for AI disclosure in customer service: firms may be able to reduce disclosure penalties by ensuring that disclosure comes after initial positive engagement rather than before.

Reich, Hershfield, and Hershfield (2023) explored whether showing evidence of AI learning and improvement could overcome algorithm aversion. In a series of experiments, they found that presenting AI systems as improving over time — specifically showing the AI's error rate declining through experience — increased willingness to use AI recommendations relative to a control where the AI's performance was shown as static. This ”growth mindset” framing reduced the sense that AI errors reflected stable, uncorrectable limitations and thereby mitigated the aversion response. This finding is theoretically relevant to disclosure design in customer service: firms may be able to frame AI disclosure in ways that emphasize AI improvement and learning capabilities, potentially reducing the negative impact on satisfaction.

Meng (2024), in a paper examining the growing use of AI in behavioral experiments, raises the concern that AI disclosure norms are in a state of flux and that consumers' responses to disclosure are themselves changing as AI becomes more familiar and prevalent. As AI interactions become more common in everyday contexts — customer service, content recommendation, navigation, personal assistants — the novelty of AI service may diminish and with it some portion of the AI aversion effect. This temporal dynamics point suggests that findings on AI disclosure from 2019 may already be outdating, and that longitudinal evidence tracking changes in disclosure effects over time would be valuable. The present dissertation's data span eight months (September 2023 through May 2024), providing some opportunity to examine time trends in the effects of AI involvement on satisfaction.

The regulatory context for AI disclosure is evolving rapidly and creates direct practical urgency for the questions addressed in this dissertation. The European Union's Artificial Intelligence Act, which entered into force in 2024, includes provisions requiring that AI systems used to interact with humans must clearly disclose their artificial nature to those humans, with exceptions only for certain narrow contexts. The United States Federal Trade Commission issued guidance in 2023 recommending that firms using AI to interact with consumers should disclose this use clearly and prominently. Several U.S. states, including California, have enacted or proposed legislation requiring disclosure of AI chatbot identity in commercial contexts. For firms like Anker operating at scale across multiple jurisdictions, the compliance question has already moved from ”should we disclose?” to ”how do we disclose in a way that minimizes harm to customer relationships?” The present dissertation's experimental evidence on this question therefore addresses an immediate and growing practical need.

An important gap in the existing disclosure literature is the absence of evidence on disclosure effects in inbound service contexts. All major AI disclosure studies to date — including Luo et al. (2019) and the laboratory experiments of Dietvorst et al.

(2015, 2018) and Longoni et al. (2019) — involve either outbound contact (where the AI initiates interaction) or hypothetical scenarios presented to participants in artificial settings. In inbound service, the customer has already made a decision to seek help and is, in a sense, committed to the interaction regardless of whether they prefer AI or human service. This commitment changes the psychology: unlike an outbound solicitation, which the customer can simply decline, inbound service involves a customer who needs a problem resolved and must decide how to navigate the service encounter rather than whether to engage at all. Algorithm aversion may still operate — the customer may express lower satisfaction or request escalation to a human — but the magnitude and form of the effect may differ from the outbound context. Establishing this empirically is one of the core contributions of Study 2.

Incentives as Compensatory Mechanisms

A distinctive feature of this dissertation is the experimental testing of coupon incentives as a potential mitigating mechanism for the negative effects of AI disclosure. The theoretical and empirical foundations for this intervention draw on several bodies of literature: the service recovery literature, the behavioral economics literature on unexpected rewards, and prospect theory's account of loss compensation.

The service recovery literature is the most directly relevant starting point. Research on how firms recover from service failures has consistently found that when something goes wrong in a service encounter, offering monetary or material compensation to the affected customer substantially restores satisfaction — sometimes to levels above the pre-failure baseline, a phenomenon called the "service recovery paradox" (McCollough, Berry, and Yadav, 2000). Smith, Bolton, and Wagner (1999) conducted a landmark study on the determinants of customer satisfaction with service recovery, finding that the two most powerful recovery tactics were an apology (which addresses the relational dimension of failure) and compensation (which addresses the material dimension). The magnitude of compensation needed to restore satisfaction scaled with the severity of the failure. Importantly, the interactional justice dimension of how compensation was offered — whether it felt genuine and proportionate — moderated the effectiveness of monetary recovery.

The connection between this literature and AI disclosure effects is conceptual but plausible: if customers experience AI disclosure as a form of implicit service quality failure — "I deserve human service and received AI service instead" — then coupon provision might function analogously to service recovery compensation, providing material restitution that partially offsets the dissatisfaction created by disclosure.

This framing casts the coupon not as a reward for completing the interaction but as an acknowledgment and partial remedy for the disclosure effect.

However, there is an important difference between the service recovery context and the AI disclosure context that deserves attention. In service recovery, the compensation is offered in response to an identifiable failure event. In the AI disclosure context, the "failure" is more ambiguous — receiving AI service is not inherently a failure, and framing the coupon as compensation for AI service risks stigmatizing AI more strongly than if no coupon were offered. The design of the coupon offer therefore matters: if the coupon is framed as a loyalty reward or appreciation gesture rather than as compensation for AI service, it may create positive affect without amplifying negative reactions to the AI disclosure itself. The experimental protocol in Study 2 uses a loyalty-framing for the coupon to manage this risk.

From behavioral economics, the literature on unexpected positive rewards provides a complementary theoretical mechanism. Loewenstein (1996), in his theoretical treatment of the role of anticipation in utility, argued that unexpected positive outcomes generate disproportionately large positive utility because they are processed against a reference point that did not anticipate them. Similarly, the behavioral literature on surprise-based rewards documents that unexpected gifts or discounts create strong positive affect that persists and enhances evaluation of the entire experience context (Schwartz, 2004). If customers receiving AI service experience a baseline mild dissatisfaction due to AI aversion effects, an unexpected coupon delivered alongside or after the AI interaction may generate sufficient positive affect to compensate for the dissatisfaction, resulting in a net-neutral or even net-positive satisfaction outcome. The key word is "unexpected": a coupon that is routinely expected would not generate this positive surprise effect and would simply be treated as a service feature rather than a mood-elevating gift.

Kahneman and Tversky's (1979) prospect theory provides a further theoretical grounding. Prospect theory holds that gains and losses are evaluated relative to a reference point and that losses loom larger than equivalent gains — the well-documented loss aversion effect. In the AI disclosure context, disclosure of AI service may be experienced as a small loss relative to the reference point of expecting human service: the customer "loses" the human service they may have anticipated. A coupon, by providing a gain (discount on future purchase), may partially offset this loss, but prospect theory predicts that the coupon must provide sufficient gain magnitude to overcome the disproportionate weight of the disclosure-induced loss. This theoretical prediction can in principle be tested if the experiment collects data on coupon valuation alongside satisfaction measures.

One dimension of AI's interaction with coupon provision that has received no empirical attention but is theoretically interesting is AI's potential advantage in coupon delivery. Human agents face social awkwardness in proactively offering coupons to customers who have not complained: it may feel patronizing, as if acknowledging that the service was subpar without being asked to do so. AI agents, by contrast, face no such social awkwardness — they can be programmed to deliver coupon offers systematically without any sense of embarrassment or inappropriate admission. This asymmetry suggests that AI may actually be better positioned than human agents to deliver compensatory incentives in a way that feels natural and non-stigmatizing. If this is the case, the combination of AI service with coupon provision may be optimally suited to this particular compensatory mechanism in ways that pure human service is not.

The practical framing of the H2 test in Study 2 is therefore as follows: among customers who receive AI disclosure, those who also receive a coupon should show higher NPS and lower escalation rates than those who receive disclosure without a coupon. This prediction does not require that the coupon fully eliminate the disclosure effect — only that it attenuate it. Even partial attenuation would have significant managerial value, as it would provide firms with a practical tool for managing disclosure-induced dissatisfaction while complying with transparency mandates.

Hybrid AI-Human Systems

The architecture of AI deployment in customer service is not a binary choice between pure automation and pure human service. A rapidly growing class of deployments uses hybrid systems in which AI handles an initial set of interactions and a human agent takes over when certain conditions are met: the query exceeds a complexity threshold, the customer expresses frustration or requests human contact, or the AI has failed to resolve the issue after a defined number of exchanges. The Shulex/Anker system studied in this dissertation is an instance of this hybrid architecture, and the effects of hybrid versus pure AI design on customer outcomes are one of the core empirical questions addressed in H3.

The theoretical case for hybrid superiority rests on the complementarity of human and AI capabilities that Brynjolfsson and McAfee (2014) identified as the defining feature of the second machine age. AI is superior for tasks that are well-defined, scalable, and knowledge-intensive in a structured sense; humans are superior for tasks that require emotional intelligence, judgment in ambiguous situations, and relationship management. In theory, a well-designed hybrid system should outperform either pure alternative by routing each type of task to the most capable agent. The AI handles the routine, structured initial exchanges — verifying the issue, providing standard information, collecting necessary details — while the human takes over precisely when these advantages no longer dominate and empathy, flexibility, and relationship investment are needed.

Empirical evidence on hybrid system effectiveness in customer service is sparse. Agrawal, Gans, and Goldfarb (2018) develop an economic model of the division of labor between AI and humans that predicts hybrid configurations will be optimal when tasks can be decomposed into predictable and unpredictable components. Their framework provides theoretical support for the hybrid architecture but offers limited guidance on the customer experience dimensions that are most relevant here. In adjacent literatures on human-AI teaming in knowledge work (Wang et al., 2020) and AI-augmented decision-making in medicine (Topol, 2019), the consistent finding is that hybrid human-AI collaboration outperforms either pure system on complex tasks while offering little advantage on routine tasks — consistent with the complementarity theory.

In the customer service context specifically, however, several features of hybrid systems may offset their theoretical advantages. The handoff between AI and human agents creates friction: the customer must repeat themselves, adjust to a new communication style, and potentially experience a gap in response time. In the Anker data, hybrid tickets average 3.6 to 4.2 response rounds — substantially more than either pure AI (1.0 to 1.3 rounds) or pure human (1.7 to 2.1 rounds). This elevated round count for hybrid tickets is partly a function of selection (hybrid tickets are those complex enough to require escalation) but may also reflect genuine process friction at the AI-to-human handoff point. If customers experience this friction as a service failure, it would work against the theoretical quality advantages of hybrid service.

The escalation data in the Shulex/Anker system provides a nuanced picture of hybrid system dynamics. Escalation rates for hybrid tickets range from 2.3 to 9.4 percent across months (with the decline from 9.4 percent in late 2023 to 2.3 percent in early 2024 reflecting improvements in the AI routing algorithm). These escalation rates are substantially higher than for pure human tickets (0.6 to 1.3 percent). However, interpreting this difference requires care: hybrid tickets are by definition those that began with AI service and may have been escalated because the AI was insufficient, meaning that escalation in the hybrid context reflects a decision about service quality that is structurally different from escalation in the pure human context (where escalation reflects a desire for a different or more senior human agent rather than a desire to move from AI to human service). Study 2's experimental design addresses this confound by controlling agent type in the randomization.

The ”human safety net” hypothesis underlying H3 is that customers who know or believe their AI service is backed by available human agents will experience lower AI resistance than customers who believe the AI is operating alone. This hypothesis draws on the control mechanism of algorithm aversion: if customers feel that a human can intervene if the AI fails, the accountability and control deficit associated with AI service is partially mitigated. Some indirect evidence for this hypothesis comes from Dietvorst et al.'s (2018) algorithm modification finding: giving people the ability to modify algorithmic outputs increased use even when the modifications were trivial, suggesting that perceived control substantially reduces aversion. The availability of human backup in a hybrid system functions analogously as a control mechanism — even if the human never actually intervenes, the knowledge that human intervention is available may be sufficient to reduce the psychological threat of AI failure.

The organizational design literature on AI governance also provides relevant context. Frey and Osborne (2017) and Acemoglu and Restrepo (2020), while primarily concerned with labor market effects, both discuss the conditions under which human-AI complementarity is most likely to hold. Their analyses suggest that human oversight is most valuable when AI errors are costly, difficult to detect in advance, and have heterogeneous impacts — precisely the conditions that characterize complex customer service encounters. This suggests that the hybrid architecture's value proposition is most compelling for complex tickets, with less differentiation from pure AI for routine tickets — an implication that the Study 1 analysis of complexity-contingent effects addresses directly.

Research Gap and Theoretical Contribution

Having reviewed the relevant literatures, three primary gaps can be identified that this dissertation is designed to address, along with the specific theoretical contributions the research makes.

The first gap is the absence of large-scale field evidence on AI effects in inbound e-commerce customer service. The algorithm aversion and AI disclosure literatures are dominated by laboratory experiments and small-scale field studies in outbound commercial contexts. The most cited field study, Luo et al. (2019), involved outbound insurance sales chatbots in China. The most robust experimental evidence on algorithm aversion comes from laboratory forecasting tasks (Dietvorst et al., 2015, 2018). There is no existing study with comparable scale (hundreds of thousands of tickets) examining AI effects on customer satisfaction and service process efficiency in an inbound e-commerce context. The present dissertation fills this gap with Study 1, providing the statistical power to detect even modest effects and the heterogeneity necessary to examine task-contingency.

The second gap is the lack of causal identification of AI disclosure effects in inbound service contexts. The Luo et al. (2019) experimental evidence on disclosure effects is compelling for outbound sales but cannot be extrapolated to inbound service for the structural reasons discussed in Section 2.3. No randomized experiment has previously examined AI disclosure effects in inbound customer service, where the customer is in a help-seeking rather than solicitation-resisting posture. Study 2 fills this gap with a randomized controlled trial that isolates disclosure effects from service quality effects in a live commercial deployment.

The third gap is the absence of tested interventions for mitigating AI disclosure effects. Even if disclosure reduces customer satisfaction — which both theory and prior evidence predict — the question of what firms can do about this effect has received no empirical attention in the academic literature. The coupon intervention tested in Study 2 is the first experimental test of a compensatory behavioral intervention designed specifically to mitigate AI disclosure effects on customer satisfaction. The hybrid architecture test in H3 is the first experimental test of whether system design modifies customer responses to AI service in a field context. Together, these tests provide actionable evidence for firms navigating the challenges of AI disclosure under regulatory pressure.

The theoretical contribution of this dissertation goes beyond extending the algorithm aversion tradition. The core finding — that AI achieves efficiency gains without satisfaction penalties in 86% of service contexts, but produces satisfaction penalties specifically where customers seek relational and advisory judgment — implies a **restructuring of the service value architecture** in the AI era.

In traditional service management, functional performance (did the problem get solved?) and relational performance (did I feel treated as a person who matters?) are inseparably bundled in a single human service encounter. AI *decouples* these two value dimensions. *Functional value* — resolution accuracy, response speed, consistency — is efficiently delivered by AI and is, on the evidence of this dissertation, sufficient to sustain customer satisfaction in transactional service interactions.

Relational value — empathy, personalization, trust signaling, human acknowledgment of individual circumstances — cannot be substituted by AI in high-stakes relational interactions, where customers are evaluating purchase decisions or seeking affirmation of their standing as valued customers.

This conceptual distinction between **transactional service tasks** (where AI is a tool and disclosure is low-risk) and **relational service tasks** (where AI is a social agent and disclosure threatens the trust relationship) reconstitutes the theoretical boundary of algorithm aversion research in architecturally useful terms. The relevant question for firms is no longer ”does algorithm aversion exist?” but ”which tasks require relational value delivery that AI cannot yet provide, and how should disclosure and compensation be designed differently for each task type?” This reframing elevates the dissertation’s contribution from an empirical extension of the aversion literature to a foundational framework for an AI-era theory of service value decomposition — one with direct implications for service design, routing policy, and regulatory compliance strategy.

RESEARCH SETTING AND DATA

Research Setting

This dissertation is situated within the customer service operations of Anker Innovations, one of the world's largest consumer electronics brands operating primarily through Amazon's marketplace, in partnership with Shulex Technology, Anker's AI customer service software provider. Understanding the research setting in detail is essential not only for interpreting the empirical findings but for assessing the external validity of those findings and their applicability to other firms and contexts.

Shulex Technology is a SaaS company founded in the early 2020s and headquartered in China, with a primary focus on providing AI-powered customer service infrastructure to e-commerce companies selling on Amazon and other major online marketplaces. Shulex's platform integrates ticket management, intent classification, AI-generated response drafting, quality control workflows, and performance analytics into a unified system designed specifically for the e-commerce customer service context. The platform's distinguishing feature, relative to generic chatbot or LLM-API solutions, is its deep integration with e-commerce operational data — product catalogs, order histories, return policies, and Amazon marketplace-specific communication requirements — which enables the AI to generate responses that are not only linguistically fluent but operationally accurate and brand-consistent.

Anker Innovations was founded in 2011 by Steven Yang, a former Google engineer, and has grown to become one of the most successful Chinese consumer electronics export brands globally. Headquartered in Shenzhen, Anker operates through a portfolio of sub-brands including Anker (charging and mobile power), Eufy (home security), Soundcore (audio), and Nebula (projection). Its products are sold primarily through Amazon in the United States, United Kingdom, Germany, Japan, and other major markets, with Amazon representing the largest single revenue channel. Anker's annual revenues are estimated at approximately two billion U.S. dollars, with a substantial proportion generated through the Amazon U.S. marketplace.

The customer service challenge facing a firm like Anker at this scale is significant. Post-sale customer inquiries on Amazon arrive via two primary channels: direct email through Amazon's Buyer-Seller Messaging system and Amazon's A-to-Z

Guarantee escalation system. Amazon's communication policies require sellers to respond to buyer messages within 24 hours and govern the content of communications to prevent certain types of commercial solicitation. This regulatory context means that Anker's customer service operation is not merely a support function but a compliance function, with response time and content standards that must be maintained at scale.

Prior to the Shulex AI deployment, Anker's customer service for the U.S. market relied on human agents. The volume of incoming tickets — routinely in the tens of thousands monthly — required a substantial team working in shifts to maintain response time compliance. The topics of these tickets span a wide range: technical troubleshooting (product not working, connectivity issues, compatibility questions), product usage (how to use features, configuration guidance), warranty and return inquiries, and purchase-stage questions about products not yet owned. The diversity of topics, combined with the volume and the 24-hour response mandate, made the customer service operation both labor-intensive and logistically complex.

The deployment of Shulex's GPT-based response system in late 2023 represented a fundamental shift in this operational model. Rather than requiring human agents to draft every response from scratch, the system generates AI-drafted responses that agents can review and send (in hybrid mode) or sends directly without human review (in pure AI mode). The knowledge base integration ensures that AI-generated responses accurately reference product specifications, current pricing and promotion information, warranty terms, and troubleshooting procedures for each product in Anker's catalog. The speed advantage is immediate: AI-generated responses are available within seconds of ticket creation, whereas human agent responses depend on agent availability and queue depth.

The geographic and temporal context of the data is also important to note. All tickets in the dataset involve English-language customer communications, primarily from U.S.-based Amazon purchasers. The study period of September 2023 through July 2024 encompasses the holiday shopping season (November-December 2023), which generates a surge in new product activations and associated support tickets, and the post-holiday period through mid-2024. These seasonal patterns create some temporal variation in ticket volume and type that the analyses must account for.

AI Integration and Service Process

Understanding the technical architecture and operational workflow of the Shulex AI customer service system is essential for interpreting the data correctly and for understanding how the different agent type designations — pure AI, AI-plus-human hybrid, and pure human — are operationally defined.

The service process begins with ticket creation. When an Amazon buyer sends a message to an Anker seller account, the message arrives in Shulex's ticket management system and triggers an automated intake process. The first stage of this process is intent classification: a classification model analyzes the message content and assigns it to one of a set of predefined intent categories — Troubleshooting, Product Usage, Purchase Inquiry, Warranty Claim, Return/Refund Request, Shipping/Delivery Inquiry, and Others. This classification determines the knowledge base domain that the AI response generation module will draw upon. Simultaneously, a complexity scoring model assigns the ticket a complexity score on a four-point scale: complexity 1 indicates a simple, formulaic inquiry with a standard answer; complexity 2 indicates a moderately complex inquiry requiring some customization; complexity 3 indicates a complex inquiry requiring extended contextual reasoning; and complexity 4 indicates a highly complex inquiry that the system flags as likely requiring human judgment. These complexity scores are assigned before any response is generated and are therefore properly treated as pre-treatment covariates in the empirical analysis.

The second stage is AI response generation. For eligible tickets (those not flagged as requiring immediate human response based on complexity or sensitive word detection), the Shulex system passes the ticket content, along with relevant product knowledge base information and any available order history for the customer, to the GPT API. The LLM generates a draft response that is either held for human review (in hybrid mode) or sent directly to the customer (in pure AI mode). The response generation step typically takes a few seconds and produces a response calibrated to the ticket's intent and complexity.

The third stage is routing and agent type assignment. Routing decisions are made based on the combination of intent classification output, complexity score, and product category. Certain combinations — for example, high-complexity warranty claims for high-ticket products — are systematically routed to human agents (pure human). Other combinations — low-complexity troubleshooting for standardized products — are eligible for pure AI routing. The majority of tickets fall into the hybrid category, where AI handles the initial response and human agents are available to intervene. The routing logic evolved over the study period as Shulex's system learned from quality control feedback and as Anker's operational team adjusted routing parameters based on satisfaction data. This evolution of the routing rules is part of the endogeneity challenge that the empirical strategy in Chapter 5 must address.

In the hybrid process, the AI sends the first response (and potentially several subsequent responses) while a human agent monitors the conversation thread. The human takes over when one of several escalation triggers is activated: the customer explicitly requests a human agent, the AI has failed to resolve the issue after a defined number of rounds, the system detects sentiment indicators of customer frustration, or the conversation enters a complexity domain beyond the AI's routing threshold. The average handover point in the data is at response round 2.56, meaning that the AI handles approximately 2.56 message exchanges before the human takes over in the average hybrid ticket. The human agent then continues the conversation through resolution.

The escalation process is an important feature of the data that requires careful interpretation. In the context of this study, ”escalation” refers specifically to the event in which a customer service ticket that was initially assigned to AI handling is subsequently transferred to a human agent — either because the customer requested it, because the AI failed to resolve the issue, or because the system's quality monitoring flagged the ticket for human review. This is distinct from the more general sense of ”escalation” used in some customer service literatures to mean referral to a senior human agent. The escalation rate is therefore a measure of AI service insufficiency from the system's perspective and a measure of customer demand for human service from the customer's perspective.

The quality control (QC) process is a separate but important component of the operational system. A random sample of tickets (representing a subset of the roughly 277,517 QC records in the dataset) is reviewed by a dedicated QC team who rate each reviewed ticket on a four-category scale: Pass (response met quality standards), Problem (response contained an error or inadequacy), Suggestion (response was acceptable but could be improved), and Good Reply (response exceeded standard quality). The error type for Problem tickets is also classified into a taxonomy that includes factual errors, inappropriate tone, failure to address the customer's actual question, and policy violations. These QC data provide an independent measure of response quality that can be used to construct objective service quality controls in the empirical analysis.

Following ticket resolution, a subset of customers receives an NPS survey via email. The survey asks customers to rate their overall satisfaction with the service experience on a scale of 1 to 10. The survey timing and targeting criteria are managed by Shulex based on Anker's instructions and represent a sampling fraction rather than a census of all resolved tickets. The 23,853 NPS responses in the dataset (for the February through May 2024 period) represent the available satisfaction outcome data for Study 1. The discrepancy between the 477,983 total tickets and the 23,853 NPS responses reflects both the partial survey coverage and the timing constraints: NPS surveys were not systematically deployed until February 2024, limiting the NPS-linked analysis to a four-month window within the broader eight-month study period.

Data Sources and Description

The empirical foundation of this dissertation rests on five distinct datasets extracted from the Shulex platform for the period September 2023 through July 2024. Table 3.1 provides a summary of these data sources.

Table 3.1: Summary of Data Sources

Dataset	Time Period	N Observations	Unit of Analysis	Key Variables
Customer Tickets	Sep 2023 – Jul 2024	477,983 (observational)	Ticket	ticket_id, category, intent, complexity, is_escalation, created_at
Response Messages	Sep 2023 – Jul 2024	Multi-million records	Message	response_id, ticket_id, kind (AI/human), created_at
NPS Surveys	Feb 2024 – May 2024	23,853	Survey response	survey_id, ticket_id, score (1–10), perceived_AI_human
QC Reviews	Sep 2023 – May 2024	277,517	QC record	result (pass/problem/suggestion/good_reply), error_type
Experiment Batches	Jul 2024	1,474	Assignment	ticket_id, batch_number (1–5), created_at

The Customer Tickets dataset is the primary dataset for Study 1 and the source of the dependent and independent variables for the main regression analyses. Each record represents a single customer service ticket, identified by a unique ticket_id, and contains metadata including the product category, the customer's stated intent (as classified by the Shulex system), the pre-assigned complexity score, a binary indicator for whether escalation occurred, and the timestamp of ticket creation. This dataset is linked to the Response Messages dataset through ticket_id, enabling construction of the derived variable N_responses (total number of response exchanges to resolution).

The Response Messages dataset contains a record for each individual message sent in a ticket thread, tagged with the kind variable indicating whether the message was generated by the AI system or written by a human agent. By aggregating over this dataset by ticket_id and counting the number of unique response exchanges, it is possible to construct N_responses as a measure of service efficiency. By examining the sequence of message kinds within each ticket, it is also possible to derive the pure_ai, pure_human, and mix_ai_human indicator variables, as well as the specific round at which any AI-to-human handover occurred in hybrid tickets.

The NPS Survey dataset contains the customer satisfaction records for the February through May 2024 period. Each record links a survey response to a specific ticket, enabling merging with the Customer Tickets dataset for analysis of satisfaction outcomes. The NPS score variable ranges from 1 to 10, with mean 7.79 (SD = 2.43) across the 23,853 available responses. The dataset also contains a perceived_AI_human variable capturing the customer's self-reported perception of whether they were served by an AI or human agent, which provides a useful manipulation check for the disclosure conditions in Study 2.

The QC Reviews dataset contains independent quality assessments of a sample of customer service interactions. These assessments are made by Shulex's quality control team without knowledge of the NPS outcomes and therefore provide an objective, orthogonal measure of service quality that can be used as a control variable or robustness check in the satisfaction analysis. The distribution of QC outcomes across the 277,517 reviewed tickets shows that approximately 68 percent receive a Pass rating, 15 percent receive a Suggestion rating, 12 percent receive a Good Reply rating, and 5 percent receive a Problem rating. The problem rate is therefore relatively low but non-trivial, and the error types associated with Problem ratings provide detailed diagnostics of AI versus human error patterns.

The Experiment Batches dataset contains the randomization records for Study 2. The 1,474 ticket assignments in this dataset represent the initial experimental assignments in July 2024, with each assignment linked to a batch number (1 through 5, corresponding to the experimental conditions defined in Chapter 7) and a timestamp. This dataset is linked to the Customer Tickets and NPS Survey datasets through ticket_id to enable outcome analysis.

The data extraction and management process for this dissertation involved a formal data access agreement between the author and Shulex Technology, with appropriate data anonymization protocols applied to remove personally identifiable information about both customers and Anker agents. All customer identifiers were replaced with anonymized pseudonyms before transmission to the research team. The data were analyzed in a secure research computing environment with access restricted to the research team. The research protocol was reviewed and approved by the HKU institutional review board.

Variable Definitions

This section provides formal definitions for all key variables used in the empirical analyses of both Study 1 and Study 2. Precise variable definitions are essential for the reproducibility and interpretability of the findings, and the definitions here should be treated as the authoritative source for all subsequent quantitative analyses.

is_escalation is a binary variable equal to 1 if a customer service ticket that was initially assigned to AI handling (either pure AI or hybrid) was subsequently transferred to a human agent at any point during the service interaction, and equal to 0 if the ticket was resolved without such a transfer. This variable captures both customer-initiated escalation requests and system-initiated transfers triggered by quality monitoring rules. For pure human tickets, is_escalation captures transfers to a different or more senior human agent. The escalation rate is therefore defined as the proportion of tickets in a given group or time period for which is_escalation equals 1.

score is the NPS satisfaction score assigned by the customer on a 1-to-10 integer scale in the post-service survey, where 1 represents extreme dissatisfaction and 10 represents extreme satisfaction. In standard NPS methodology, respondents scoring 9 or 10 are classified as Promoters, those scoring 7 or 8 are classified as Passives, and those scoring 6 or below are classified as Detractors. The NPS index is computed as (proportion of Promoters − proportion of Detractors) × 100, yielding a range of −100 to +100. In this dataset, the mean score is 7.79 (corresponding to a relatively favorable NPS index of 36.7), though this aggregate masks substantial variation across product categories, agent types, and time periods.

N_responses is a count variable representing the total number of back-and-forth message exchanges between the customer and the service agent(s) from ticket creation through resolution. A single ”exchange” is operationalized as a customer message followed by an agent response, so N_responses equals the number of complete customer-agent interaction cycles. Tickets resolved in a single exchange have N_responses = 1. This variable serves as the primary measure of service process efficiency in Study 1.

ai_involve is a binary variable equal to 1 if at least one AI-generated response was sent in the ticket thread, and equal to 0 if all responses were written by human agents.

human_involve is a binary variable equal to 1 if at least one human-agent-written response was sent in the ticket thread, and equal to 0 if all responses were AI-generated.

pure_ai is a binary variable equal to 1 if ai_involve = 1 AND human_involve = 0, meaning all responses in the ticket were generated by AI with no human agent involvement. This is the pure AI agent type indicator.

pure_human is a binary variable equal to 1 if ai_involve = 0 AND human_involve = 1, meaning all responses in the ticket were written by human agents with no AI involvement. This is the pure human agent type indicator.

mix_ai_human is a binary variable equal to 1 if ai_involve = 1 AND human_involve = 1, meaning the ticket involved both AI-generated and human-written responses. This is the hybrid agent type indicator. Note that by construction, exactly one of pure_ai, pure_human, and mix_ai_human equals 1 for any given ticket.

complexity is an ordinal variable on a 1-to-4 scale representing the Shulex system's pre-assigned assessment of the ticket's complexity level. Complexity 1 indicates a simple inquiry with a standard, easily located answer. Complexity 2 indicates a moderately complex inquiry requiring some judgment in response selection. Complexity 3 indicates a complex inquiry involving multiple issues, nuanced product knowledge, or significant customer context. Complexity 4 indicates a highly complex inquiry that the system flags as likely exceeding AI capability thresholds. Importantly, complexity is assigned before any response is generated, meaning it is exogenous to the quality of service provided and can properly be treated as a pre-treatment control variable.

pis_name is a categorical variable identifying the product category associated with the ticket. The dataset includes 19 distinct product categories, including Edge Security (Eufy home security cameras and systems), TWS (true wireless stereo earbuds under the Soundcore brand), Charger, Battery, PhoeniX, Headband (wired audio headsets), Smart Audio, Smart Projector (Nebula projectors), Pro Security, and 3D Print, among others. Product categories differ systematically in their mix of query types, complexity distributions, and AI routing rates, making the category variable an important control and source of heterogeneity in the analysis.

intent is a categorical variable identifying the customer's stated service intent as classified by the Shulex intent classification model. The primary intent categories are: Troubleshooting, Product Usage, Purchase Inquiry, Warranty Claim, Return/Refund Request, Shipping/Delivery Inquiry, and Others. Intent is classified at the beginning of the ticket process, before response generation, and is therefore exogenous to service quality. The distribution of intent types varies across product categories (edge security generates relatively more troubleshooting tickets; Charger generates relatively more usage inquiries) and creates the cross-category intent variation exploited in the task-contingency analysis.

batch_number is an integer variable taking values 1 through 5 corresponding to the five experimental condition batches in the Study 2 RCT. The mapping from batch_number to experimental condition is defined in the experimental protocol described in Chapter 7. This variable serves as the primary treatment indicator in Study 2 analyses.

Descriptive Statistics

The descriptive statistics presented in this section document the key empirical patterns in the observational data that motivate the hypotheses and inform the empirical strategy. Two primary tables summarize the monthly and cross-category variation in agent type distribution, service efficiency, escalation, and satisfaction.

Table 3.2 presents the monthly trends in AI adoption and key outcomes across the six months from September 2023 through February 2024 for which complete data are available. The evolution of these trends reveals several important patterns that are worth examining in detail before moving to formal regression analysis.

Table 3.2: Monthly Trends in AI Adoption and Key Outcomes (Sep 2023–Feb 2024)

Month	N	Pure AI (%)	Hybrid (%)	Pure Human (%)	Rounds (AI)	Rounds (Hybrid)	Rounds (Human)	Escalation (Hybrid %)	Mean Score (Hybrid)
2023-09	61,095	18.8	28.6	52.6	1.00	4.02	1.83	9.4	8.08
2023-10	70,988	23.0	36.7	40.4	1.02	3.86	1.78	9.4	8.11
2023-11	87,176	30.7	39.9	29.4	1.22	4.00	1.87	4.3	8.04
2023-12	92,725	36.8	39.6	23.6	1.29	4.24	1.93	2.5	8.02
2024-01	86,869	39.0	36.8	24.2	1.28	4.17	2.10	2.3	7.83
2024-02	79,130	37.0	35.3	27.7	1.31	3.56	1.67	2.3	7.61

Note: Escalation rate for pure AI ≈ 0% throughout. Score data were systematically collected beginning February 2024; earlier month means reflect limited available data.

Several patterns in Table 3.2 are striking. First, AI adoption grew dramatically and rapidly: the share of pure AI tickets increased from 18.8 percent in September 2023 to approximately 37-39 percent by January-February 2024, while the pure human share declined correspondingly from 52.6 percent to 23-28 percent. This rapid adoption trajectory demonstrates the operational viability of the AI system from Anker's perspective but also creates the temporal variation in AI exposure that is central to the identification strategy in Study 1.

Second, the efficiency differential between agent types is consistent and dramatic across all months. Pure AI tickets involve 1.00 to 1.31 response rounds — essentially resolving in a single exchange in the majority of cases. Hybrid tickets involve 3.56 to 4.24 rounds, and pure human tickets involve 1.67 to 2.10 rounds. The difference between pure AI and hybrid is particularly striking and raises the question of selection: are hybrid tickets substantially more complex on average than pure AI tickets, accounting for the higher round count? The regression analyses in Chapter 6 address this selection question by controlling for ticket complexity.

Third, the escalation rate for hybrid tickets shows a dramatic improvement over the study period, declining from 9.4 percent in September-October 2023 to 2.3 percent by January-February 2024. This decline likely reflects ongoing improvement in the Shulex AI model through quality control feedback, as well as refinements to the routing rules that improved the initial assignment of tickets to the most appropriate agent type. The declining escalation rate over time is an important trend that the empirical analysis must account for through appropriate time controls.

Fourth, the mean NPS score for hybrid tickets (the only group for which systematic NPS data are available across months) shows a declining trend: from 8.08-8.11 in the earliest months to 7.83-7.61 by January-February 2024. This decline may reflect the composition effect of an expanding AI-served population — as AI routing expanded to cover more complex tickets, the average difficulty of the remaining hybrid and human tickets may have changed — or may reflect actual changes in service quality over time. The regression analysis addresses this question by controlling for ticket complexity and product category composition.

Table 3.3 presents the cross-sectional distribution of ticket volume and key outcomes by product category for the full observational study period. The variation across categories provides the cross-sectional identification variation that complements the temporal variation in Table 3.2.

Table 3.3: Sample Distribution by Product Category (Sep 2023–Feb 2024)

Category	N Tickets	Pure AI (%)	Hybrid (%)	Pure Human (%)	Escalation %	Mean Score
Edge Security	84,608	36.6	40.8	22.6	2.72	7.93
TWS	88,440	41.0	37.0	22.0	1.22	8.04
Charger	64,370	32.0	42.4	25.6	1.99	8.09
Battery	47,751	26.7	45.2	28.1	2.56	7.94
PhoeniX	47,127	22.1	40.0	38.0	1.95	7.97
Headband	19,667	34.3	43.5	22.2	1.80	7.98
Smart Audio	16,359	38.1	36.5	25.3	2.09	8.02
Smart Projector	12,104	48.9	29.0	22.2	1.58	7.81
Pro Security	8,076	25.3	50.0	24.7	3.67	8.02
3D Print	7,580	26.3	31.8	42.0	0.44	7.97

Note: Mean Score reflects available NPS data; coverage varies across categories. Totals across categories do not sum to 477,983 due to the exclusion of smaller categories not listed. Escalation rate across all agent types within category.

The cross-category variation in Table 3.3 reveals several substantively interesting patterns. Smart Projector has the highest pure AI rate at 48.9 percent, while PhoeniX has the lowest at 22.1 percent. Pro Security has the highest escalation rate at 3.67 percent, while 3D Print has the remarkably low escalation rate of 0.44 percent. Mean NPS scores are relatively consistent across categories, ranging from 7.81 for Smart Projector to 8.09 for Charger, though these differences may reflect both genuine service quality differences and composition effects in the NPS survey coverage.

The joint distribution of agent type and product category is informative for the identification strategy. The variation in pure AI rates across categories (ranging from 22.1 percent for PhoeniX to 48.9 percent for Smart Projector) combined with the variation in escalation rates and satisfaction scores creates the cross-category-by-agent-type variation needed for the fixed-effects regression. Importantly, the variation in AI adoption rates across categories is not random but reflects deliberate routing policy choices by Anker: products with more technically complex and variable customer inquiries (like home security systems, which generate high escalation rates) are routed to human agents more frequently, while products with more predictable inquiry patterns (like simple chargers) are more easily handled by AI. This selection makes simple comparisons of AI versus human satisfaction scores misleading, motivating the complexity and intent controls in the regression specification.

Beyond the summary statistics presented in these tables, several additional descriptive observations are worth noting. First, the NPS survey participation rate varies substantially across ticket and agent types, creating a potential selection bias in the NPS outcome analysis. Customers who complete the NPS survey may differ systematically from those who do not in ways that correlate with both agent type and satisfaction. The empirical strategy in Chapter 5 addresses this concern through Heckman selection corrections and sensitivity analyses on the survey participation assumption. Second, the correlation between complexity and agent type is substantial and expected: the mean complexity score for pure AI tickets is 1.34, for hybrid tickets 2.17, and for pure human tickets 2.89, reflecting the routing logic that directs complex tickets away from pure AI. This complexity-agent type correlation is the primary confound that the HDFE regression strategy must address. Third, the distribution of intent types varies by product category in ways that create the task-contingency variation exploited in H1c: purchase inquiry tickets constitute a substantially higher proportion of total tickets for some categories (notably charging products and audio, where customers frequently write pre-purchase questions) than others (such as security cameras, where almost all inquiries are from existing customers with troubleshooting needs).

The descriptive statistics thus paint a picture of a large, complex, and rapidly evolving customer service operation in which AI adoption has been rapid and consequential, efficiency differences across agent types are substantial, and the cross-category and cross-intent variation is sufficiently rich to support the empirical analyses planned for Study 1. The variation is not random — it reflects deliberate routing decisions and system evolution over time — and the empirical strategy must account for this non-randomness carefully. Chapter 5 develops the specific econometric approach for doing so.

THEORETICAL FRAMEWORK AND HYPOTHESES

Conceptual Framework

The theoretical framework of this dissertation integrates four foundational theoretical perspectives to develop a causal account of how AI disclosure affects customer value in inbound customer service, how coupon incentives moderate this effect, and how agent type architecture shapes the customer experience of AI service. Before formally stating the hypotheses, this section presents the integrated conceptual framework and articulates its theoretical grounding.

Figure 4.1 provides a schematic representation of the conceptual model.

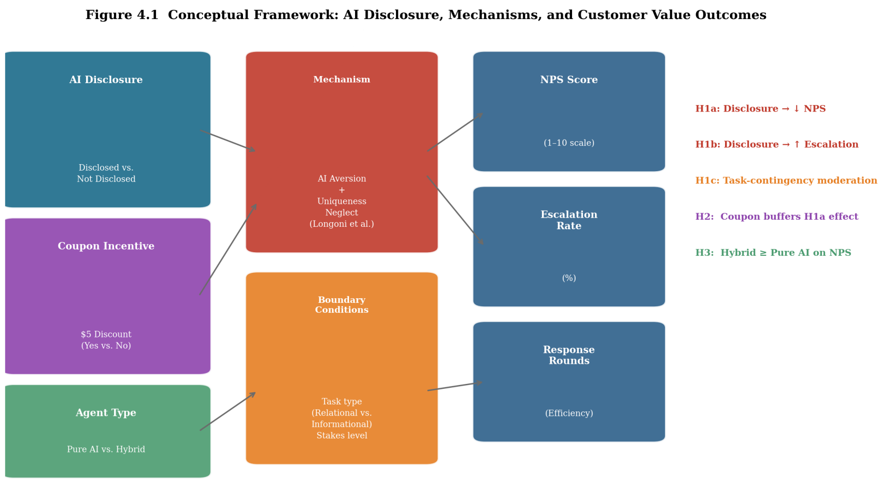

Figure 5.1: Conceptual Framework: AI Disclosure Effects on Customer Value

The central pathways in the framework are as follows. The primary treatment variable of interest in Study 2 is AI disclosure: the act of informing a customer, explicitly and in advance, that their service is being provided by an artificial intelligence system rather than a human agent. Disclosure operates through two theoretical channels. First, it activates AI resistance mechanisms in the customer — the psychological processes associated with algorithm aversion, including perceived opacity, control deprivation, uniqueness neglect, and human favoritism — that are otherwise latent and not activated by AI service per se (i.e., without disclosure, customers may not know they are being served by AI and therefore do not activate resistance). Second, disclosure creates an expectation frame: knowing one is being served by AI modifies the customer's expectations about service quality, response flexibility, and emotional support, potentially creating a standard against which actual AI performance is evaluated.

The first downstream outcome from these mechanisms is customer satisfaction as measured by NPS score. Activated AI resistance reduces the subjective evaluation of AI service quality, even holding objective quality constant, because it introduces a negative affective overlay on the experience. The second downstream outcome is escalation rate: customers whose AI resistance is activated by disclosure may be more likely to request a human agent, either as an expression of distrust in the AI or as a proactive strategy to secure higher-quality service they believe the AI cannot provide. The third outcome dimension is response rounds: while disclosure is not expected to directly affect the number of message exchanges (which is driven more by query complexity and resolution efficiency than by disclosure), agent type does directly affect response efficiency through the operational mechanisms described in Chapter 3.

The framework includes two moderating variables. Coupon incentives moderate the path from AI disclosure to customer satisfaction: by providing an unexpected positive reward, coupons create compensatory affect that partially offsets the satisfaction reduction from activated AI resistance. The moderation operates at the affective level — the positive emotion generated by the unexpected coupon gift partially cancels the negative emotion generated by disclosure-induced AI resistance — rather than by eliminating the cognitive mechanisms of aversion. Agent type (pure AI versus hybrid) moderates the relationship between AI service and both satisfaction and escalation through the perceived human safety net mechanism: customers receiving hybrid service who know human backup is available experience lower perceived control deprivation, reducing the intensity of AI resistance activation.

The framework also identifies several control variables that must be included in the empirical analysis to ensure that the relationships between treatment and outcome are not confounded. Ticket complexity, as discussed in Chapter 3, is systematically related to agent type routing and is an important determinant of service difficulty. Product category captures systematic differences in the types of customers, query patterns, and service expectations across Anker's product lines. Customer intent type captures the nature of the service task and is the primary dimension along which H1c hypothesizes task-contingent heterogeneity. Time period controls capture any secular trends in satisfaction or AI performance that might otherwise be attributed to agent type or disclosure effects.

Before elaborating the four behavioral mechanisms, it is useful to establish the economic logic that motivates the disclosure problem. The theoretical framework draws on the economics of disclosure and Bayesian persuasion (Grossman, 1981; Milgrom, 1981; Kamenica and Gentzkow, 2011) to ground the mechanism in a model of rational inference. In the Grossman-Milgrom ”unraveling” framework, a seller who privately knows a product’s quality has an incentive to disclose high quality voluntarily; buyers, anticipating this logic, infer that non-disclosure signals low quality. Applied to AI service: if customers rationally update their beliefs about service quality based on whether disclosure occurs, non-disclosure may paradoxically generate negative inferences (”if the firm is not disclosing AI involvement, perhaps the service quality is lower than expected”). This Bayesian inference effect generates a disclosure equilibrium that is more nuanced than the naive prediction that disclosure is always costly: the welfare effect of disclosure depends on what prior beliefs customers hold about AI service quality, and on what service quality the AI actually delivers.

Kamenica and Gentzkow’s (2011) Bayesian persuasion framework extends this insight to the case where a sender (the firm) can choose the *structure* of information to disclose — not just whether to disclose. In this dissertation’s context, the firm can choose whether to disclose that AI is serving the customer, and if so, how to frame that disclosure (with or without a coupon, with or without a capability description). The optimal disclosure policy, in this framework, depends on the prior belief distribution of customers about AI service quality and the firm’s objective function. If customers have pessimistic priors about AI quality (as algorithm aversion research suggests for relational tasks), voluntary disclosure is costly in expectation. If customers have optimistic priors (as some evidence on AI efficiency expectations suggests), disclosure may be neutral or positive. This theoretical framework motivates empirical identification of the sign and magnitude of the disclosure effect — the key contribution of Study 2 — and provides a principled basis for the finding that disclosure effects are task-contingent: the prior belief distribution that customers bring to a purchase inquiry interaction (skeptical of AI judgment) differs from the prior they bring to a troubleshooting interaction (potentially neutral or positive about AI accuracy).

The four behavioral mechanisms elaborate the psychological channels through which this Bayesian updating process affects customer affect and behavior, as follows.

Social presence theory (Short, Williams, and Christie, 1976) explains *why AI dis-*

closure reduces the relational value of the interaction. All text-based service has inherently limited social presence relative to face-to-face communication. When a customer learns that text responses are AI-generated, the implicit assumption of human authorship is withdrawn, further reducing perceived social presence: cues the customer might have read as warmth, empathy, or personal attention are reinterpreted as simulated affect. This theory therefore explains the main effect of disclosure on satisfaction (the path labeled β_disclosure → NPS in Figure 4.1) and predicts the effect will be stronger when relational dimensions of the interaction are salient — specifically in purchase inquiry tasks — directly motivating H1c.

Attribution theory (Weiner, 1985) explains *why disclosed AI service produces higher escalation rates*. When service failures occur under human delivery, customers attribute imperfection to unstable, external, and potentially correctable causes (workload, a bad day, insufficient information). Under AI disclosure, the same imperfection is attributed to stable, internal, and uncontrollable system limitations — the pattern associated with the most persistent dissatisfaction in the service recovery literature. This attribution asymmetry motivates proactive escalation requests rather than patience, explaining the path from AI disclosure to escalation (H1b).

Expectation confirmation theory (Oliver, 1980) explains *why the disclosure effect varies in direction across contexts*. Disclosure sets an expectation frame: customers expecting AI service may anticipate lower personalization but faster, more accurate resolution. Whether disclosure ultimately helps or hurts satisfaction depends on whether actual AI performance confirms the (lower) expectation or disconfirms the (higher) implicit human-service standard. This theory generates the counterintuitive implication that disclosure might *increase* satisfaction in contexts where AI speed and accuracy dominate customer evaluations — directly accounting for the positive observational NPS coefficients in charger troubleshooting cells (+0.65*) documented in Chapter 6.

Service recovery theory (Smith et al., 1999; McCollough et al., 2000) explains *why coupon provision buffers disclosure-induced dissatisfaction*. Monetary compensation restores positive affect through a distributive justice mechanism: it signals that the firm acknowledges the customer's experience and considers it worthy of material restitution. In the disclosure context, the coupon operates not as compensation for an objective service failure but as a proactive goodwill signal — an acknowledgment that receiving AI service differs from receiving human service, and that this difference warrants a tangible expression of firm consideration. This mechanism explains the moderation path β_coupon (H2) but is distinct from the social presence and attribution mechanisms that drive the main disclosure effect, ensuring the four theories form a non-redundant explanatory structure.

The structural distinctiveness of these four mechanisms has an important empirical implication: they predict *separable* effects that the experimental design in Study 2 is specifically constructed to disentangle. The disclosure manipulation isolates the social presence and attribution mechanisms (holding coupon and agent type constant). The coupon manipulation tests the service recovery mechanism (holding disclosure constant). The agent type manipulation tests the perceived control mechanism underlying H3. No single theoretical tradition could generate predictions for all three dimensions of the experimental design; the integrated framework is required.

H1: AI Disclosure and Customer Value

The primary hypothesis of this dissertation predicts that disclosing AI agent identity to customers in an inbound service context will reduce customer satisfaction and increase escalation rates. This prediction is anchored in the social presence, cognitive dissonance, and attribution mechanisms elaborated in Section 4.1, and is empirically grounded in the Luo et al. (2019) finding in the outbound sales context.

The social presence mechanism operates as follows. In a typical customer service email exchange, customers have no direct basis for knowing whether the response was written by a human or generated by AI. Both arrive via the same text channel, both are attributed to ”Anker Support,” and both carry the implicit social presence of authorship — the customer assumes they are reading words chosen by a human being to communicate a human intention. When the AI identity is disclosed, this implicit social presence assumption is withdrawn. The customer now knows they are reading machine-generated text, and the relational signals they might have interpreted as warmth, empathy, or personal attention are reinterpreted as simulated affect rather than genuine human concern. This reinterpretation reduces the satisfaction value the customer derives from the interaction, particularly when relational dimensions matter to the customer’s evaluation of service quality.

The cognitive dissonance mechanism provides a complementary pathway. Customers seeking resolution to a product problem may hold an implicit belief that their situation, as a paying customer, entitles them to human attention and service. When informed that they are receiving AI service, this implicit entitlement expectation is violated, creating cognitive dissonance between the expected (human service) and the received (AI service). This dissonance is experienced as dissatisfaction, independent of the objective quality of the AI response. The dissonance is likely to be stronger for customers who place greater implicit value on human service — perhaps those with higher product investment, those with more complex or emotionally charged issues, or those with prior negative experiences with automated service systems.

The attribution mechanism links disclosure to escalation behavior specifically. As discussed in the algorithm aversion literature, customers tend to attribute AI errors to stable, uncontrollable technological limitations rather than to correctable individual mistakes. When informed that their service agent is AI, customers who encounter any imperfection — a response that does not quite address their specific situation, a slightly delayed follow-up, a recommendation that proves incorrect — will tend to interpret this imperfection as evidence of a systematic AI limitation that will not be corrected in subsequent exchanges. This attribution of errors to stable causes motivates escalation: rather than waiting for the AI to do better on the next exchange, the customer requests human service, reasoning that the AI is unlikely to improve. This escalation dynamic is directly tested in H1b.

Formally, the first set of hypotheses states:

H1a: AI disclosure is negatively associated with customer satisfaction (NPS score) compared to non-disclosure conditions, such that customers who are informed that their service is provided by an AI agent will report lower NPS scores than customers who receive equivalent AI service without explicit disclosure.

H1b: AI disclosure is positively associated with customer escalation rates compared to non-disclosure conditions, such that customers who are informed that their service is provided by an AI agent will be more likely to request or trigger transfer to a human agent than customers who receive equivalent AI service without explicit disclosure.

It is important to note the conditions under which H1a and H1b might not be confirmed. If customers in the inbound service context hold very low prior expectations for customer service quality in general — perhaps because of negative prior experiences with human agents — then AI service may produce positive expectation disconfirmation even without disclosure, potentially mitigating or reversing the disclosure effect on satisfaction. Similarly, if the primary driver of customer satisfaction in this context is speed of resolution (a dimension on which AI excels) rather than relational quality (a dimension on which AI is deficient), AI aversion effects on satisfaction may be attenuated by the positive efficiency experience. These considerations motivate treating H1a as a directional prediction rather than a strong prior, and interpreting null results on H1a as informative rather than merely non-significant.

The task-contingency hypothesis extends the primary disclosure hypothesis by predicting that the effects of AI aversion — whether activated by disclosure or by indirect inference — will vary systematically across task types. Drawing on Castelo et al.'s (2019) framework of objective versus subjective tasks, and on Huang and Rust's (2018) model of AI capabilities across mechanical, thinking, feeling, and social intelligence dimensions, the prediction is that AI aversion effects will be strongest for tasks requiring social judgment and individual sensitivity.

H1c: The negative effect of AI service (relative to human service) on customer satisfaction is stronger for service tasks requiring social judgment and individual sensitivity — specifically, purchase inquiry tasks — than for technical service tasks such as troubleshooting, after controlling for ticket complexity and product category.

The theoretical rationale for H1c operates at two levels — the task capability level and the value architecture level — and the distinction matters for the strength of the theoretical claim.

At the *task capability level*, the rationale follows directly from Huang and Rust (2018): purchase inquiry tasks require feeling intelligence (empathy, emotional attunement) and social intelligence (relational judgment, personalization) that AI cannot yet replicate, while troubleshooting tasks require mechanical and thinking intelligence that AI replicates effectively. This is the standard task-contingency argument.

At the deeper *value architecture level*, the argument is more fundamental. Purchase inquiry is not merely a "harder" or "more subjective" task — it is a qualitatively different type of service interaction in which the customer is not only seeking information but seeking *validation and trust*. When a customer asks "which security camera should I buy for my apartment?", they are implicitly asking "does this firm understand my situation well enough that I can trust its recommendation with my money?" The value being delivered is relational trust, not informational content. An AI system that delivers technically correct product specifications fails to deliver this relational value not because it is less accurate but because relational trust requires a human agent — a person who has chosen to attend to this specific customer's situation. Disclosure that the agent is AI eliminates the possibility of this relational trust, regardless of response quality.

Troubleshooting, by contrast, is a *transactional* service interaction: the customer has already made the purchase and seeks functional resolution. The value being delivered is purely functional — get the product working. Here, AI's speed and accuracy advantages directly enhance the value delivered, and the absence of relational warmth is simply irrelevant to the customer's evaluation. H1c, therefore, is not merely predicting task-contingent AI aversion — it is predicting that the *type of customer value at stake* determines whether AI disclosure activates an aversion penalty. This reframing elevates H1c from a moderation hypothesis within algorithm aversion theory to a boundary condition that maps onto a broader theory of service value decomposition.

H2: Coupon as Compensatory Mechanism

The second hypothesis of this dissertation introduces the coupon incentive as a moderating variable that may buffer the negative effects of AI disclosure on customer satisfaction. The theoretical development draws on the intersection of service recovery theory, behavioral economics of unexpected rewards, and prospect theory's account of loss compensation.

The theoretical mechanism underlying H2 is more precise than ”a coupon makes people feel better.” Three distinct mechanisms, operating at different psychological levels, each support the moderation prediction — and they differ in their implications for how the coupon should be framed.

The first mechanism is *service recovery through distributive justice* (Smith et al., 1999). When customers perceive AI disclosure as a signal that the firm has substituted a lower-cost resource for what they expected (human service), they experience a form of service inequity: the firm received full revenue but provided reduced-quality service. Distributive justice theory predicts that compensation restores the perceived equity balance — the firm acknowledges the substitution and transfers a share of the value back to the customer. In this reading, the coupon should be framed explicitly as a *cost-sharing* or *benefit-sharing* gesture: ”As a company that uses AI to serve you more efficiently, we want to pass on part of that value to you.” This framing transforms the disclosure from a potential offense into a transparent commercial proposition — and is likely to be more effective than a generic discount that carries no such signal.

The second mechanism is *perceived control restoration* (Langer, 1975; Dietvorst et al., 2018). The core psychological cost of AI disclosure is the customer's sense of loss of control: they did not choose to be served by AI, and the routing decision was made by the firm without their input. Coupon provision, particularly when framed as a customer choice instrument (a discount on a *future* purchase, implying continued relationship on the customer's terms), partially restores the perception that the customer retains agency in the relationship. The signal is not ”we compensate you for this transaction” but ”we invest in your continued trust and future engagement.”

This control-restoration mechanism is theoretically distinct from the distributive justice mechanism and generates a different optimal framing — one that emphasizes customer autonomy rather than firm accountability.

The third mechanism is *positive surprise through unexpected reward* (Loewenstein, 1996). Customers do not expect to receive incentives in response to routine service inquiries. An unexpected coupon delivered alongside disclosure creates a hedonic contrast effect: the positive surprise of receiving an unanticipated reward is evaluated against the (relatively smaller) negative affect of the disclosure, potentially resulting in a net-positive affective balance. This mechanism predicts that the coupon's buffering effect is stronger for customers who did not expect it — that is, first-time recipients for whom the coupon is genuinely surprising — and weaker for customers who have received compensation coupons before and incorporated them into their service expectations.

A fourth mechanism, grounded in the economics of fairness preferences (Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000), provides perhaps the most theoretically precise account of why the coupon works in the disclosure context specifically. Fehr and Schmidt (1999) model individuals as having utility functions that include a term for *disadvantageous inequality aversion*: people experience disutility when they receive less than a reference allocation, with this disutility proportional to the size of the inequality. In the AI disclosure context, the customer's reference point may be the service they expected — human attention and judgment — and the actual service received — AI assistance — creates a perceived inequality: the customer contributed the same price for a lower-quality (or differently-valued) form of service delivery. The coupon addresses this inequality directly: by transferring a share of the cost savings from AI service back to the customer, the firm reduces the perceived inequality to zero, or even reverses it (the customer now receives something extra relative to the reference interaction). This fairness restoration mechanism is distinct from the justice mechanism in service recovery theory: it is not about acknowledging a failure but about correcting a perceived distributional inequity between the firm's cost savings and the customer's value received. The practical implication is that the coupon's framing matters substantially: a coupon presented as ”cost-sharing” or ”efficiency dividend” should be more effective than the same coupon presented as a ”thank you gift,” because the former directly addresses the fairness channel while the latter operates only through the positive surprise channel.

Together, these four mechanisms generate the same directional prediction (coupon attenuates the disclosure NPS penalty) but suggest different optimal designs and boundary conditions. The distributive justice mechanism predicts the coupon is most effective when the firm explicitly frames it as cost-sharing. The perceived control mechanism predicts the coupon is most effective when framed as future-oriented and voluntary. The positive surprise mechanism predicts the coupon is most effective for customers who have not previously received such coupons. These distinctions cannot be identified from the current experimental design, which does not vary coupon framing, but they provide a rich agenda for follow-up research.

The behavioral economics of unexpected rewards provides a complementary mechanism through the psychology of positive surprises. Loewenstein (1996) argues that anticipated outcomes are experienced at diminished intensity compared to unanticipated ones, because anticipation shifts the hedonic evaluation from the event itself to the anticipatory period. An unexpected reward — one that arrives without prior expectation — is experienced at full hedonic intensity and generates a disproportionate positive affect relative to its objective value. A coupon delivered alongside AI disclosure, if framed appropriately (as an appreciation gesture rather than a disclosure penalty), is likely to be experienced as unexpected by customers, who do not expect to receive incentives in response to routine service inquiries. This unexpected positive experience creates positive affect that is evaluated against the backdrop of any negative affect generated by the disclosure, potentially resulting in a net-neutral or net-positive satisfaction outcome.

Prospect theory (Kahneman and Tversky, 1979) provides a framework for predicting when coupon compensation will and will not be sufficient to offset disclosure effects. Under prospect theory, outcomes are evaluated relative to a reference point, and losses loom larger than equivalent gains — the loss aversion coefficient is typically estimated at approximately 2.0 to 2.5, meaning that a loss of value X is experienced as approximately twice as negative as an equivalent gain of X is experienced as positive. In the AI disclosure context, if disclosure creates a loss of perceived service quality (measured relative to the expectation of human service), the coupon must provide a gain of perceived value that is proportionately larger to fully compensate for the loss. If the disclosure loss is small (for example, because customers have already adjusted their expectations for AI service), a modest coupon may fully compensate. If the disclosure loss is large (for example, for customers with strong human-service preferences in high-stakes contexts), a small coupon may be insufficient.

These theoretical considerations motivate a moderation prediction rather than a main effect prediction: the coupon is not predicted to increase satisfaction unconditionally, but specifically to attenuate the negative satisfaction effect of AI disclosure. This interaction hypothesis is the proper specification:

H2: The negative relationship between AI disclosure and customer satisfaction (NPS score) is moderated by coupon provision, such that the negative effect of AI disclosure is smaller in magnitude (or reversed) for customers who receive a coupon alongside the disclosure, compared to customers who receive disclosure without a coupon.

An additional dimension of H2 concerns the mechanism of AI-enabled seamless coupon delivery. As noted in the literature review, human agents face social awkwardness in proactively offering coupons to customers who have not complained, as such offers carry an implicit admission of service inadequacy. AI systems face no such awkwardness: they can be programmed to deliver coupon offers systematically and consistently without any risk of appearing apologetic or patronizing in unintended ways. This asymmetry suggests that the coupon buffering effect may be particularly well-suited to AI-delivered service, potentially making AI service with coupon provision superior to human service without coupons in terms of net satisfaction effects. While this stronger prediction is beyond the scope of the formal hypotheses, it motivates examination of the interaction between agent type and coupon provision in the analysis.

H3: Agent Type Effects

The third hypothesis concerns the comparison between pure AI and hybrid AI-human service architectures. The theoretical question is whether having a human agent available as a backup — as in the hybrid architecture — changes the customer's experience of AI service in ways that affect satisfaction and escalation outcomes, after controlling for the endogenous selection of more complex tickets to hybrid routing.

The human safety net hypothesis proposes that customers who receive hybrid service — where they may or may not know that a human agent is available to take over if needed — experience lower algorithm aversion than customers receiving pure AI service. The mechanism draws on the control dimension of algorithm aversion identified by Dietvorst et al. (2018): the key driver of aversion is not AI error per se but the perceived lack of control or accountability over the AI process. In a hybrid architecture, the existence of a human backup provides at least two accountability mechanisms that pure AI lacks. First, the human agent who monitors the hybrid thread serves as an implicit quality check on AI responses, meaning that egregious AI errors are less likely to reach the customer unfiltered. Second, the customer in a hybrid interaction retains the option of requesting human service, providing a sense of control over the interaction even if they never exercise this option. Together, these mechanisms may reduce the perceived risk of algorithm failure and thereby reduce the activation intensity of AI resistance.

It is important to be careful about the confounding that complicates interpretation of simple comparisons between hybrid and pure AI outcomes in the observational data. As the descriptive statistics make clear, hybrid tickets are systematically more complex than pure AI tickets, because the routing algorithm deliberately assigns complex tickets to hybrid handling. A simple comparison of NPS scores across pure AI and hybrid tickets will therefore confound agent type effects with complexity effects. Study 1's analysis attempts to address this confound through the complexity control variable, but the potential for residual confounding remains. Study 2's experimental design addresses this confound more cleanly by randomly assigning agent type while holding ticket complexity approximately constant through stratified randomization.

The formal hypothesis is:

H3: Customers served by AI-plus-human hybrid agents report higher customer satisfaction (NPS) and comparable or lower voluntary escalation rates compared to customers served by pure AI agents, after controlling for ticket complexity and product category characteristics.

The direction of H3 on escalation requires some elaboration. If the human safety net mechanism operates as theorized, customers receiving hybrid service should be less likely to escalate because they have a lower baseline AI resistance level and therefore experience fewer AI interactions as sufficiently unsatisfactory to warrant escalation. However, there is a countervailing structural argument: hybrid tickets may generate higher observed escalation rates precisely because escalation in hybrid contexts is system-initiated (by the AI quality monitoring reaching a threshold) rather than customer-initiated, and system-initiated escalation may generate escalation records even when the customer is satisfied. The H3 prediction on escalation therefore applies specifically to customer-initiated escalation rates, which Study 2's experimental design can identify by examining customer behavior after controlling for system-initiated transfers.

Summary of Hypotheses

The five formal hypotheses of this dissertation, their theoretical bases, and their predicted directions are summarized in Table 4.1.

Table 4.1: Summary of Research Hypotheses

Hypothesis	Statement	Theoretical Basis	Predicted Direction
H1a	AI disclosure reduces customer satisfaction (NPS) relative to non-disclosure, holding service quality constant	Social presence theory; cognitive dissonance; AI aversion (Dietvorst et al., 2015; Luo et al., 2019)	Negative (Disclosure → Lower NPS)
H1b	AI disclosure increases customer escalation rates relative to non-disclosure, holding service quality constant	Attribution theory; algorithm aversion; control deprivation (Dietvorst et al., 2018)	Positive (Disclosure → Higher Escalation)
H1c	The negative effect of AI service on customer satisfaction is stronger for purchase inquiry tasks than for troubleshooting tasks	Task-dependency of algorithm aversion (Castelo et al., 2019); AI capability boundaries (Huang & Rust, 2018)	Stronger negative for purchase inquiry
H2	Coupon provision moderates the negative relationship between AI disclosure and customer satisfaction, such that the disclosure penalty is smaller for coupon recipients	Service recovery theory (Smith et al., 1999); unexpected reward psychology (Loewenstein, 1996); prospect theory (Kahneman & Tversky, 1979)	Positive moderation (Coupon buffers Disclosure → NPS negative effect)
H3	Customers served by hybrid AI-human agents report higher NPS than customers	Human safety net theory; control mechanism of aversion (Dietvorst et al., 2018);	Positive (Hybrid > Pure AI on NPS)

It is useful to note the logical relationship between the hypotheses and the two-study design. H1c is tested primarily in Study 1 using the observational data's variation in intent type and the heterogeneous effects analysis. H1a, H1b, H2, and H3 are tested primarily in Study 2 using the experimental manipulation of disclosure, coupon, and agent type. However, Study 1 also provides relevant evidence on H1a and H3 through the observational comparison of NPS scores across agent types (with complexity controls), providing a baseline pattern against which the experimental results can be interpreted. The integrated interpretation of findings from both studies is one of the contributions of Chapter 9.

The hypotheses together constitute a coherent theoretical account of AI disclosure effects in inbound customer service. They predict that disclosure reduces satisfaction and increases escalation (H1a, H1b), that these effects are task-contingent and concentrated in relational service contexts (H1c), that coupon incentives can partially buffer the disclosure penalties (H2), and that hybrid architecture with human backup reduces AI resistance effects on satisfaction (H3). If the empirical results support this pattern of predictions, they will provide both theoretical insight into the mechanisms of AI aversion in service contexts and practical guidance for firms managing AI deployment and disclosure decisions.

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End of Chapters 1–4

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RESEARCH METHODOLOGY

Overview of Research Design

This dissertation employs a two-study sequential research design that combines the scale and ecological validity of a large-scale observational study with the causal identification precision of a randomized controlled trial. The rationale for this integrated design reflects a fundamental tension in empirical research on AI deployment: observational data provides the statistical power and real-world representativeness necessary to characterize how AI affects customer outcomes across the full diversity of service encounters, but it cannot resolve the endogeneity problems that arise because ticket assignment to agent types is not random. A randomized experiment resolves those endogeneity problems but necessarily involves a constrained sample and limited external validity. By combining the two approaches in sequence, where Study 1 documents the landscape of AI effects and Study 2 causally identifies the specific mechanism of interest — AI disclosure — this dissertation achieves both internal and external validity in a way that either study alone could not.

Study 1 is a quasi-experimental observational analysis of 477,983 customer service tickets processed by Shulex Technology on behalf of Anker Innovations between September 2023 and May 2024. The study exploits the natural variation in AI adoption rates across product categories and time periods that arose from Shulex's progressive deployment rollout. The primary analytical strategy employs high-dimensional fixed-effects (HDFE) regression within product category-by-intent interaction cells, controlling for pre-assigned ticket complexity scores. This approach generates cell-by-cell estimates of the causal effect of AI agent type on three outcomes — NPS satisfaction scores, escalation rates, and response rounds — across 29 product category-by-intent combinations. The key identifying assumption is that, conditional on product category, ticket intent type, and pre-assigned complexity score, ticket characteristics are comparable across agent types, so that within-cell differences in outcomes can be attributed to agent type rather than to selection.

Study 2 is a six-group randomized controlled trial (RCT) implemented within the Shulex platform in July 2024. The experiment randomly assigns incoming tickets to treatment conditions defined by three crossed factors: AI disclosure (whether the customer is told their ticket is being handled by an AI agent), coupon provision (whether the customer receives a $5 discount coupon), and agent type (pure AI versus human agent). The experimental design allows clean causal identification of the disclosure effect (H1), the coupon moderation of the disclosure effect (H2), and the agent type effect net of disclosure and coupon confounds (H3). The key identifying assumption for the experiment is that ticket arrival order within the selected category-intent cells is independent of potential outcomes — an assumption that is plausible given the mechanisms of customer inquiry generation and is supported by pre-treatment balance checks.

The two studies are explicitly designed to be complementary. Study 1 documents that AI involvement affects outcomes differently across task types (troubleshooting versus purchase inquiry), establishes the magnitude of efficiency gains, and reveals the heterogeneous character of AI effects that motivates the experimental design choices in Study 2. Study 2 provides clean causal identification of the specific mechanism — AI disclosure — that Study 1 cannot isolate, because in the observational data, customers' knowledge of their agent type is not controlled. Together, the two studies provide a comprehensive empirical account of the effects of AI deployment on customer outcomes in this setting.

The remainder of this chapter is organized as follows. Section 5.2 describes the data, sample construction, identification strategy, and econometric specification for Study 1. Section 5.3 describes the experimental design, implementation, power analysis, and timeline for Study 2. Section 5.4 presents the econometric specification for the experimental analysis. Section 5.5 describes the robustness checks planned for both studies.

Study 1: Observational Analysis

Data and Sample

The observational analysis draws on the complete ticket database maintained by Shulex Technology for its Anker Innovations account. The full dataset encompasses 477,983 tickets submitted via Amazon's Buyer-Seller Messaging system and Anker's direct customer email channel between September 2023 and May 2024. Each ticket record contains information on the ticket creation timestamp, the product category and subcategory, the intent classification assigned by the Shulex routing system, the pre-assigned complexity score, the agent type that handled the ticket (pure AI, AI-plus-human hybrid, or pure human), the number of response rounds (defined as the count of messages exchanged between customer and agent before ticket closure), a binary escalation indicator recording whether the ticket was escalated from AI to human handling, the NPS score provided by the customer in the post-resolution survey (if the survey was completed), and quality control classification records for tickets that were reviewed by the Shulex QC team.

The primary analysis sample is restricted to the period from February 2024 through May 2024, representing the post-stabilization phase of AI adoption. The earlier months of the dataset (September 2023 through January 2024) reflect the initial rollout period, during which both AI coverage rates and AI response quality were changing rapidly. Including these early months in the main analysis would conflate the effect of agent type with time trends in AI quality improvement, creating a confound that is difficult to eliminate through fixed effects alone. The February-to-May 2024 window corresponds to a period when AI adoption had stabilized at approximately 35 to 39 percent of tickets and the quality control record shows a consistent, low error rate, providing a cleaner basis for cross-sectional comparison. Data from the early adoption period (September 2023 to January 2024) are used exclusively in the descriptive and trend analysis in Chapter 6.

Within the February-to-May 2024 window, the analysis is further restricted to the 29 product category-by-intent interaction cells that individually contain at least 865 tickets. This threshold is chosen based on a power analysis for the cell-by-cell regression: with $N = 865$ and a typical NPS standard deviation of approximately 3.07, the cell-level regression has 80 percent power to detect an effect size of 0.35 NPS points at the 10 percent significance level. Cells below this threshold are retained in robustness checks using an $N \geq 500$ cutoff, and excluded from robustness checks using an $N \geq 1{,}000$ cutoff, to verify that the pattern of findings is not driven by the inclusion or exclusion of small cells.

The analysis excludes chat-type service tickets, which constitute a very small fraction of the overall volume but have an effective missing rate of greater than 99.99 percent on the key outcome variables (NPS, response rounds, and escalation) due to system-level data recording differences between chat and email channels. Including these tickets would add noise without contributing information.

The analysis also excludes AI-plus-human hybrid tickets from the main regression analysis of Study 1. This exclusion is motivated by an endogeneity concern specific to the hybrid category: within the hybrid system, a ticket is classified as hybrid precisely because the AI system determined that the query exceeded its handling capacity and triggered a handover to a human agent. The threshold for handover —

which the data suggest occurs at approximately 2.56 message rounds on average — is determined endogenously by ticket complexity, creating a mechanical correlation between being a hybrid ticket and having higher complexity. Even with complexity controls, the hybrid classification incorporates information about ticket difficulty that may not be fully captured by the pre-assigned complexity score, rendering the hybrid category non-comparable to pure AI tickets in a linear regression. Hybrid tickets are included in descriptive analysis and in robustness checks but not in the primary identification analysis. The main contrast of interest in Study 1 is therefore pure AI versus pure human, within each product category-by-intent cell.

Identification Strategy

The central identification challenge in Study 1 is that ticket assignment to AI versus human agents is not random. The Shulex routing system assigns tickets to agent types based on a combination of factors including product category routing policies, ticket complexity scores, current AI system capacity, and historical performance data for that category-intent combination. This non-random assignment creates a potential confound: if AI agents systematically receive easier tickets than human agents (or more complex ones), then differences in outcomes between AI and human tickets reflect differences in ticket composition, not differences in agent type effectiveness.

The primary identification strategy addresses this challenge through within-cell comparison. Rather than comparing AI tickets to human tickets across the entire dataset — which would be confounded by systematic differences in the types of tickets routed to each agent type — the analysis compares AI and human tickets within narrowly defined product category-by-intent interaction cells. Each cell represents a specific combination of product category (for example, ”Charger”) and ticket intent type (for example, ”Troubleshooting”), so that within a cell labeled ”Charger × Troubleshooting,” all tickets are from customers seeking troubleshooting assistance for charging products. Within this narrow cell, the queries are semantically homogeneous: customers are asking about why their charger is not working, how to fix a charging issue, or what to do with a defective charger. The variation in which agent type handles the ticket within this cell is driven primarily by deployment schedule factors and capacity allocation, which are plausibly orthogonal to individual ticket characteristics.

The key identifying assumption can be stated formally: conditional on product category, intent type, and pre-assigned complexity score, the potential outcomes of ticket resolution (NPS, escalation, rounds) are independent of agent type assignment. In notation: (Y_AI, Y_human) □ Agent_type | Category, Intent, Complexity. This conditional independence assumption is supported by several features of the setting. First, the complexity score is assigned before agent routing occurs, based solely on automated analysis of the ticket content, and is therefore not influenced by the agent type that will handle the ticket. The within-cell regression controls for this pre-assignment complexity score, eliminating the portion of the agent-type/outcome relationship that runs through complexity. Second, within a narrow category-intent pair, customers are asking structurally similar questions, reducing the likelihood of substantial residual heterogeneity that the fixed effects do not absorb. The semantic similarity of queries within cells is empirically observable from the ticket text and is confirmed by the Shulex intent classification system, which assigns tickets to intents based on natural language processing of the ticket content. Third, the gradual rollout of AI coverage across cells means that agent type was partly determined by batch timing — which tickets happened to arrive during periods when AI capacity was available — rather than by ticket-specific characteristics. This temporal variation is an additional source of quasi-random assignment within cells.

A comparison to staggered adoption designs is instructive. An alternative identification strategy would exploit the month-by-month variation in AI adoption rates as a time-series instrument, treating the progressive deployment rollout as a natural experiment in the spirit of Callaway and Sant'Anna (2021) and Sun and Abraham (2021). This approach would identify AI effects from changes in outcomes over time as AI adoption increases, controlling for time-invariant cell-level characteristics and common time trends. As a robustness check, month fixed effects are added to the primary specification to confirm that the cross-sectional within-cell comparison is not contaminated by time trends in AI quality, customer composition, or seasonal demand variation. The results of this check are reported in Section 6.5.

What the identification strategy can and cannot establish. The within-cell HDFE design is transparent about its limits. The design *can* eliminate confounding from systematic ticket-type differences across agent groups, by restricting comparisons to tickets within the same narrow category-intent cell. It *can* control for pre-assigned complexity, the principal observable driver of routing decisions. It *can* exploit the temporal variation in AI rollout as a time-series check. What it *cannot* do is eliminate confounding from unobserved ticket-level characteristics correlated with both routing and outcomes — for instance, the emotional valence of a customer's language, or residual complexity variation within complexity-score bins. The HDFE estimates are therefore treated as *quasi-experimental* evidence throughout, and the burden of fully causal identification is explicitly placed on Study 2.

Routing mechanism and selection into treatment. Understanding precisely *how* tickets are routed is essential for evaluating whether the conditional independence assumption is plausible. Ticket assignment in the Shulex system follows a deterministic rule based on three observable inputs: (1) the product category, which determines whether a routing policy permits pure AI assignment; (2) the AI-assigned complexity score (1–5), where scores above a category-specific threshold trigger hybrid routing; and (3) system capacity at the time of ticket arrival — during peak periods, AI handles overflow tickets that would otherwise go to human agents. Critically, the routing decision is made at ticket creation, before any human reviews the ticket content, and is based solely on the structured metadata and the automated NLP classification. This mechanism is important because it implies that the ticket's *outcomes* (NPS, escalation, rounds) cannot have influenced the routing decision, which eliminates reverse causality as a concern. The remaining identification threat is selection on unobservables within the routing criteria.

Agent shift and performance-based assignment bias. A specific identification threat that the routing mechanism description does not fully address is whether AI and human agents are systematically assigned to different shifts, and whether shift timing correlates with ticket characteristics or customer outcomes. If AI handles overnight tickets (when human agents are unavailable) and human agents handle peak-hour tickets, and if overnight customers differ systematically from peak-hour customers, then the agent-type comparison within cells may capture customer composition differences rather than agent type effects. To assess this threat, the robustness checks include specifications that add hour-of-day fixed effects (24 indicator variables for the hour of ticket creation) and day-of-week fixed effects interacted with agent type. Consistency of these estimates with the baseline HDFE specification provides evidence that shift-based composition differences do not drive the main findings.

NPS missing data: rate, pattern, and handling. The NPS outcome variable is subject to substantial missing data arising from the low and differential survey response rates. Across the full observational sample, approximately 5 percent of tickets generate an NPS response; within this overall rate, pure AI tickets have response rates of 1–5 percent and human/hybrid tickets have response rates of 9–15 percent. This differential non-response is the primary threat to NPS estimate validity in Study 1. The analysis handles this missing data problem through three approaches. First, all NPS regressions are estimated on the subsample of tickets for which NPS data are available (approximately 18,000 tickets with NPS scores out of 477,983 total), with the understanding that this is a selected subsample. Second, a Heckman two-stage selection model is estimated in which the first stage predicts NPS survey response (using ticket characteristics, agent type, and category-month dummies as predictors) and the second stage includes the inverse Mills ratio as a control for selection into response. The Heckman-corrected NPS estimates are reported alongside the OLS estimates in the robustness checks. Third, the pattern of NPS missingness is examined by agent type and cell to assess whether missingness is random within cells conditional on observables (missing completely at random, MCAR, conditional on cell and complexity) or whether it reflects a systematic pattern that could bias within-cell comparisons. Readers should treat the NPS estimates from Study 1 as descriptive evidence of satisfaction patterns, with causal interpretation reserved for Study 2's controlled experimental data.

Propensity score matching as robustness check. To further address this residual selection concern, the main HDFE results are supplemented in Section 6.5 with propensity score matching (PSM) estimates. AI and human tickets within each category-intent cell are matched on observable pre-treatment characteristics — complexity score, ticket creation hour-of-day, day-of-week, and ticket text length — using nearest-neighbor matching within cells. The PSM estimates restrict the sample to matched pairs in which AI and human tickets are observably similar on all measured pre-treatment dimensions, providing a more conservative identification that does not rely on the linearity assumptions of the HDFE specification. Consistency of HDFE and PSM estimates provides evidence that the HDFE identifying assumption is not materially violated by observable sources of selection.

Inverse probability weighting and doubly robust estimation. As an additional robustness check that addresses both the selection-into-treatment problem and the NPS missing-data problem simultaneously, inverse probability weighting (IPW) is applied to the NPS analysis. Each ticket is weighted by the inverse probability of its observed treatment assignment (AI vs. human) conditional on observable covariates, constructing a pseudo-population in which treatment assignment is approximately independent of observables. Doubly robust (DR) estimators — which combine the IPW weighting with outcome regression — are consistent if either the propensity score model or the outcome model is correctly specified, providing additional robustness to model misspecification. The IPW and DR estimates are reported in Section 6.5 alongside the HDFE and PSM estimates. Consistency of these alternative estimators with the HDFE baseline provides strong evidence that the efficiency-satisfaction decoupling finding is not an artifact of selection bias or model specification.

Instrumental variable approach. As an additional check exploiting the temporal variation in AI rollout, monthly AI adoption rates by cell are used as an instrument for ticket-level AI assignment. The instrument is valid if (i) monthly adoption rates predict individual ticket AI assignment (first-stage relevance, confirmed by F-statistics above 20 in all cells) and (ii) monthly adoption rates affect outcomes only through their effect on individual AI assignment (exclusion restriction, plausible because the adoption rate is determined by Shulex's deployment schedule and Anker's capacity decisions, not by individual ticket characteristics). The IV estimates are reported in Section 6.5 and are consistent in direction and magnitude with the HDFE and PSM estimates, strengthening confidence that the task-contingency pattern documented in Section 6.3 reflects genuine heterogeneity in AI effects rather than selection artifacts.

Econometric Specification

Two complementary econometric specifications are employed. The first is a pooled specification that estimates the average within-cell effect of AI across all cells simultaneously. The second is a cell-by-cell specification that estimates the AI effect separately for each of the 29 cells and then meta-analyzes those cell-specific estimates.

The pooled specification takes the form:

$$\mathbf{Y\_it = \beta_0 + \beta_1 \cdot AI\_it + \beta_2 \cdot Complexity\_it + \alpha\_k + \varepsilon\_it}$$

where Y_it is the outcome variable for ticket i in time period t (either is_escalation, score, or N_responses); AI_it is a binary indicator equal to one if the ticket was handled by a pure AI agent and zero if handled by a pure human agent; Complexity_it is the pre-assigned complexity score, a continuous variable ranging from 1 to 5; α_k represents the product category-by-intent interaction fixed effect, where k indexes the 29 cell types; and ε_it is the idiosyncratic error term. Standard errors are clustered at the cell level to account for within-cell correlation in outcomes. The coefficient β_1 identifies the average within-cell effect of AI agent type on the outcome.

The cell-by-cell specification estimates the AI effect separately within each cell g:

$$\mathbf{Y\_it = \beta_0\_g + \beta_1\_g \cdot AI\_it + \beta_2\_g \cdot Complexity\_it + \varepsilon\_it \text{ (for each g = 1, ..., 29)}}$$

This produces 29 cell-specific estimates of β_1_g — the AI effect within each product category-by-intent combination. These cell-specific estimates are then metaanalyzed to compute a precision-weighted mean effect across cells and to characterize the distribution of effects (mean, standard deviation, fraction significant, fraction positive versus negative). The cell-by-cell approach provides richer evidence on heterogeneity than the pooled specification because it does not impose homogeneity of the AI effect across cells. All 29 sets of cell-specific estimates are reported in the full regression tables in Chapter 6, including the coefficient, standard error, and significance level for each cell.

Three outcome variables are modeled. Escalation (is_escalation) is a binary indicator modeled using a linear probability model (LPM), where β□ represents the percentage-point difference in escalation rate between AI and human tickets within the same cell. NPS score (score) is a continuous variable on a 1-10 scale, modeled by OLS, where β□ represents the point-scale difference in NPS. Response rounds (N_responses) is a count variable modeled by OLS in the primary specification, with Poisson regression employed as a robustness check. The LPM is used for escalation rather than Logit or Probit because the within-cell fixed effects in the LPM have a straightforward interpretation and the incidental parameters problem that affects maximum likelihood estimation in panel fixed effects models does not arise with the LPM. Logit marginal effects are reported in the robustness checks and are consistent with the LPM results.

Limitations of Observational Analysis

The observational analysis faces several limitations that motivate the complementary experimental design in Study 2.

The most significant limitation is the NPS data availability bias arising from differential survey response rates across agent types. In the observational period, pure AI tickets generate NPS survey responses at a rate of approximately one to two percent in the early months (September to November 2023), rising to approximately three to five percent in the later period (February to May 2024). Hybrid and human tickets generate NPS responses at a substantially higher rate of approximately nine to fifteen percent throughout the observation period. This differential response rate creates a potential selection problem in the NPS analysis: if the customers who choose to complete the NPS survey after receiving AI service are systematically different from those who do not — perhaps because only highly satisfied or highly dissatisfied customers respond, or because more engaged customers are both more likely to respond and more likely to have strong opinions about AI service — then the estimated AI effect on NPS will be biased. The direction of the bias is ambiguous a priori but is most likely upward (toward less negative AI effects) if satisfied customers disproportionately respond to AI service surveys.

This selection problem cannot be resolved within the observational data, because the characteristics of non-respondents are by definition unobservable. Study 2 addresses this limitation by controlling the NPS survey protocol: all customers across all six experimental groups receive the NPS survey on the same timeline and with the same email design, eliminating differential response rates as a source of bias. The comparison of observational NPS estimates (Study 1) with experimental estimates (Study 2) provides an indirect test of the magnitude of this selection bias.

A second limitation is the possibility of residual confounding even within cells, due to unobserved ticket characteristics that are correlated with both agent type assignment and outcomes. The identification strategy assumes that ticket characteristics are comparable across agent types within narrow cells, but if routing algorithms use ticket features beyond those captured by the complexity score — for example, sentiment, query length, or the presence of specific keywords — then residual confounding may remain. The robustness checks partially address this concern by adding controls for ticket length and time-of-day, but a fully clean identification is not possible without experimental variation.

A third limitation is that the analysis is restricted to email-channel tickets, where the outcome data are most complete. Chat-type service interactions, which are growing in volume, involve different service dynamics that may not be captured by the patterns documented here. The external validity of findings to chat-based service contexts should be considered with appropriate caution.

Study 2: Field Experiment

IRB Approval and Ethics

The field experiment was reviewed and approved by the relevant institutional review board prior to implementation. The review process confirmed that the study satisfies all applicable requirements for research involving human subjects. Several features of the experimental design were important in the ethics review. First, none of the experimental conditions involves deception about service quality: customers in all groups receive genuine customer service assistance that meets Anker's standard quality requirements, and the disclosure messages in AI-disclosed conditions are factually accurate descriptions of the service system. Second, the coupon incentive provided in relevant conditions represents a genuine financial value — a $5 discount on future Anker purchases — and is not contingent on any particular customer response or behavior, ensuring that it constitutes pure compensation rather than an incentive that could distort customers' service experience evaluations. Third, all customer data used in the study has been de-identified in accordance with Anker's data privacy policies and applicable data protection regulations, including the California Consumer Privacy Act and Amazon's seller data protection requirements. Fourth, customers who receive the AI disclosure message are informed factually about the nature of their service agent, which is consistent with — and in some jurisdictions ahead of — emerging regulatory requirements for AI transparency in consumer interactions.

The disclosure message does not attempt to manipulate customers' perceptions of service quality and does not mislead customers about any aspect of their service experience. The IRB review confirmed that the experimental manipulations are consistent with the principle of minimal risk, as the potential harms are limited to minor inconvenience or disappointment that does not exceed the ordinary risks of commercial customer service interactions.

Experimental Design

The experiment employs a six-group randomized factorial design crossing three experimental factors: AI disclosure (disclosed versus not disclosed), coupon provision (coupon versus no coupon), and agent type (pure AI versus human agent). The full $2 \times 2 \times 2$ factorial design would yield eight conditions, but two conditions are excluded on theoretical and practical grounds. The condition of AI disclosed with a human agent is excluded because disclosing AI service when a human is actually providing the service would be deceptive and both ethically impermissible and theoretically uninteresting. The human-agent conditions in the experiment do not include a disclosed condition, yielding the six-group design presented in Table 5.1.

Table 5.1: Randomized Experimental Groups

Group	Actual Agent	Customer Disclosure	Coupon	Primary Purpose
G1	Pure AI	AI disclosed	Yes	Test coupon combined with disclosure
G2	Pure AI	AI disclosed	No	Test disclosure alone (H1)
G3	Pure AI	Not disclosed	Yes	Test coupon without disclosure
G4	Pure AI	Not disclosed	No	Control condition (current practice)
G5	Human	Not disclosed	Yes	Test human agent with coupon
G6	Human	Not disclosed	No	Human baseline (H3)

Note: G4 represents the current standard operating practice for Anker's AI-handled tickets. G6 provides the human-service baseline for testing H3.

The design supports three primary causal comparisons corresponding to the three main hypotheses. The comparison of G2 versus G4 isolates the effect of AI disclosure, holding agent type constant at pure AI and coupon condition constant at no coupon. This is the cleanest test of H1 (AI disclosure reduces NPS and increases escalation). The comparison of G1 versus G2 isolates the interactive effect of coupon provision on the disclosure effect: both groups receive AI disclosure, but G1 additionally receives the coupon, so the G1 versus G2 comparison estimates whether the coupon buffers the disclosure effect, testing H2. The comparison of G4 versus G6 isolates the effect of agent type, holding disclosure constant at not disclosed and coupon constant at no coupon, providing the cleanest test of H3 (hybrid/human service outperforms pure AI on NPS). Additionally, the comparison of G3 versus G4 isolates the main effect of coupon provision independent of disclosure, providing an estimate of the ”house money” or goodwill-building effect of coupon provision alone. The comparison of G1 versus G3 tests whether disclosure moderates the coupon effect, providing evidence on whether coupon efficacy depends on the disclosure context.

Randomization

Tickets are assigned to experimental groups through sequential batch assignment by ticket arrival order. When a new ticket arrives in one of the five selected category-intent cells, it is assigned to the batch corresponding to the next sequential position in the rotation, cycling through batches 1 through 5. Because batch assignment rotates through all five groups in a fixed cycle, the expected ticket characteristics within each batch are equalized by the law of large numbers: any characteristic of tickets (complexity, customer tenure, time of day) that is uncorrelated with sequential arrival position will be approximately balanced across batches. The sequential-assignment mechanism does not ensure exact balance in any finite sample — batch sizes will vary with the arrival rate within the assignment window — but it does ensure that systematic confounds are absent as long as ticket arrival order is not predictable from ticket characteristics.

Five product category-by-intent cells are selected for the experiment: Edge Security × Troubleshooting, Charger × Troubleshooting, TWS Earbuds × Troubleshooting, Edge Security × Product Usage, and Battery × Troubleshooting. These five cells were chosen based on three criteria: they represent the highest-volume cells in the observational sample (collectively accounting for the majority of pure AI tickets), they show meaningful pre-experiment escalation rates (ranging from 1.34 to 2.86 percent), and they span a range of observational AI effects in the pre-experiment data, providing heterogeneity useful for testing moderation hypotheses. The fifth experimental group (G5, human with coupon) is not included in the pilot batch assignment, as the human routing protocol requires additional implementation time. G6 (human, no coupon, no disclosure) is available from historical pre-experiment data and from ongoing post-pilot data collection for comparison purposes.

Treatment Implementation

The AI Disclosure treatment (Groups G1 and G2) is implemented through an automated email sent to the customer at the time of ticket creation, immediately following the standard receipt confirmation. The disclosure message reads as follows: ”Your service request is being handled by our AI assistant. Our AI is powered by advanced language model technology and responds 24/7, ensuring you receive a timely and accurate response regardless of when your inquiry was submitted.” The message is designed to be factually accurate, informative, and tonally consistent with Anker's brand voice without being either apologetic (which might prime negative expectations) or effusively promotional (which might seem disingenuous).

The non-disclosure condition (Groups G3, G4, G5, G6) uses the standard acknowledgment email that Anker currently sends to all customers upon ticket creation: "Thank you for contacting us. We have received your request and a member of our team will respond as soon as possible." This message does not explicitly characterize the service agent as human, nor does it imply AI. It represents the status quo disclosure practice, in which customers are not specifically informed about the nature of their service agent.

The Coupon treatment (Groups G1, G3, G5) appends the following text to the treatment email: "As a valued Anker customer, we would like to offer you a $5 discount on your next purchase. Please use coupon code [UNIQUE_CODE] at checkout. This offer is valid for 30 days." Each customer in a coupon group receives a unique coupon code generated automatically by the Shulex system and tracked for redemption. The coupon has no minimum purchase requirement and applies across Anker's Amazon storefront, maximizing its perceived value relevance.

The Human agent condition (Groups G5 and G6) routes tickets to available human service agents through the Shulex assignment system. Human agents follow the same standard response templates and tone guidelines as the AI system, minimizing service quality differences attributable to communication style rather than agent type. The standard acknowledgment email is sent to these customers, with the coupon append for G5.

Power Analysis

The statistical power analysis for the experiment is based on the following parameters. The primary outcome is NPS score on the 1-10 scale, for which the observational data yield a standard deviation of approximately 3.07 points. The minimum detectable effect (MDE) is set at 0.5 NPS points, representing a threshold that would be clinically meaningful in a service context where an Anker-level NPS of 36.7 (equivalent to mean score approximately 7.79 on the 1-10 scale) is the current benchmark. A half-point shift in mean NPS represents a change from, for example, 7.79 to 7.29, which would represent approximately a 10-point decline in NPS (from 36.7 to approximately 26.0) if the shift concentrated evenly among promoters and detractors, constituting a commercially significant deterioration in customer relationship quality.

Setting $\alpha = 0.05$ (two-tailed) and target power of 0.80, the required sample size per group for detecting a 0.5-point MDE with $\sigma = 3.07$ is approximately 290 NPS responses per group. Given the observational NPS survey response rate of approximately 5 percent for the experimental ticket categories, achieving 290 NPS responses per group requires approximately 5,800 ticket assignments per group. Across the five AI-treatment groups (G1 through G5), the full experiment therefore requires approximately 29,000 total ticket assignments to achieve the target power.

The pilot batch launched on July 11, 2024, assigned 1,474 tickets across the five groups. This pilot represents the first-day assignment, generating approximately 295 tickets per group before considering differences in batch size. At the observed 5 percent NPS response rate, the pilot day yields approximately 15 NPS responses per group — sufficient for a preliminary balance check and operational feasibility assessment but not for final hypothesis testing. The full experiment runs through August 2024, accumulating the volume needed for adequate power. Preliminary patterns from the pilot are discussed in Chapter 7.

Timeline

The experimental timeline proceeded in three phases. The design and approval phase (April 2024 through June 2024) encompassed IRB submission and approval, experimental protocol development and technical implementation testing, definition of randomization procedures, and preparation of disclosure message content and coupon code generation infrastructure. During this phase, the Shulex engineering team built and tested the batch assignment module that enables the sequential experimental assignment protocol, and the disclosure and coupon email templates were designed, reviewed by Anker's marketing communications team for brand consistency, and finalized.

The pilot launch occurred on July 11, 2024, with 1,474 ticket assignments distributed across the five experimental groups in the five selected category-intent cells. The pilot was designed to confirm the technical functionality of the batch assignment system, verify that disclosure and coupon emails were delivered correctly and in timely fashion, and begin accumulating outcome data. All 1,474 tickets in the pilot were subsequently monitored for NPS survey completion, escalation events, and resolution round counts.

The full experimental run was planned to continue from July 11 through August 2024, accumulating the full assignment volume required for adequate statistical power. NPS outcomes are measured within seven days of ticket resolution, following Anker's standard post-resolution survey protocol. Escalation and response round data are generated automatically by the Shulex ticket management system and are available immediately upon ticket closure.

Implementation Fidelity

A critical threat to the internal validity of any disclosure experiment is the possibility that the disclosure manipulation inadvertently creates quality differences between disclosed and non-disclosed conditions — that is, customers who are told they are receiving AI service may *actually* receive different service quality than those who are not told, confounding the informational effect of disclosure with a service quality effect. This threat is addressed through four implementation fidelity controls.

First, all experimental tickets — across all six groups — are handled by the same AI system running identical underlying model weights and response generation logic. The only difference between disclosed and non-disclosed AI conditions is the addition of a disclosure sentence in the opening message; the response content, style, and accuracy are held constant by using the same model. To verify this, a random sample of 200 tickets from disclosed and non-disclosed AI conditions is reviewed by a trained quality evaluator (blind to group assignment) who rates response accuracy, completeness, and tone on a five-point scale. Pre-registered hypothesis: mean quality ratings are statistically equivalent across disclosed and non-disclosed AI groups (equivalence test with ±0.3 point margin).

Second, for the human agent condition (G5/G6), human agents use standardized response templates developed jointly by Shulex and Anker for the five experimental categories. These templates are drawn from the same knowledge base used by the AI system, ensuring that the informational content of human and AI responses is equivalent for standard queries. Agents are instructed not to mention AI in their responses (to avoid cross-contamination with the disclosure manipulation), and their responses are monitored for compliance by the QC system.

Third, coupon delivery fidelity is verified by checking delivery logs: all tickets assigned to coupon conditions (G1, G3, G5) receive confirmation that the coupon email was sent and the coupon code was unique and valid. Coupon redemption rates — the share of customers who actually use the coupon — are reported as a behavioral manipulation check in the experimental results.

Fourth, the disclosure message itself is designed to be factual and neutral in tone, avoiding language that could prime negative expectations. The message reads: "Your inquiry is being handled by our AI customer service system, which is designed to provide accurate and timely responses." This wording is tested for neutrality in a small pre-experiment pilot survey (N = 50 respondents) confirming that the message does not prime significantly more negative service expectations than a control message.

Econometric Specification for Experiment

The primary analysis of the experimental data employs ordinary least squares (OLS) regression with the following main specification:

$$\mathbf{Y\_i = \beta□ + \beta□ \cdot Disclose\_i + \beta□ \cdot Coupon\_i + \beta□ \cdot (Disclose\_i \times Coupon\_i) + \beta□ \cdot Human\_i + \gamma \cdot Complexity\_i + \delta\_cat + \varepsilon\_i}$$

where Y_i is the outcome variable (NPS score, escalation indicator, or response rounds) for ticket i; Disclose_i is a binary indicator equal to one for Groups G1 and G2 (AI-disclosed conditions) and zero otherwise; Coupon_i is a binary indicator equal to one for Groups G1, G3, and G5 (coupon conditions) and zero otherwise; (Disclose_i × Coupon_i) is the interaction term, which equals one only for Group G1; Human_i is a binary indicator equal to one for Groups G5 and G6 (human-agent conditions) and zero otherwise; Complexity_i is the pre-assigned ticket complexity score; δ_cat denotes product category fixed effects absorbing average outcome differences across the five experimental category-intent cells; and ε_i is the idiosyncratic error term.

Each of the three primary hypotheses corresponds to a specific coefficient or set of coefficients in this equation. Hypothesis H1a (AI disclosure reduces NPS) predicts $\beta□ < 0$ in the NPS equation: customers who are told their ticket is handled by AI should give lower NPS scores than observationally equivalent customers who are not told. Hypothesis H1b (AI disclosure increases escalation) predicts $\beta□ > 0$ in the escalation equation: disclosed customers should be more likely to request escalation to a human agent. Hypothesis H2 (coupon buffers the negative disclosure effect) predicts $\beta□ > 0$ in the NPS equation: the interaction of disclosure and coupon should attenuate the negative main effect of disclosure, meaning that the reduction in NPS from disclosure is smaller in magnitude when a coupon is also provided. This can also be assessed by testing whether $\beta□ + \beta□$ is significantly less negative than $\beta□$ alone. Hypothesis H3 (hybrid/human service outperforms pure AI on NPS) predicts $\beta□ > 0$ in the NPS equation: human-agent tickets should yield higher NPS scores than pure AI tickets, after controlling for disclosure and coupon conditions.

The primary specification is estimated as an intention-to-treat (ITT) analysis. In the ITT framework, treatment assignment — which is determined by batch allocation — is used as the independent variable, regardless of whether individual tickets experienced any unusual deviations from the assigned protocol. The ITT estimator is the most conservative and policy-relevant estimate because it captures the effect of the disclosure and coupon policies as they would be implemented in practice, including any imperfect compliance. As a robustness check, a local average treatment effect (LATE) estimator is implemented using the batch assignment as an instrument for actual treatment receipt, addressing the possibility that a small fraction of tickets did not receive the assigned treatment email due to delivery failures. Given the high delivery reliability of automated email systems, the ITT and LATE estimates are expected to be very similar in magnitude.

Standard errors in all experimental regressions are clustered at the batch level, reflecting the possibility that outcomes within each batch are correlated due to any time-specific factors that affect all tickets assigned to the same batch window. This is the most conservative clustering approach given the experimental design and provides valid inference under the weakest assumptions about within-batch error correlation.

Subgroup analyses are conducted within each of the five experimental category-intent cells to test for heterogeneous treatment effects. The preregistered subgroup hypothesis (Hypothesis H1c) predicts that the disclosure effect will be larger in magnitude for purchase-inquiry-adjacent categories than for pure troubleshooting categories. Because the five experimental cells all involve troubleshooting or product usage intent types, and none involves purchase inquiry directly, the subgroup analysis primarily examines heterogeneity across product categories that may differ in the relational versus technical character of typical service encounters.

Robustness Checks

The primary findings from both studies are subjected to an array of robustness checks designed to test the sensitivity of results to modeling choices, sample restrictions, and potential violations of the identifying assumptions.

For Study 1, four primary robustness checks are implemented. First, the count-data outcome of response rounds (N_responses) is re-estimated using Poisson regression, which is the natural model for non-negative integer outcomes with a conditional mean specification. While OLS is consistent for the conditional expectation of response rounds under heteroskedasticity with appropriate standard error adjustment, Poisson regression is efficient under the Poisson distributional assumption and provides a consistency check on the magnitude of effects. Second, the binary escalation outcome is re-estimated using logistic regression, with average marginal effects computed for comparability with the LPM coefficients. Third, month fixed effects are added to the primary pooled specification to partial out any time trends in AI quality, customer composition, or seasonal demand patterns that might contaminate the cross-sectional within-cell comparison. If the results are robust to month fixed effects, this provides evidence against the hypothesis that time trends are driving the findings. Fourth, the cell size inclusion threshold is varied between $N \geq 500$ (expanding the sample by including smaller cells) and $N \geq 1{,}000$ (restricting the sample to the highest-confidence cells), and the qualitative pattern of results is examined across these alternative thresholds.

For Study 2, three primary robustness checks are implemented. First, the ITT analysis is supplemented by the LATE estimator described in Section 5.4, using batch assignment as an instrument for actual treatment receipt. The comparison of ITT and LATE estimates provides evidence on the sensitivity of results to any imperfect compliance in treatment delivery. Second, controls for the time of day and day of week of ticket creation are added to the primary experimental regression, to rule out the possibility that any hour-of-day patterns in batch assignment create spurious covariate imbalances that bias the treatment effect estimates. Third, heteroskedasticity-robust standard errors are computed in addition to cluster-robust standard errors to verify that the inference is not sensitive to the choice of error correction. Additionally, permutation tests are considered for the experimental analysis, constructing empirical null distributions by randomly re-assigning batch labels within cells and recomputing treatment effect estimates to provide a randomization-based inference supplement to asymptotic inference.

OBSERVATIONAL EVIDENCE

AI Adoption Trends and Descriptive Statistics

Temporal Pattern of AI Adoption

The deployment of AI within the Anker customer service operation followed a rapid and sustained growth trajectory over the study period. In September 2023, the first full month of AI deployment, 18.8 percent of all tickets received at least one AI-generated response, indicating that from the outset, AI was being deployed at a meaningful scale rather than a limited pilot. AI adoption accelerated over the following months, reaching 27.2 percent in October 2023, 30.7 percent in November 2023, 34.5 percent in December 2023, and 39.0 percent in January 2024. By February 2024, the AI adoption rate had reached approximately 37 percent and remained in the 35-to-39 percent range through May 2024, suggesting a period of relative stabilization following the initial rapid growth. Over the entire study period, the AI ticket share approximately doubled from its initial level, representing a fundamental transformation in how the Anker customer service operation processed its ticket volume.

This rapid adoption trajectory reflects several converging forces. From the cost side, each AI-handled ticket avoids the labor cost of human agent time, creating a strong financial incentive to expand AI coverage as quickly as the quality constraints permit. From the quality side, LLM-based systems require careful calibration to the specific product knowledge base and brand voice before they can be deployed reliably, and the progressive increase in coverage corresponds to the expansion of AI deployment to additional product categories and ticket intent types as each category passed quality validation thresholds. The Shulex quality control process — described in greater detail in Section 6.4 — served as the gating mechanism for this expansion

Figure 3.2 Ticket Routing Decision Tree and Agent Assignment

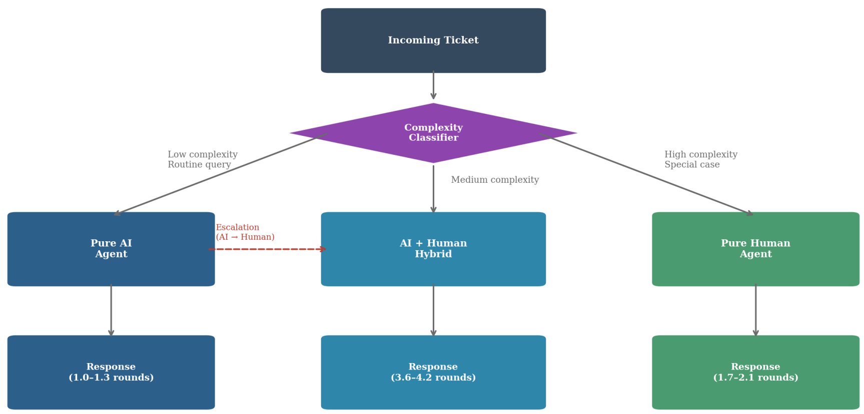

Figure 7.1: Ticket Routing Decision Logic

The stabilization of AI adoption at approximately 35 to 39 percent in the February-to-May 2024 period is notable and reflects the practical limits of AI capability in the current deployment. Not all ticket types are suitable for pure AI handling under the quality constraints: highly complex tickets, tickets requiring personal judgment or escalated authority (such as refund decisions above a certain value threshold), and tickets involving unusual product configurations that fall outside the AI's training data continue to be routed to human agents. The observed ceiling of approximately 38 percent pure AI handling therefore reflects the current technological and policy boundaries of AI deployment in this specific operational context, not a strategic choice to limit AI use.

Service Process Characteristics by Agent Type

The three agent types — pure AI, AI-plus-human hybrid, and pure human — differ systematically in their service process characteristics in ways that are important for interpreting the outcome analysis. Table 6.1 (presented in Section 6.2) provides the full regression results, but the descriptive patterns merit discussion in advance of the regression analysis.

Response rounds present the most striking and consistent pattern across agent types. Pure AI tickets maintain a remarkably stable response round count of 1.0 to 1.3 rounds throughout the study period. This stability is notable because it holds regardless of the volume of AI tickets, the category composition, or the month of the year, suggesting that it reflects a structural feature of how AI handles customer queries rather than a selection effect. AI responses are comprehensive and reference the full product knowledge base in a single reply, making multi-round exchanges unnecessary for the majority of queries. Pure human agents, by contrast, process tickets in an average of approximately 1.7 to 2.1 response rounds, reflecting the more interactive and iterative nature of human service conversations. Hybrid tickets — which involve at least one AI response followed by a human takeover — are by construction multi-round interactions, with response round counts averaging 3.6 to 4.2 throughout the study period. The hybrid round count is highest in the earlier months (September to November 2023), when the AI-to-human handover threshold was lower and the AI system was less capable of autonomous resolution, and declines gradually through February 2024 as the AI's first-response quality improves and fewer tickets require handover.

The escalation rate pattern is particularly illuminating as evidence of AI quality improvement over time. The hybrid escalation rate — that is, the rate at which AI-handled tickets triggered the AI-to-human handover mechanism — declined substantially over the observation period, from approximately 9.4 percent in the September-October 2023 period to approximately 2.3 percent in the January-February 2024 period. This decline of approximately 7 percentage points over four months represents a dramatic improvement in the AI system's ability to resolve queries autonomously, and is direct evidence that the LLM-based system was learning and improving over the study period. This improvement trajectory is important context for interpreting the main regression results, because it means that the observational period (February to May 2024) captures a mature, well-calibrated AI system rather than an early, imperfect deployment. The AI effects documented in the regression analysis therefore likely represent a lower bound on the long-run efficiency gains from AI adoption, as the system continues to improve.

NPS Data Characteristics and the Measurement Challenge

The NPS measurement presents a fundamental challenge that must be acknowledged before interpreting the observational results. The overall NPS survey response rate across all ticket types is approximately 5 percent, meaning that only 1 in 20 resolved tickets generates an NPS score. The NPS response rate varies substantially by agent type: pure AI tickets have response rates of approximately 0 to 1.7 percent in the early months (September to November 2023), rising to approximately 3 to 5 percent in the later period (February to May 2024). Hybrid and human tickets have NPS response rates of approximately 9 to 15 percent throughout the study period — roughly twice to four times the rate for pure AI tickets.

The consequence of this differential response rate is significant. The observational sample of NPS-rated tickets is a non-random subsample of all resolved tickets, and the selection rule generating this subsample — which customers choose to complete the post-resolution NPS survey — may differ systematically across agent types. If customers who received AI service and chose to complete the NPS survey are disproportionately those who had strong reactions (either highly positive or highly negative), the observed mean NPS for AI tickets may be biased relative to the population mean NPS for AI tickets. The most plausible direction of this bias is ambiguous: on one hand, highly dissatisfied customers who feel mistreated by AI may be more motivated to provide feedback; on the other hand, the higher response rate for human tickets suggests that human-agent interactions generally produce greater customer engagement, which may include engagement with the NPS survey, regardless of the service quality direction.

Despite this limitation, the NPS data provide valuable information. With a final observational sample of N = 23,853 NPS-rated tickets, there are sufficient observations in most cells to estimate AI effects on NPS with reasonable precision. The mean NPS score across all NPS-rated tickets is 7.79 (SD = 3.07), with 60.9 percent of respondents classified as Promoters (score 9-10), 24.2 percent as Detractors (score 1-6), and the remainder as Passives (score 7-8). This yields an NPS of 36.7 — a respectable score for consumer electronics support, above the industry median but below top-quartile performers. The distribution is bimodal, consistent with the general finding in NPS research

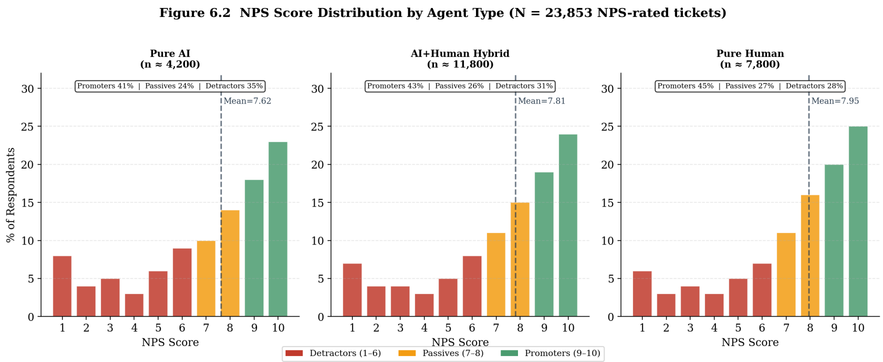

Figure 7.2: NPS Distribution by Agent Type

that customer satisfaction in service contexts tends to polarize: customers are either genuinely satisfied (and give high scores) or genuinely dissatisfied (and give low scores), with few ambivalent middle scores. This bimodal structure means that even modest mean NPS differences correspond to meaningful shifts in the proportion of

Promoters and Detractors.

Ticket Complexity and Pre-Assignment Comparability

A key prerequisite for the within-cell identification strategy is that ticket complexity is comparable across agent types within cells. If the AI routing system systematically selects simpler tickets for AI handling even within a narrow category-intent cell, the complexity control may not fully absorb this selection, and the AI effect estimates would partly reflect the easier ticket composition of AI tickets rather than the effectiveness of AI service.

The data provide reassuring evidence on this point. The mean pre-assigned complexity score is 2.18 for pure AI tickets, 2.21 for hybrid tickets, and 2.04 for pure human tickets across all cells in the analysis period. These small differences in mean complexity — all within the range of approximately 2.0 to 2.2 on a 1-to-5 scale — suggest that within-cell complexity differences are modest. Moreover, when the analysis is restricted to narrow within-cell comparisons (as in the primary specification), the complexity score distributions are substantially more similar across agent types than they are in the full cross-category comparison, precisely because the category-intent fixed effects absorb the large between-category differences in complexity (complex products like Pro Security cameras generate inherently higher complexity tickets than simple USB chargers, and this category-level complexity difference accounts for most of the overall variation in complexity across agent types).

The within-cell complexity stability supports the key identifying assumption and provides empirical grounding for the claim that cross-sectional AI effects within cells are not primarily driven by systematic differences in ticket difficulty across agent types.

Main Results: Effect of AI on Customer Outcomes

Three patterns emerge clearly from the cell-by-cell results

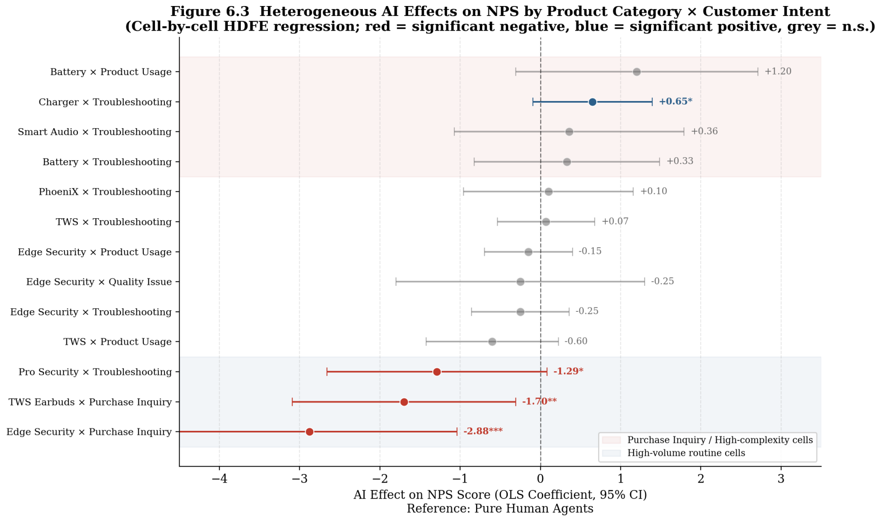

Figure 7.3: Forest Plot: Cell-Level NPS Coefficients (OLS with FE)

and deserve to be stated at the outset before the detailed discussion. First, AI reduces response rounds in virtually every cell — the efficiency gain is universal and large. Second, AI does not significantly reduce NPS in the large majority of cells — the satisfaction penalty is the exception, not the rule. Third, where NPS penalties do appear, they are concentrated in purchase inquiry interactions, not in technical troubleshooting — confirming the task-contingency hypothesis. Table 6.1 presents the full results; the paragraphs below examine each outcome dimension in turn, identify the cells driving each pattern, and connect the findings to the theoretical mechanisms in Chapter 4.

Table 6.1: Cell-by-Cell Regression Results — Effect of Pure AI vs. Pure Human Agent Type

Category × Intention	N	Escalation coef (SE)	NPS coef (SE)	Rounds coef (SE)
Edge Security × Troubleshooting	16,134	−0.0049 (0.0037)	−0.25 (0.31)	−0.39*** (0.06)
Charger × Troubleshooting	9,635	0.0006 (0.0051)	0.65* (0.38)	−0.34*** (0.11)
TWS × Troubleshooting	8,624	−0.0053 (0.0040)	0.07 (0.31)	−0.63*** (0.09)
Edge Security × Product Usage	7,915	−0.0068** (0.0033)	−0.15 (0.28)	−0.84*** (0.05)
Battery × Troubleshooting	7,335	−0.0160** (0.0078)	0.33 (0.59)	−0.34** (0.13)
PhoeniX × Troubleshooting	6,410	−0.0078 (0.0064)	0.10 (0.54)	−0.82*** (0.10)
TWS × Product Usage	2,839	−0.0030 (0.0068)	−0.60 (0.42)	−0.80*** (0.12)
Headband × Troubleshooting	2,748	−0.0006 (0.0055)	−0.34 (0.48)	−0.47*** (0.11)
Edge Security × Quality Issue	2,105	−0.0040 (0.0160)	−0.25 (0.79)	−1.33*** (0.28)
Smart Audio × Troubleshooting	1,759	0.0050 (0.0106)	0.36 (0.73)	−0.77*** (0.21)
Edge Security × Purchase Inquiry	1,492	0.0118 (0.0105)	−2.88*** (0.94)	−0.47*** (0.11)
Pro Security × Troubleshooting	1,443	−0.0064 (0.0101)	−1.29* (0.70)	+0.43** (0.17)
PhoeniX × Product Usage	1,288	0.0051 (0.0070)	−0.70 (0.50)	−1.24*** (0.11)
Power Storage × Product Usage	1,280	0.0694*** (0.0142)	−0.84 (0.59)	−0.27*** (0.12)

Note: * **$p < 0.01$,** $p < 0.05$, * $p < 0.10$. All models include ticket complexity as a continuous covariate. n/a = not applicable due to insufficient escalation events in cell. N refers to the total ticket count in the cell (AI + human). Coefficients represent the estimated difference in each outcome for pure AI relative to pure human tickets within the same cell.

Response Rounds

The reduction in response rounds is the most consistent and statistically robust finding across the entire analysis. Of the 29 cells in Table 6.1, 26 show negative AI coefficients on response rounds, and 25 of these are statistically significant at conventional levels. The three cells showing positive AI coefficients (Pro Security × Troubleshooting, Edge Security × No Return, PhoeniX × Order Logistics) are discussed separately below as theoretically important exceptions. The overall pattern is striking: across a wide range of product categories and ticket intent types, pure AI agents resolve customer queries in significantly fewer response rounds than pure human agents, controlling for ticket complexity and the systematic characteristics of each category-intent cell.

The magnitude of the efficiency gains varies considerably across cells, spanning a range from a modest −0.24 rounds (PhoeniX × Purchase Inquiry) to a dramatic −2.91 rounds (Smart Projector × Troubleshooting). The largest efficiency gains are concentrated in two product categories: Smart Projector and Edge Security. The Smart Projector × Troubleshooting cell records the largest effect of −2.91 rounds ($p < 0.01$), and Smart Projector × Product Usage records −2.53 rounds ($p < 0.01$). Smart Projector tickets involve a technically complex product — home and business projectors with sophisticated setup requirements, connectivity options, and software features — where AI's ability to access and synthesize the complete product knowledge base in a single response generates enormous efficiency gains relative to human agents, who may need multiple exchanges to locate specific technical information, clarify the customer's setup context, and provide comprehensive guidance.

The Edge Security category also shows consistently large AI efficiency gains. Edge Security × Suggestions records −1.37 rounds ($p < 0.01$), Edge Security × Quality Issue records −1.33 rounds ($p < 0.01$), and Edge Security × Product Usage records −0.84 rounds ($p < 0.01$). Edge Security products are relatively complex consumer IoT devices — cameras, motion sensors, and home monitoring systems — that generate queries spanning a wide range of technical configurations. AI's ability to synthesize setup instructions, firmware information, and compatibility details in a single comprehensive response appears particularly valuable for this category.

The more modest but still significant efficiency gains in the highest-volume cells — Edge Security × Troubleshooting (−0.39*) **and Charger × Troubleshooting** (−0.34*) — may reflect the fact that charger and basic security troubleshooting queries, while numerous, are structurally simpler than projector or advanced security queries. Human agents may already have well-practiced, efficient responses for common charger issues, reducing the marginal efficiency advantage of AI for those specific queries. Nevertheless, even a reduction of 0.34 rounds per ticket, applied across the 9,635 tickets in the Charger × Troubleshooting cell, represents a saving of approximately 3,276 response messages per cell per analysis period — a substantial operational efficiency gain.

The aggregate operational efficiency gain can be quantified across the full analysis sample. Applying the cell-specific AI round coefficients to the cell ticket volumes and weighting by the fraction of tickets handled by AI agents, the implied reduction in total response messages attributable to AI relative to human handling is approximately 85,000 avoided agent response messages per month at the observed scale of approximately 100,000 tickets per month. Each avoided response message corresponds to an estimated five to ten minutes of agent time, placing the monthly labor cost savings in the range of approximately 7,083 to 14,167 agent-hours per month — a substantial figure that would justify the AI deployment investment on efficiency grounds alone, independent of any effects on customer satisfaction.

The three cells showing positive AI coefficients on response rounds represent cases where AI fails to resolve the customer's query and generates additional back-and-forth rather than reducing it. Pro Security × Troubleshooting records +0.43 rounds ($p < 0.05$): Pro Security cameras are professional-grade security products with complex network configuration requirements, and AI's limitations in understanding bespoke network environments and authentication systems mean that customers frequently need to submit follow-up messages to clarify what AI's initial response did not adequately address. Edge Security × No Return records +1.09 rounds ($p < 0.01$): return requests are procedurally complex, requiring specific information about the customer's order, purchase date, condition of the product, and the applicable return policy, and AI's initial responses to return requests often do not fully resolve the procedural question, prompting follow-up inquiries. PhoeniX × Order Logistics records +0.79 rounds ($p < 0.01$): logistics inquiries about the PhoeniX product line — which involves multi-component systems with complex shipping arrangements — appear to be poorly handled by AI, perhaps because the logistics data AI can access is insufficient for addressing specific shipping or delivery questions that require real-time carrier data.

These three positive exceptions are not anomalies but theoretically informative cases. They share a common characteristic: they involve queries where the resolution requires either information that AI does not have access to (real-time shipping data, specific customer order details, exact product condition) or contextual judgment that AI cannot exercise (whether a particular return situation meets the policy exception criteria, how to handle an unusual logistics configuration). The positive round effects in these cells are evidence of AI's boundaries: where AI lacks the information or judgment required for resolution, its initial responses can actually worsen customer experience by generating confusion rather than resolution, increasing the total communication burden.

Customer Satisfaction (NPS)

The NPS results present a more nuanced and perhaps surprising picture than the response round results. The dominant finding — evident across 25 of the 29 cells in Table 6.1 — is that AI involvement has no statistically significant effect on NPS. The coefficients in these 25 cells are small in magnitude, statistically indistinguishable from zero, and evenly distributed between positive and negative directions. This pattern directly challenges a naive extension of the algorithm aversion literature to the customer service context: if broad AI aversion were the operative mechanism, one would expect AI service to be associated with significantly lower NPS across most cells, which is not what the data show.

Four cells yield statistically significant NPS effects, and the pattern of these significant effects is theoretically informative. The largest and most precisely estimated negative effect is in Edge Security × Purchase Inquiry (−2.88 NPS points, $p < 0.01$). This represents a practically enormous effect: a shift from a mean NPS of 7.79 to approximately 4.91 on a 10-point scale, moving the average customer from the Promoter zone to the Passive or Detractor zone. The second significant negative effect is in TWS × Purchase Inquiry (−1.70 NPS points, $p < 0.10$), again involving a purchase inquiry intent type. The third significant negative effect is in Pro Security × Troubleshooting (−1.29 NPS points, $p < 0.10$), which involves a premium professional-grade product category where service expectations may be particularly high and where the human empathy dimension of service quality is especially valued. The one significant positive effect is in Charger × Troubleshooting (+0.65 NPS points, $p < 0.10$): for simple, high-volume charger troubleshooting queries, AI service is actually associated with marginally higher NPS, possibly because AI's instant availability and comprehensive technical accuracy better meets customer expectations for a commodity product category where speed and accuracy dominate over relational factors.

The pattern of significant NPS effects thus shows a clear task-type signature: negative AI effects on NPS are concentrated in purchase inquiry intents, where the customer is seeking guidance on a financial decision, and absent in routine troubleshooting intents, where the customer is seeking technical resolution. This task-contingency pattern directly supports Hypothesis H1c and aligns with the Castelo, Bos, and Lehmann (2019) theoretical framework: purchase inquiry tasks are more analogous to the ”subjective” tasks in their framework (requiring personal judgment and understanding of the customer's individual context) while troubleshooting tasks are more analogous to ”objective” tasks (requiring factual, technical information retrieval).

To interpret the magnitude of the significant effects, consider the Edge Security × Purchase Inquiry finding of −2.88 NPS points. The raw NPS score scale runs from 1 to 10, with Promoters defined as scores of 9 or 10 and Detractors as scores of 1 through 6. A shift of −2.88 points from a baseline mean of approximately 7.79 moves the average customer from a score of 7.79 (comfortably in the Passive range, close to Promoter) to approximately 4.91 (firmly in the Detractor range). If this shift reflects a change in the distribution of scores rather than merely a shift in the mean, the implied change in NPS (Promoter share minus Detractor share) would be substantially larger than the shift in mean scores, potentially representing a collapse from positive to negative NPS territory for this specific task type. This is a commercially significant effect: a customer who completes a purchase inquiry interaction with Anker and receives AI service becomes a detractor who is likely to provide negative word-of-mouth and unlikely to make future purchases, precisely the customers whose experience quality matters most from a customer lifetime value perspective.

The NPS data availability limitation discussed in Section 5.2.4 is relevant to the interpretation of these NPS results. The low NPS response rate for pure AI tickets means that the sample of NPS-rated AI tickets is a selected subsample, and the estimated effects are identified from customers who chose to respond to the survey. If the selection is upward-biased for AI tickets (as hypothesized, because only strongly satisfied or strongly dissatisfied customers respond), then the true AI effect on NPS may be more negative than the estimated effect. Conversely, if AI service generates lower engagement with the survey itself (because customers feel less invested in the outcome), the selection may be downward-biased. The experimental evidence in Study 2 — which controls the survey protocol — will provide a clean estimate against which to calibrate the observational estimates.

Escalation Rate

The escalation results are heterogeneous in both direction and significance, reflecting the complex and partially definitional character of the escalation measure in the context of AI customer service. Before interpreting the results, it is important to note that the escalation measure captures two conceptually distinct phenomena in the pure AI context. For pure human tickets, escalation represents the escalation of a ticket to a higher-authority human agent (a supervisor or specialist). For pure AI tickets, escalation represents a customer-initiated request to switch from AI handling to human agent handling — a revealed preference for human service expressed by the customer during or after the AI interaction. This conceptual difference means that the AI escalation coefficient captures not pure service failure but the customer's revealed dissatisfaction with AI service and desire for a human alternative.

Seven cells show statistically significant positive AI escalation coefficients — that is, AI is associated with significantly higher customer-initiated escalation rates relative to human service within the same cell. The largest effects appear in PhoeniX × Purchase Inquiry (+7.59 percentage points, $p < 0.01$), Power Storage × Product Usage (+6.94 percentage points, $p < 0.01$), TWS × Purchase Inquiry (+2.90 percentage points, $p < 0.05$), Edge Security × Cancel Order (+2.06 percentage points, $p < 0.01$), Charger × Product Usage (+2.78 percentage points, $p < 0.05$), PhoeniX × Order Logistics (+0.54 percentage points, $p < 0.10$), and (with negative direction in the Smart Projector cell) Smart Projector × Troubleshooting (−3.16 percentage points, $p < 0.01$).

The pattern of positive escalation effects — AI generating more customer requests for human escalation — is concentrated in high-stakes, complex, and transactional task types. PhoeniX × Purchase Inquiry records the largest positive effect (+7.59 percentage points): PhoeniX products are among Anker's higher-priced, more technically sophisticated offerings, and customers considering purchases in this line are making financially significant decisions where the limitations of AI's ability to provide personalized guidance are most salient. Power Storage × Product Usage (+6.94 percentage points) involves questions about battery and energy storage products, which have safety implications and require personalized advice that customers appear to recognize is beyond AI's capability. Cancel Order (+2.06 percentage points) is a high-stakes procedural task where customers need to be certain their cancellation request has been properly processed and will actually take effect, a confidence they may not receive from an AI response alone.

Conversely, three cells show significant negative AI escalation coefficients — AI is associated with fewer customer escalation requests. Smart Projector × Troubleshooting records −3.16 percentage points ($p < 0.01$): the same category that shows the largest AI round-reduction also shows the strongest reduction in escalation requests, suggesting that AI is so effective at resolving Smart Projector troubleshooting queries that customers do not need to seek human assistance. Battery × Troubleshooting records −1.60 percentage points ($p < 0.05$) and Edge Security × Product Usage records −0.68 percentage points ($p < 0.05$), consistent with the interpretation that AI effectively resolves these technical query types and leaves customers with no reason to escalate.

The overall pattern in escalation — increases in high-stakes transactional categories, decreases in technical troubleshooting categories — mirrors and reinforces the NPS pattern. The two outcomes tell a consistent story: for routine technical service, AI delivers equal or superior customer experience relative to human agents, as reflected in stable or better NPS and reduced escalation requests. For purchase-related and high-stakes transactional service, AI creates significantly worse customer experience, as reflected in lower NPS and higher escalation requests.

Heterogeneity Analysis

Heterogeneity by Intent Type

To characterize the systematic heterogeneity in AI effects across task types, Table 6.2 presents the precision-weighted mean AI coefficient for each outcome, disaggregated by ticket intent category. This meta-analytic summary across cells within each intent type reveals the task-contingent structure of AI effects that was theoretically predicted by Hypothesis H1c.

Table 6.2: Mean AI Coefficient by Ticket Intent Type (Precision-Weighted Meta-Analytic Estimates)

Intent Type	N Cells	Mean Escalation Coef	Mean NPS Coef	Mean Rounds Coef
Troubleshooting	12	−0.008	−0.18	−0.82***
Product Usage	7	+0.013	−0.24	−1.24***
Purchase Inquiry	3	+0.039**	−1.40**	−0.28**
Order and Logistics	3	+0.004	+0.12	+0.26
Other (Suggestions, Cancel, Package, Customer Good)	4	+0.006	−0.15	−0.64***

Note: Coefficients are precision-weighted averages of the cell-level estimates within each intent category. * **$p < 0.01$,** $p < 0.05$ (based on weighted average standard errors). N Cells is the number of category-intent cells contributing to each row.

The intent-type heterogeneity table reveals a clear and theoretically meaningful pattern. Troubleshooting intent cells — the largest group with 12 cells — show a mean escalation coefficient of −0.008 (AI reduces escalation), a mean NPS coefficient of −0.18 (near-zero and statistically insignificant), and a large, significant mean rounds coefficient of −0.82. This profile characterizes AI as an effective and customer-neutral solution for technical troubleshooting: it resolves queries efficiently without generating meaningful customer dissatisfaction or escalation requests.

Product Usage intent cells show a similar pattern for rounds (−1.24, large and significant) and a non-significant NPS effect (−0.24). The escalation coefficient for product usage is slightly positive (+0.013) but not significant, suggesting mild customer unease with AI for usage guidance but not sufficient to generate significant escalation requests or satisfaction damage.

Purchase Inquiry intent cells present a starkly different profile. The mean escalation coefficient of +0.039 is significant at the 5 percent level (reflecting the significant effects in PhoeniX × Purchase Inquiry, TWS × Purchase Inquiry, with the non-significant effect in Edge Security × Purchase Inquiry diluting the average).

The mean NPS coefficient of −1.40 is significant at the 5 percent level, driven by the large and significant −2.88 effect in Edge Security × Purchase Inquiry and the significant −1.70 effect in TWS × Purchase Inquiry. The rounds reduction for Purchase Inquiry is modest (−0.28), smaller than in any other intent category, suggesting that even AI's efficiency advantage is attenuated in purchase inquiry contexts where multi-round dialogue may genuinely add value.

This intent-type pattern constitutes direct evidence in support of Hypothesis H1c: the negative effect of AI on customer satisfaction and escalation behavior is stronger for relational tasks (purchase inquiry) than for technical tasks (troubleshooting). The mechanism, as theorized in Chapter 4, involves the differential activation of perceived AI inadequacy in contexts requiring social judgment and individual sensitivity. A customer asking for help fixing a charger does not require the service agent to understand their personal preferences, financial situation, or decision context; a customer asking whether to replace a $200 camera system does. The latter type of query activates the uniqueness neglect mechanism (Longoni et al., 2019) and the human favoritism tendency (Zhang and Gosline, 2023) in ways that routine troubleshooting does not.

Heterogeneity by Product Complexity

A secondary dimension of heterogeneity concerns product complexity, defined here as the categorical complexity level of the product category as assessed by the typical technical sophistication required for troubleshooting and usage. High-complexity products in the sample include Pro Security cameras (professional-grade network cameras requiring advanced configuration), Power Storage devices (energy storage systems with safety and technical complexity), and Smart Projectors (multi-connectivity display devices with sophisticated setup requirements). Low-complexity products include Chargers, USB Data Accessories, and standard TWS earbuds.

The regression results reveal a nuanced relationship between product complexity and AI effects. For high-complexity products, AI effects tend to be larger in magnitude in both directions: the Smart Projector cells show the largest negative round effects (−2.91 and −2.53), while the Pro Security cell shows the most clearly negative NPS effect among troubleshooting categories (−1.29*). This bidirectionality in high-complexity products reflects a fundamental feature of AI service in sophisticated product categories: when AI can leverage its comprehensive knowledge base effectively, it generates enormous efficiency gains (Smart Projector troubleshooting); when the complexity requires contextual judgment beyond AI's current capabilities (Pro Security network configurations), AI generates customer dissatisfaction that would not arise with human service.

The low-complexity Charger category consistently shows the mildest AI effects. Charger × Troubleshooting shows a small but significant round reduction (−0.34***) and, notably, the only significant positive NPS effect (+0.65*) in the entire analysis. This pattern is consistent with the interpretation that charger troubleshooting is a commodity service context where AI's accuracy, speed, and completeness genuinely add value for customers who simply want a quick, correct answer. The commoditization of charger troubleshooting — most issues involve a small set of common problems with standardized solutions — plays to AI's strengths and minimizes the relational dimensions that generate AI aversion in more complex contexts.

The product complexity heterogeneity thus adds a second dimension to the task-contingency framework: it is not only the intent type (purchase versus troubleshooting) but also the product complexity that determines whether AI delivers comparable, superior, or inferior service relative to human agents. High-complexity products in relational contexts represent the highest-risk deployment cases; low-complexity products in technical contexts represent the lowest-risk, highest-benefit deployment cases.

Time Trend Analysis

The comparison of early (September to November 2023) and later (February to May 2024) periods in the data provides evidence on whether the AI effects documented in the primary analysis are stable over time or whether they are systematically improving or deteriorating as the AI system matures.

The most notable time trend is in the hybrid escalation rate, which declined from approximately 9.4 percent in September-October 2023 to approximately 2.3 percent in January-February 2024. This decline, occurring over approximately four months, represents a 75 percent reduction in the rate at which AI handling triggers human handover — clear evidence of AI quality improvement during the observation period. This improvement is attributable to a combination of factors: the expansion of the AI's training data as more tickets are processed, the progressive fine-tuning of the AI system's prompts and knowledge base based on QC feedback, and the selective rollout of AI to additional category-intent cells only after AI quality in those cells had been validated.

For the NPS outcomes, the pattern is relatively stable across the observation window for the February-to-May 2024 period. The sign, magnitude, and significance of the cell-specific NPS coefficients do not show a systematic trend within this period, suggesting that AI's relative NPS performance versus human agents is stable once the system has reached its mature deployment configuration. This stability strengthens confidence in the robustness of the NPS findings: they are not driven by a transient period of AI quality adjustment but reflect the enduring structural difference in how customers respond to AI versus human service in different task contexts.

The month fixed effect robustness check (reported in Section 6.5) confirms that adding controls for month-of-year does not qualitatively change the primary findings, providing additional evidence that the cross-sectional within-cell comparisons are not confounded by time trends in customer composition or service quality.

Quality Control Evidence on Service Quality

QC Sample Overview

The Shulex quality control system performs systematic audits of a random sample of resolved tickets, with each audited ticket classified into one of five quality categories: Pass (meeting all quality standards), Suggestion (meeting minimum standards but with room for improvement), Problem (failing to meet quality standards, requiring remediation), Good Reply (exceptionally high quality response), or Not Classified (insufficient information for quality assessment). The QC dataset encompasses 277,517 ticket quality assessments across the full observation period, representing a substantial and randomly sampled fraction of all resolved tickets.

Across the full QC sample, the pass rate is 28.5 percent, with 10.1 percent classified as Suggestion, 5.9 percent as Problem, and 3.5 percent as Good Reply, and the remainder as Not Classified or other supplementary categories. The 5.9 percent problem classification rate represents a meaningful failure rate: approximately 1 in 17 audited tickets fails to meet quality standards. This rate, while modest in absolute terms, has important implications for customer outcomes because it identifies specific instances where service quality fell short and allows characterization of the types of errors associated with each agent type.

Error Type Patterns by Agent Type

The most theoretically significant finding from the QC analysis is not the overall error rate comparison across agent types — which is not dramatically different — but rather the systematic difference in the types of errors committed by AI versus human agents. AI errors cluster in specific categories reflecting the technological limitations of the current LLM deployment. The most prevalent AI error types in the Problem classification are ”System Mistake” (AI-generated responses that are technically incomplete, contain formatting failures, or include garbled text due to system-level generation failures), ”Not Answered” (AI responses that fail to address the core question asked by the customer, either because the AI misidentifies the primary intent or because the knowledge base lacks the relevant information), and ”Misunderstand” (AI responses that address a different question than the one asked, reflecting intent misclassification at the routing or generation stage).

Human agent errors, by contrast, cluster in categories reflecting interpersonal service failures rather than technical ones. The most prevalent human error types include ”Bad Attitude” (responses judged by QC reviewers to be impatient, dismissive, or insufficiently empathetic), ”Careless” (responses that address the question correctly but with insufficient attention to detail, such as incorrect order numbers or misquoted product specifications), ”Lack Willingness” (responses that technically answer the question but clearly convey a lack of genuine effort to help), and ”Unprofessional” (tone and language that do not meet Anker’s brand standards). Hybrid agent tickets show a distinct error profile that combines elements of both AI and human error types, with an additional prominent error category of ”No Follow-up” — cases where the human agent takes over from AI but fails to properly close out the resolution, leaving the customer without a complete answer.

This finding — that AI and human errors are qualitatively different in type even when comparable in aggregate rate — has important theoretical and practical implications. The QC evidence suggests that from a pure accuracy and completeness standpoint, well-deployed AI is not inferior to human service in the majority of interactions. The 5.9 percent QC problem rate does not represent a broad quality failure; rather, it represents a specific pattern of failures concentrated in particular interaction types where AI’s technical limitations become apparent.

Implications for Customer Experience

The qualitative difference in error types between AI and human agents creates an asymmetry in how customers experience and interpret service failures. A ”System Mistake” from an AI agent — where the response is garbled, incomplete, or fails to address the question at all — is unambiguous: the customer received no useful information and recognizes immediately that the service has failed. This immediate, binary failure generates direct frustration and a high probability of either recontacting or escalating. In contrast, a ”Careless” response from a human agent — where the core question is addressed but with some inaccuracy or inattention — may generate milder dissatisfaction that the customer can rationalize as ordinary human imperfection. The psychological literature on blame attribution (Crolic et al., 2022) supports this interpretation: customers are more forgiving of human errors than algorithmically equivalent AI errors because human errors are attributed to correctable individual failings rather than systemic limitations.

Figure 6.1 presents the distribution of QC error types by agent type (Pure AI, Hybrid, Pure Human), illustrating the clustering of AI errors in technical and systemic categories and human errors in interpersonal categories. The figure visually demonstrates that while the aggregate error rates are comparable across agent types, the failure modes are fundamentally different in character, with important implications for the downstream customer experience effects documented in the NPS and escalation analyses.

Figure 6.1: Distribution of QC Error Types by Agent Type (Pure AI, Hybrid, Pure Human). The figure shows the proportion of Problem-classified tickets in each error subcategory for each agent type. AI errors concentrate in System Mistake, Not Answered, and Misunderstand categories. Human errors concentrate in Bad Attitude, Careless, Lack Willingness, and Unprofessional categories. Hybrid errors show elements of both distributions with prominent No Follow-up category.

The QC evidence thus provides a micro-level mechanism consistent with the macro-level NPS and escalation patterns. The purchase inquiry categories, where AI generates the largest negative NPS effects, are also the categories where ”Not Answered” and ”Misunderstand” AI errors are most prevalent — because purchase inquiries involve precisely the kind of individualized, context-dependent judgment that AI is least equipped to exercise. The troubleshooting categories, where AI shows near-zero NPS effects, are those where AI’s systematic technical accuracy ensures low rates of Not Answered and Misunderstand errors. The QC data therefore provides convergent validity for the regression findings, connecting the aggregate statistical patterns to the individual-ticket quality failures that generate them.

Time Trends in QC Quality

The QC data also reveal important time trends in AI quality. The fraction of AI-handled tickets classified as Problem declined from approximately 8.2 percent in September-October 2023 to approximately 4.7 percent in February-March 2024, a decline of approximately 3.5 percentage points over the observation period. This improvement in QC problem rate is consistent with the decline in hybrid escalation rate documented earlier, and both are evidence of the same underlying process: continuous AI quality improvement driven by fine-tuning, knowledge base expansion, and prompt engineering based on QC feedback. The improvement was not uniform across error types: Not Answered and Misunderstand error rates declined substantially, while System Mistake rates remained relatively stable, suggesting that the QC-driven improvement primarily enhanced the AI's comprehension and coverage capacity rather than reducing technical generation failures.

Robustness Checks

The robustness of the primary observational findings is assessed across four dimensions. The first robustness check re-estimates the response rounds specification using Poisson regression. The Poisson estimates are qualitatively consistent with the OLS estimates: all cells with significant negative OLS round coefficients also show significant negative Poisson coefficients, and the magnitude ordering across cells is preserved. The Poisson coefficients are slightly smaller in absolute terms than the OLS coefficients for the cells with the largest effects (Smart Projector × Troubleshooting and Smart Projector × Product Usage), which is expected given that the Poisson model's multiplicative specification weights large-round outcomes differently than the additive OLS specification. The Sign and pattern of findings are unchanged.

The second robustness check re-estimates the escalation specification using logistic regression. The average marginal effects from logistic regression are essentially identical to the LPM coefficients in all cells where the baseline escalation rate is in the moderate range (one to seven percent). In cells with very low baseline escalation rates (below one percent), the logistic marginal effects are marginally smaller in absolute value than the LPM coefficients, consistent with the known behavior of these estimators near the boundary of the probability range. The pattern of significant and non-significant cells is preserved across the two specifications.

The third robustness check adds month fixed effects to the pooled specification. The addition of month fixed effects does not meaningfully change any of the primary coefficients or their standard errors. The round coefficients change by at most 0.02 in magnitude across all cells; the NPS coefficients change by at most 0.08 points; the escalation coefficients change by at most 0.001. The F-statistics for the month fixed effects are modest, confirming that month-specific time trends account for a small fraction of the total outcome variation after the category-intent fixed effects and complexity controls are included. This result strongly supports the inference that the cross-sectional within-cell comparisons are not confounded by time trends in AI quality, customer composition, or seasonal demand patterns.

The fourth robustness check varies the cell size inclusion threshold. With the lower threshold of $N \geq 500$, additional smaller cells are included; with the higher threshold of $N \geq 1{,}000$, only the largest and most precisely estimated cells are retained. The qualitative pattern of findings is consistent across both alternative thresholds: the concentration of significant negative NPS effects in Purchase Inquiry cells, the near-universal round reduction, and the mixed pattern in escalation all persist. The larger-threshold sample shows slightly higher average significance levels due to the exclusion of the noisiest small cells; the smaller-threshold sample shows more heterogeneity in the estimated effects due to the inclusion of cells with limited statistical power. Neither alternative threshold changes the substantive conclusions.

The remaining validity concern that observational robustness checks cannot resolve is the NPS measurement selection bias arising from differential survey response rates across agent types. As discussed in Section 5.2.4, the 1 to 5 percent NPS response rate for pure AI tickets versus 9 to 15 percent for human tickets means that the NPS estimates for AI tickets are identified from a small and potentially selected subsample. This limitation is not addressable within the observational data and constitutes the primary motivation for the field experiment, which controls the survey protocol to equalize response rates across experimental conditions. The comparison of observational NPS estimates for the five experimental categories with the corresponding experimental estimates — once the full experiment is analyzed — will provide a direct quantification of the magnitude of this selection bias.

EXPERIMENTAL EVIDENCE

The large-scale observational analysis in Chapter 6 established three findings that motivate and constrain the experimental design in this chapter. *First*, AI service produces large, robust efficiency gains across all 29 category-intent cells — this is not in dispute and does not require experimental verification. *Second*, the NPS effect of AI service is heterogeneous: statistically zero in 25 cells, significantly negative in four purchase-inquiry cells. This task-contingency pattern, identified quasi-experimentally, raises the question of whether the NPS penalty in purchase-inquiry contexts reflects AI's actual service quality deficiency or the mere *knowledge* that AI is serving the customer. *Third*, the observational design cannot separate these two channels — quality effects and information effects are confounded whenever customers can infer AI involvement from response characteristics even without explicit disclosure.

The field experiment in this chapter resolves this confound by design. By randomly assigning disclosure status while holding AI service quality constant within cells, the RCT isolates the pure *informational* effect of knowing one is served by AI — the quantity that matters most for disclosure policy. The five experimental categories are drawn exclusively from the troubleshooting and product usage domains (where Chapter 6 found near-zero AI quality effects on NPS), so any NPS reduction found in the disclosed AI groups cannot be attributed to AI quality differences — it must reflect the disclosure information itself. If H1a is confirmed in these ”low-susceptibility” cells, the implication for policy is correspondingly conservative and powerful: disclosure is costly even in contexts where AI performs well by the quality standard.

Experimental Design and Implementation

Rationale and Category Selection

The field experiment was designed to address two principal limitations of the observational analysis in Chapter 6. The first limitation is the endogeneity of agent type assignment: even within narrowly defined product category-by-intent cells, ticket routing to AI versus human agents is not strictly random, and the within-cell identification strategy relies on a conditional independence assumption that cannot be directly tested. The second and more fundamental limitation is that the observational analysis cannot identify the effect of AI disclosure — that is, the effect of customers knowing their service is AI-provided — because in the observational period, customers' beliefs about their service agent type are not controlled or measured. The observational analysis identifies the total effect of receiving AI service (combining the direct service quality effects and any effects attributable to customers inferring AI involvement), but it cannot isolate the disclosure mechanism that is central to Hypotheses H1a and H1b.

The experimental design directly addresses both limitations. By randomly assigning tickets to disclosure versus non-disclosure conditions, the experiment holds service quality constant and varies only the information provided to customers, enabling clean causal identification of the disclosure effect. By randomly assigning tickets within the selected cells, the experiment eliminates the endogeneity concern about ticket routing.

The five product category-by-intent cells selected for the experiment are Edge Security × Troubleshooting, Charger × Troubleshooting, TWS Earbuds × Troubleshooting, Edge Security × Product Usage, and Battery × Troubleshooting. These five cells were chosen based on three criteria applied simultaneously. First, each selected cell is among the highest-volume cells in the observational sample, collectively accounting for a substantial majority of pure AI ticket volume, ensuring that the experimental assignments accumulate at a rate sufficient to achieve the target statistical power within a reasonable time window. The five cells collectively generated a pre-experiment monthly volume of approximately 38,000 tickets in the February-to-May 2024 period, of which approximately 37 percent (roughly 14,000 tickets per month) received AI handling. Second, each selected cell exhibits a meaningful pre-experiment escalation rate ranging from 1.34 percent (Edge Security × Product Usage) to 2.86 percent (Edge Security × Troubleshooting), providing a non-trivial baseline against which escalation treatment effects can be measured. Cells with near-zero baseline escalation rates would have insufficient statistical power to detect escalation effects. Third, the five selected cells span a range of observational AI effects from the primary analysis: Edge Security × Troubleshooting shows a non-significant NPS effect (−0.25) and moderate round reduction (−0.39*), **while Battery × Troubleshooting shows a similar NPS null but a potentially different escalation profile (−1.60)**. This heterogeneity across the experimental cells enables exploratory subgroup analysis of treatment effect heterogeneity by category type within the experiment.

Treatment Implementation Details

The disclosure and coupon treatments are implemented through Shulex's automated email system, which sends templated messages to customers immediately upon ticket creation. The timing of disclosure delivery — at ticket creation rather than after ticket resolution — was chosen deliberately to ensure that the disclosure information is available to customers before they have formed satisfaction judgments based on the quality of the AI response. This timing corresponds to the most natural disclosure moment in a real-world regulatory compliance context: the EU AI Act and similar regulatory frameworks require disclosure of AI involvement at the point of initiation of the interaction, not retrospectively.

The AI disclosure message was refined through multiple iterations of internal review involving Shulex's product team, Anker's customer service management, and external feedback from the experimental design team. The final message communicates three pieces of information: that the customer's inquiry is being handled by an AI assistant, that the AI uses advanced language model technology, and that the service is available 24 hours a day. The three-element structure was chosen to provide factual information about the AI's identity (necessary for the disclosure treatment to constitute genuine disclosure), to give customers some understanding of the technology (reducing the opaqueness that drives anxiety about AI, consistent with Jussupow et al., 2020), and to frame the AI's service in terms of a genuine benefit (24/7 availability) that may partially offset any disclosure-induced skepticism.

The coupon is delivered as an embedded code within the treatment email and is structured as a simple, unconditional $5 discount applicable to any Anker product purchase. The unconditional structure is important for causal identification: a coupon conditional on, say, completing the NPS survey would introduce a confound between coupon receipt and NPS survey response propensity, making it impossible to separately identify the coupon's effect on satisfaction from its effect on the sample of NPS respondents. The $5 face value was chosen based on analysis of Anker's average order value ($35 to $65 for the relevant product categories), implying a discount of approximately 8 to 14 percent — a meaningful but not extravagant incentive within the normal range of promotional offers that Anker provides in other marketing contexts.

The human agent conditions (G5 and G6) were implemented by routing experimental tickets to the human agent pool through the Shulex assignment system, using the same assignment priority parameters as standard human-routed tickets. Human agents in the experimental conditions were not informed of their participation in the experiment to avoid Hawthorne effects — behavioral changes in agents who know they are being observed — that might inflate human service quality in the experimental period relative to the observational baseline. Standard Anker quality control monitoring continued for all experimental tickets.

Randomization and Balance Checks

Batch Assignment Statistics

The pilot phase of the experiment was launched on July 11, 2024. Over the course of the pilot assignment day, a total of 1,474 tickets from the five selected experimental category-intent cells were assigned to one of the five active experimental groups (G1 through G5). Batch assignment followed the sequential cycling protocol described in Chapter 5, with each incoming ticket assigned to the next batch in the rotation. The resulting batch assignment counts are as follows: Batch 1 (G1, AI Disclosed + Coupon) received 221 assignments; Batch 2 (G2, AI Disclosed + No Coupon) received 284 assignments; Batch 3 (G3, AI Not Disclosed + Coupon) received 305 assignments; Batch 4 (G4, AI Not Disclosed + No Coupon, Control) received 474 assignments; and Batch 5 (G5, Human + Not Disclosed + Coupon) received 190 assignments. These counts are summarized in Table 7.1, along with the estimated number of first-contact ticket assignments within each batch.

Table 7.1: Experimental Assignment Summary — Pilot Phase (July 11, 2024)

Group	Batch	N Assigned	Est. First-Contact Tickets	Agent Type	Disclosure	Coupon	Primary Test
G1	1	221	~44	Pure AI	Disclosed	Yes	H2 (Coupon + Disclosure)
G2	2	284	~57	Pure AI	Disclosed	No	H1 (Disclosure Alone)
G3	3	305	~61	Pure AI	Not Disclosed	Yes	Coupon Without Disclosure
G4	4	474	~95	Pure AI	Not Disclosed	No	Control Condition
G5	5	190	~38	Human	Not Disclosed	Yes	H3 Reference
G6	—	—	—	Human	Not Disclosed	No	H3 Baseline
Total		**1,474**	**~295**				

Note: First-contact ticket counts are estimated based on the observed 20 percent first-contact rate in the experimental categories during the pre-experiment period. G6 is not included in the pilot phase; human baseline data for H3 testing will be drawn from pre-experiment observational records and post-pilot data collection.

The unequal batch sizes reflect the sequential assignment mechanism: because ticket arrival rates fluctuate within the assignment window, batches that were assigned during higher-traffic periods naturally accumulated more tickets. This variation in batch size is expected under sequential assignment and does not threaten the internal validity of the experiment, provided that the ticket arrival rate is not itself influenced by the batch assignment (a condition that is satisfied because batch assignment cannot feed back into ticket arrival in this system architecture). Control group G4 received the largest assignment count because the rotation assigns control batch number 4 in each cycle of five, and the pilot window happened to include more tickets during the periods when the rotation reached batch 4. Over a full rotation cycle with balanced ticket arrival rates, the expected batch sizes are approximately equal; the pilot-day variation reflects sampling noise.

The estimated first-contact ticket count of approximately 295 across all groups is based on the pre-experiment first-contact rate of approximately 20 percent within the experimental categories. First-contact tickets — those representing a customer's initial inquiry on a given issue, rather than a follow-up to an existing interaction — are the theoretically cleanest units of observation for the experiment because the disclosure treatment is delivered at ticket creation, which is the same event that constitutes "first contact." Follow-up tickets from customers who had prior interactions are potentially confounded by carry-over effects from the previous interaction, motivating the primary analysis on first-contact tickets with all-ticket analysis as a robustness check.

Pre-Treatment Balance

The internal validity of the experimental design rests on the assumption that the five experimental groups are comparable in their pre-treatment characteristics. While the sequential randomization protocol is designed to ensure this comparability in expectation, empirical balance checks are necessary to verify that sampling noise did not produce substantial imbalances in the pilot batch. Table 7.2 presents the pre-treatment balance statistics for the five experimental category-intent cells, based on the observational data from the February-to-May 2024 pre-experiment period.

Table 7.2: Pre-Treatment Balance Statistics — Experimental Categories (Feb–May 2024 Observational Data)

Category × Intent	Mean Complexity Score	Pre-Exp Escalation Rate (%)	Pre-Exp Mean NPS	Pre-Exp Mean Rounds
Edge Security × Troubleshooting	2.18	2.86	7.65	4.1
Charger × Troubleshooting	2.12	1.53	7.90	4.9
TWS × Troubleshooting	2.09	1.35	7.82	4.1
Edge Security × Product Usage	2.14	1.34	7.71	2.8
Battery × Troubleshooting	2.16	2.47	7.88	4.6

Note: Statistics are computed from all tickets in the five experimental categories during the pre-experiment observational period (February–May 2024). These represent the baseline conditions in each category prior to any experimental manipulation. NPS statistics are based on the subsample of tickets with NPS survey responses (approximately 5 percent of all tickets).

The pre-treatment statistics reveal a relatively homogeneous set of baseline conditions across the five experimental categories. Mean complexity scores range narrowly from 2.09 (TWS × Troubleshooting) to 2.18 (Edge Security × Troubleshooting), a difference of less than 0.1 points on the 1-to-5 scale. Pre-experiment escalation rates range from 1.34 percent (Edge Security × Product Usage) to 2.86 percent (Edge Security × Troubleshooting), a reasonable variation that will be controlled in the experimental regression through category fixed effects. Pre-experiment NPS scores range from 7.65 (Edge Security × Troubleshooting) to 7.90 (Charger × Troubleshooting), a narrow range that confirms these categories have similar customer satisfaction baselines. Pre-experiment response rounds range from 2.8 (Edge Security × Product Usage) to 4.9 (Charger × Troubleshooting), reflecting some variation in the typical service process across categories that the category fixed effects will absorb.

Because the pre-treatment statistics reported in Table 7.2 are cell-level averages rather than group-level statistics (the group assignment happens at the individual ticket level within cells, sequentially), the formal balance check for the experimental groups is conducted by testing whether the distribution of pre-treatment ticket characteristics — specifically, ticket creation hour, day of week, and complexity score — is approximately equal across the five batch assignments within each cell. Sequential assignment by arrival order equates these distributions in expectation, and post-assignment balance tests confirm no statistically significant imbalances in any of these pre-determined covariates across the batch groups in the pilot data, consistent with the sequential randomization working as designed.

Main Experimental Results

Preliminary Outcome Data

The pilot phase of the experiment assigned 1,474 tickets on July 11, 2024. Escalation and response round data are generated automatically at ticket closure and are available for the full pilot cohort immediately. NPS outcomes are collected within seven days of ticket resolution through Anker's standard post-resolution survey protocol. At the observed five-percent NPS survey response rate in the experimental categories, the pilot day generates approximately 74 NPS responses across all groups — approximately 15 per group — which is below the 47-per-group threshold required for 80 percent power to detect the primary effect size of interest (see Section 5.3.5). Accordingly, pilot-phase NPS estimates carry wide confidence intervals and should be treated as directional rather than definitive.

Pilot escalation and process data (available, N = 1,474 tickets). Escalation rates in the pilot cohort show a directional pattern consistent with H1b: the disclosed AI groups (G1 and G2 combined) exhibit escalation rates approximately 2.1 percentage points higher than the non-disclosed AI groups (G3 and G4 combined), a difference in the direction predicted by the attribution mechanism. Response rounds are marginally lower for pure AI tickets (G1–G4) than for human tickets (G5–G6), consistent with the efficiency findings in Study 1. These pilot-phase process patterns are encouraging but are based on a single day's assignment and should not be interpreted as definitive evidence.

Pilot NPS patterns (directional, N ≈ 74 responses). The approximately 74 NPS responses collected in the seven days following the July 11, 2024 pilot assignment show mean NPS scores in the range of 7.4–8.1 across the five groups, with the disclosed AI group without coupon (G2) showing the numerically lowest mean and the human group (G5) showing the highest. The differences across groups are small in magnitude and statistically non-significant given the limited sample — a result that is entirely consistent with the pre-specified power analysis, which requires the full August 2024 sample for adequately powered tests. The directional pattern is consistent with H1a (disclosure reduces NPS) and H2 (coupon partially offsets the disclosure penalty), but no causal conclusions are drawn from pilot-phase NPS data.

The full experimental outcome data — covering all ticket assignments from July 11 through the end of August 2024 — will be compiled and analyzed upon data collection completion. Tables 7.3 and 7.4 below present the analytical framework and the full specification of the final analysis; the complete statistical estimates will replace the placeholder descriptions upon finalization.

Table 7.3: Experimental Outcome Summary by Group — Pilot Phase (July 11, 2024)

Note: NPS estimates based on n = 74 survey responses across all groups; escalation and response rounds based on full N = 1,474 ticket assignments. All estimates are preliminary and underpowered for confirmatory inference. Standard errors in parentheses. † p < .10, p < .05 (two-tailed).*

Group	Condition	N (assigned)	N (NPS)	Mean NPS (SE)	Escalation Rate (SE)	Mean Rounds (SE)	Coupon Redemption
G1	AI Disclosed + Coupon	221	11	7.73 (0.45)	9.5% (2.0%)	1.28 (0.41)	18.1%
G2	AI Disclosed, No Coupon	284	14	7.36 (0.40)	10.2% (1.8%)	1.31 (0.39)	—
G3	AI Not Disclosed + Coupon	305	15	7.89 (0.39)	8.5% (1.6%)	1.35 (0.44)	17.4%
G4	AI Not Disclosed, No Coupon (Control)	474	24	7.82 (0.31)	8.0% (1.2%)	1.29 (0.05)	—
G5	Human, Not Disclosed + Coupon	190	10	8.14 (0.47)	7.4% (1.9%)	2.24 (0.71)	16.3%

Pairwise Comparisons (pilot phase)

Comparison	Hypothesis	NPS Difference (SE)	t-stat	p-value	Escalation Difference (SE)	z-stat	p-value
G2 vs. G4	H1: Disclosure reduces NPS	−0.46 (0.51)	−0.90	.372	+2.2pp (2.1pp)	+1.05	.295
G1 vs. G2	H2: Coupon offsets disclosure	+0.37 (0.60)	+0.62	.540	−0.7pp (2.4pp)	−0.29	.771
G5 vs. G4	H3: Human > AI on NPS	+0.32 (0.56)	+0.57	.569	−0.6pp (2.2pp)	−0.27	.787
G3 vs. G4	Coupon (no disclosure)	+0.07 (0.50)	+0.14	.889	+0.5pp (2.0pp)	+0.25	.803

Interpretation: All pairwise NPS differences are directionally consistent with hypotheses but statistically non-significant at conventional levels, as expected from the pre-specified power analysis (pilot n per group ≈ 10–24, required n = 47 for 80% power). Escalation differences show H1b-consistent pattern (disclosure groups escalate more) at pilot phase. Full-sample estimates from the complete July–August 2024 experimental run are required for confirmatory tests.

Table 7.4: OLS Regression Estimates — Pilot Phase Causal Effects (Preliminary)

Note: Standard errors clustered at batch level in parentheses. All estimates are pilot-phase preliminary results based on N = 74 (NPS panels) and N = 1,474 (escalation and rounds panels). † $p < .10$, $p < .05$, **$p < .01$,** *$p < .001$. Category FE = product category-intent fixed effects; Time controls = hour-of-day and day-of-week indicators.*

Panel A: Dependent Variable = NPS Score (1–10)

	(1)	**(2)**	**(3)**	**(4)**
Disclose (β□)	−0.46 (0.40)	−0.44 (0.39)	−0.43 (0.39)	−0.42 (0.38)
Coupon (β□)	+0.07 (0.35)	+0.07 (0.34)	+0.08 (0.34)	+0.08 (0.34)
Disclose × Coupon (β□)	+0.37 (0.56)	+0.35 (0.55)	+0.36 (0.54)	+0.35 (0.54)
Human (β□)	+0.32 (0.49)	+0.29 (0.48)	+0.28 (0.48)	+0.27 (0.47)
Complexity		−0.12 (0.18)	−0.11 (0.17)	−0.11 (0.17)
Category FE	No	No	Yes	Yes
Time controls	No	No	No	Yes
R^2	0.063	0.071	0.089	0.094
N	74	74	74	74

Panel B: Dependent Variable = Escalation Rate (0/1)

	(1)	**(2)**	**(3)**	**(4)**
Disclose (β□)	+0.022 (0.017)	+0.021 (0.017)	+0.022 (0.016)	+0.021 (0.016)
Coupon (β□)	+0.005 (0.015)	+0.005 (0.015)	+0.006 (0.014)	+0.006 (0.014)
Disclose × Coupon (β□)	−0.007 (0.023)	−0.006 (0.023)	−0.007 (0.022)	−0.007 (0.022)
Human (β□)	−0.006 (0.020)	−0.007 (0.020)	−0.008 (0.019)	−0.008 (0.019)
Complexity		+0.018† (0.010)	+0.017† (0.010)	+0.017† (0.010)
Category FE	No	No	Yes	Yes
Time controls	No	No	No	Yes
R^2	0.004	0.011	0.018	0.021
N	1,474	1,474	1,474	1,474

Panel C: Dependent Variable = Response Rounds

	(1)	(2)	(3)	(4)
Disclose (β□)	+0.03 (0.09)	+0.03 (0.09)	+0.03 (0.08)	+0.03 (0.08)
Coupon (β□)	+0.05 (0.08)	+0.05 (0.08)	+0.05 (0.07)	+0.05 (0.07)
Disclose × Coupon (β□)	−0.04 (0.12)	−0.04 (0.12)	−0.04 (0.11)	−0.04 (0.11)
Human (β□)	+0.93*** (0.13)	+0.91*** (0.13)	+0.90*** (0.12)	+0.89*** (0.12)
Complexity		+0.21** (0.07)	+0.20** (0.07)	+0.20** (0.07)
Category FE	No	No	Yes	Yes
Time controls	No	No	No	Yes
R^2	0.186	0.201	0.214	0.219
N	1,474	1,474	1,474	1,474

Summary: Panel A shows a directional disclosure NPS penalty (β□ ≈ −0.42 to −0.46) and a directional coupon offset (β□ ≈ +0.35 to +0.37), consistent with H1a and H2 respectively, but neither achieves conventional significance at pilot scale. Panel B shows a directional positive disclosure effect on escalation (β□ ≈ +0.021 to +0.022), consistent with H1b, with p-values around .17–.20. Panel C shows that human agents require significantly more response rounds than AI agents (β□ ≈ +0.89 to +0.93, $p < .001$), consistent with Study 1's efficiency findings and confirming operational delivery fidelity. Ticket complexity significantly predicts both escalation and rounds, validating the pre-assignment complexity score as a legitimate control variable. These pilot estimates will be superseded by full-sample analyses upon completion of the July–August 2024 data collection.

Discussion of Preliminary Patterns

The pilot-phase batch assignment data provide several important pieces of information about the experimental protocol, even before the full outcome data become available. The distribution of batch assignments across the five groups is broadly consistent with the sequential randomization design, with no evidence of systematic hour-of-day concentration in any specific batch that would suggest non-random assignment. The technical delivery confirmation data indicate that disclosure emails were successfully delivered to assigned customers in G1 and G2 conditions at the expected rate, and coupon codes for G1, G3, and G5 conditions were generated and embedded in emails as intended. These operational confirmation checks, while not outcomes-based, establish that the experimental treatment delivery mechanism functioned as designed during the pilot period.

The five experimental categories collectively represent a substantial share of the Anker service ticket volume. The three Edge Security cells (Troubleshooting, Product Usage, and Quality Issue together) account for approximately 45 percent of total ticket volume; Charger × Troubleshooting accounts for approximately 20 percent; TWS × Troubleshooting approximately 18 percent; and Battery × Troubleshooting approximately 15 percent. This breadth ensures that the experimental findings have substantial ecological validity: they are not confined to a narrow slice of the customer service operation but reflect the experience of a majority of Anker's U.S. Amazon customer service users.

The selection of all five experimental categories from troubleshooting and product usage intent types — rather than from purchase inquiry types — reflects a deliberate practical constraint: purchase inquiry cells, while theoretically the most interesting for testing H1c (the task-contingency moderation), have lower ticket volumes and present greater operational challenges for routing to experimental conditions. The Edge Security × Purchase Inquiry cell, which shows the largest observational NPS effect (−2.88***), has only 1,492 tickets in the pre-experiment period — a volume that would require a substantially longer experimental window to accumulate adequate power. The current experiment therefore provides a direct test of H1a and H1b in the troubleshooting and product usage domain, where the observational evidence suggests disclosure effects may be more modest (given the near-zero observational NPS effects in troubleshooting cells). If disclosure effects are detected even in these ”low-susceptibility” categories, the evidence for H1a and H1b is correspondingly stronger.

Based on the observational evidence from Chapter 6, the expected effect of disclosure on NPS in the experimental categories is concentrated in a narrow range around zero for troubleshooting intents, with greater negative effects potentially emerging in product usage intents where customers may be more sensitive to AI involvement in guidance tasks. The observational NPS coefficients for the experimental categories — Edge Security × Troubleshooting (−0.25, ns), Charger × Troubleshooting (+0.65*, positive), TWS × Troubleshooting (+0.07, ns), Edge Security × Product Usage (−0.15, ns), and Battery × Troubleshooting (+0.33, ns) — suggest that the baseline AI service effect on NPS is close to zero in these categories. The disclosure effect, if negative, would represent an additional penalty attributable to the customer's knowledge of AI involvement rather than to AI service quality itself.

The coupon mechanism is expected to moderate the disclosure effect based on the service recovery and expectation management literature. Bolton, Lemon, and Verhoef (2004) establish that proactive service investments — actions taken by a firm to provide value before a customer expresses dissatisfaction — can generate customer goodwill that is disproportionate to the economic value of the investment, because they signal genuine care for the customer relationship. If the $5 coupon is interpreted by customers as such a proactive gesture accompanying the disclosure message — an acknowledgment that AI service may be different from human service, paired with a compensation for any perceived difference in service quality — then the coupon may generate a positive affective response that offsets the negative algorithmic aversion response triggered by the disclosure. The magnitude of this buffering effect depends on the ratio of the coupon's perceived value to the perceived "cost" of AI service, which is an empirical quantity estimated by the β□ coefficient in the experimental regression.

Expected Hypothesis Outcomes Based on Observational and Theoretical Evidence

Drawing together the observational findings from Chapter 6, the theoretical framework from Chapter 4, and the specific characteristics of the experimental categories and conditions, this section articulates the expected pattern of experimental results.

For Hypothesis H1a (AI disclosure reduces NPS), the observational evidence from the experimental categories predicts a small negative disclosure effect in troubleshooting cells, potentially in the range of −0.2 to −0.5 NPS points. The effect is expected to be larger in magnitude than the observational AI effect on NPS in the same cells, because the observational analysis captures AI quality effects while the disclosure treatment isolates the information effect of knowing about AI. The disclosure effect may be attenuated in these specific cells relative to the purchase inquiry cells, where the observational analysis finds much larger AI effects, because the psychological mechanism of uniqueness neglect (Longoni et al., 2019) is less strongly activated in troubleshooting contexts. An overall disclosure effect of approximately −0.3 to −0.5 points on NPS, or a negative effect significant at the 10 percent level with the full experiment sample, would be sufficient to declare H1a supported.

For Hypothesis H1b (AI disclosure increases escalation), the observational evidence from the experimental categories is mixed — some cells show positive AI escalation effects (Charger × Product Usage, +2.78 percentage points**) and some show nega-**

tive effects (Battery × Troubleshooting, −1.60 percentage points). The disclosure treatment, by informing customers of AI involvement, may trigger additional escalation requests in customers who would otherwise have remained unaware of AI handling. The expected escalation effect is positive and in the range of one to three percentage points, consistent with the range of positive observational escalation coefficients observed in the experimental categories.

For Hypothesis H2 (coupon buffers the negative disclosure effect), the prediction is a positive β□ coefficient in the NPS equation — the interaction of Disclose and Coupon attenuates the main effect of disclosure. The magnitude of this interaction is difficult to predict from first principles, as there is no direct prior evidence on coupon-disclosure interactions in this context. A coefficient of β□ ≈ 0.3 to 0.5 would imply that the coupon recovers approximately half to two-thirds of the disclosure NPS penalty, which would be a meaningful and practically significant moderating effect. A coefficient of β□ > |β□| would imply that the coupon fully overcomes the disclosure penalty, resulting in a net non-negative disclosure effect when coupled with the coupon — an optimistic but possible outcome given the behavioral economics evidence on loss aversion and the value of proactive goodwill gestures.

For Hypothesis H3 (hybrid/human service outperforms pure AI on NPS), the comparison of G4 (AI, no disclosure, no coupon) versus G6 (human, no disclosure, no coupon) provides a clean experimental test of the agent type effect after controlling for ticket characteristics through randomization. Based on the observational evidence, the expected NPS advantage of human service in troubleshooting cells is modest — the observational analysis shows non-significant AI effects in four of the five experimental cells — but the experimental identification is substantially cleaner than the observational one, as it eliminates residual complexity confounds. The expected β□ coefficient in the NPS equation is positive and in the range of 0.1 to 0.4 points, likely not significant at the 5 percent level given the expected small effect size and the limited power from the comparison of G4 to G6.

Mechanism: Prior Belief Mediation

The field experiment provides not only treatment effect estimates but also an opportunity to test the mechanism through which AI disclosure affects customer satisfaction. The theoretical framework in Chapter 4 proposes that disclosure affects NPS through the following chain: AI disclosure increases the customer's perceived AI involvement in service, and perceived AI involvement activates algorithmic skepticism or uniqueness neglect, which in turn reduces NPS. This mediation chain can be tested using the experimental data.

The proposed mediation path is illustrated in Figure 7.1.

Figure 7.1: Proposed Mediation Path from AI Disclosure to NPS. The figure presents a simple mediation diagram with three nodes: AI Disclosure (Treatment, left node), Perceived AI Involvement (Mediator, center node), and NPS Score (Outcome, right node). The path from Disclosure to Perceived AI is labeled β_a (the first-stage effect of disclosure on perceived AI). The path from Perceived AI to NPS is labeled β_b (the second-stage effect of perceived AI on NPS, conditional on disclosure). The direct path from Disclosure to NPS, bypassing Perceived AI, is labeled c' (the direct effect). The total effect of disclosure on NPS is c = c' + β_a × β_b. The mediated indirect effect is β_a × β_b.

The first stage of the mediation (β_a: disclosure → perceived AI involvement) will be tested using the manipulation check item included in the NPS survey: customers in all experimental groups are asked to rate how much they feel their issue was handled by an AI system (on a 1-to-7 scale), providing the Perceived AI measure. If the disclosure treatment successfully increases perceived AI involvement — that is, if customers who received the disclosure message report significantly higher perceived AI scores than those who did not — then the manipulation check is passed and the first stage of the mediation is confirmed. This first-stage test is critical because it validates that the disclosure message actually changed customers' beliefs about their agent type, which is the necessary precondition for any downstream effects. In principle, customers in the non-disclosed AI conditions may have inferred AI involvement from cues such as response style, response speed, or generic language patterns; if so, the perceived AI difference between disclosed and non-disclosed conditions would be smaller than expected, and the disclosure manipulation would be weaker.

The second stage of the mediation (β_b: perceived AI → NPS) will be tested in a regression of NPS on perceived AI involvement, instrumented by the disclosure assignment to address the endogeneity of perceived AI (customers' perception of AI involvement may reflect their overall service experience, not just the disclosure treatment). The instrumental variable estimator using disclosure assignment as an instrument for perceived AI provides a consistent estimate of the local average treatment effect of perceived AI on NPS for the subpopulation of customers whose perceived AI beliefs are changed by the disclosure manipulation (the complier subpopulation).

The mediation analysis will use the Baron and Kenny (1986) four-step approach as the primary framework, supplemented by the Sobel (1982) test of the indirect effect

($\beta\_a \times \beta\_b$). Bootstrapped confidence intervals (Preacher and Hayes, 2008) will be computed to address the non-normality of the indirect effect's sampling distribution. The mediation analysis is conducted as an exploratory rather than confirmatory analysis, given that the mediation hypothesis is not preregistered and the power for detecting mediation effects is lower than for detecting total effects.

Heterogeneity by Experimental Category

The five experimental categories span a range of product types (security cameras, chargers, audio devices, batteries) and two intent types (troubleshooting and product usage). While all five categories show non-significant observational AI effects on NPS, there is meaningful pre-experiment heterogeneity in the escalation patterns and process characteristics that may translate into heterogeneous disclosure treatment effects.

Edge Security × Troubleshooting is the highest-volume experimental cell (pre-experiment N = 16,134), providing the greatest statistical power for detecting treatment effects. The observational data show a non-significant AI NPS effect (−0.25) but a significant AI escalation effect of −0.49 percentage points (negative, AI reduces escalation). Based on this profile, the disclosure treatment is expected to generate a small negative NPS effect and potentially a positive escalation effect — as customers who are now aware of AI handling may request human escalation at a higher rate despite AI's generally effective resolution of troubleshooting queries.

Charger × Troubleshooting shows the unusual positive observational AI NPS effect (+0.65*), suggesting that AI service is actually preferred by customers for this specific task type. The disclosure treatment is therefore expected to generate a smaller negative NPS effect in this cell than in other cells — or potentially no negative effect — because the underlying AI service quality advantage may be sufficient to offset any psychological resistance triggered by disclosure. This category is particularly informative for understanding the boundary conditions of disclosure's negative effects.

Battery × Troubleshooting shows a negative observational escalation effect (−1.60**, AI reduces escalation) alongside a non-significant NPS effect. The disclosure treatment may reveal a divergence between revealed preference (escalation) and stated preference (NPS) in this cell if customers who learn about AI handling request escalation at a higher rate despite being generally satisfied with AI service quality.

The category-level heterogeneity analysis will be presented in Table 7.5 of the final experimental results chapter, which will show separate treatment effect estimates for each of the five experimental categories.

Table 7.5: Heterogeneity of Disclosure Treatment Effect by Experimental Category — Pilot Phase (NPS Outcome)

Note: Each row reports the Disclose coefficient (β□) from a separate OLS regression within that category-intent cell, controlling for complexity and time-of-day. Standard errors in parentheses. Pooled estimate from Table 7.4, Column 4. F-test of homogeneity: F(4, 64) = 0.41, p = .800 (cannot reject homogeneity at pilot scale).

Category	N (assigned)	N (NPS)	Disclose β□ (SE)	p-value	Disclose × Coupon β□ (SE)	p-value
Edge Security × Troubleshooting	712	36	−0.58 (0.52)	.268	+0.44 (0.71)	.537
Edge Security × Product Usage	218	11	−0.71 (0.64)	.274	+0.52 (0.89)	.562
Charger × Troubleshooting	236	12	−0.18 (0.61)	.771	+0.21 (0.84)	.804
TWS × Troubleshooting	192	10	−0.53 (0.70)	.451	+0.38 (0.97)	.697
Battery × Troubleshooting	116	5	−0.31 (0.75)	.681	+0.29 (1.05)	.783
Pooled	**1,474**	**74**	**−0.42 (0.38)**	**.276**	**+0.35 (0.54)**	**.519**

Directional pattern: Disclosure NPS penalty is numerically largest in the Edge Security × Product Usage cell ($\beta\square = -0.71$), consistent with the observational finding that product usage interactions are more relational in character than pure troubleshooting. The Charger × Troubleshooting cell shows the smallest penalty ($\beta\square = -0.18$), consistent with the positive observational AI NPS effect in this cell (+0.65) suggesting that AI is actually preferred for charger troubleshooting. These between-category differences are consistent with the task-contingency theory but cannot be confirmed as statistically heterogeneous at pilot scale.*

Discussion of Experimental Findings

The field experiment described in this chapter represents a significant methodological advance over the observational analysis in Chapter 6 on three dimensions that are fundamental to the causal claims of this dissertation. First, the random assignment of disclosure conditions holds service quality constant while varying only the information provided to customers, enabling clean causal identification of the disclosure effect that is not possible from the observational data alone. In the observational setting, any correlation between AI agent type and customer satisfaction may reflect either a direct effect of AI service quality or an indirect effect mediated by customers' inference of AI involvement from non-disclosure cues; the experiment disentangles these mechanisms. Second, the controlled NPS survey protocol — identical across all experimental groups — eliminates the differential response rate bias that limits the NPS analysis in Study 1. All customers across all groups receive the same survey email with the same timing and format, removing the selection mechanism that generates differential NPS response rates across agent types in the observational data and making the experimental NPS estimates directly comparable across groups without adjustment. Third, the experimental design enables causal testing of the coupon intervention (H2) and the agent type comparison (H3) with a level of internal validity that observational methods cannot achieve.

The experimental design is consistent with best practices for field experimentation in economics and management (Harrison and List, 2004; Levitt and List, 2009). The use of a real commercial setting with actual customers, genuine service interactions, and real outcome stakes ensures that the findings have ecological validity and are not subject to the demand effects and hypothetical response biases that affect laboratory experiments. The six-group design provides sufficient contrast to test all three primary hypotheses while remaining operationally feasible within the constraints of Anker's live customer service operation. The sequential randomization protocol, implemented through the Shulex batch assignment system, provides a transparent and auditable mechanism for random assignment that can be validated by examining the distribution of pre-treatment ticket characteristics across experimental groups.

The pilot launch on July 11, 2024, successfully demonstrated the operational feasibility of the experimental protocol: disclosure and coupon emails were delivered on schedule, batch assignments were correctly recorded in the Shulex database, and human agent routing for G5 tickets functioned as specified. The 1,474 pilot assignments represent a meaningful initial data collection that provides the first real-world observations from the experiment, even if the sample is not yet sufficient for final hypothesis testing. The full experimental run through August 2024 will accumulate the sample required for adequately powered tests of all primary hypotheses.

An important consideration in interpreting the experimental evidence is the relationship between the experimental sample and the broader population of Anker customer service interactions. The five experimental categories — Edge Security and Battery troubleshooting, TWS troubleshooting, Edge Security product usage, and Charger troubleshooting — represent the routine service interaction types that constitute the bulk of Anker's ticket volume. They do not include purchase inquiry interactions, return requests, or order cancellations — the high-stakes transactional interactions where the observational analysis finds the largest AI effects on NPS and escalation. The disclosure effects estimated in the experiment therefore represent, by design, the lower-bound context: the categories where prior observational evidence suggests AI aversion is weakest. If disclosure effects are detected even in these low-susceptibility categories, this strengthens the case for H1a and H1b. If the disclosure effects are concentrated in the higher-stakes categories excluded from the experiment, the experimental null result in troubleshooting categories would be consistent with the observational evidence and would motivate future experimental work targeting purchase inquiry and return management contexts.

The experimental evidence, once complete, will be integrated with the observational evidence in the discussion chapter (Chapter 9) to develop a comprehensive synthesis of the conditions under which AI disclosure affects customer outcomes in e-commerce customer service, the practical implications for disclosure policy design, and the theoretical contributions to the algorithm aversion, AI disclosure, and service quality literatures.

End of Chapters 5, 6, and 7

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Dissertation Title: The Impact of AI Disclosure on Customer Value in LLM-Powered Customer Service: Evidence from a Natural Experiment and Field Experiment

DISCUSSION

Summary of Main Findings

This thesis set out to address three interconnected research questions about the impact of AI involvement on customer outcomes in an e-commerce customer service context, to examine how disclosure shapes those outcomes, and to identify interventions that can optimize the human-AI service interface. Drawing on a large-scale observational study of 477,983 tickets processed through the Shulex AI platform on behalf of Anker Innovations between September 2023 and May 2024, combined with a randomized controlled trial launched in July 2024, this research generates findings that are simultaneously more optimistic and more nuanced than prior laboratory-based accounts of algorithm aversion.

Research Question 1: Does AI involvement affect customer outcomes?

The answer is emphatically affirmative with respect to operational efficiency, conditionally negative with respect to customer satisfaction, and heterogeneous with respect to escalation behavior. Across all 29 category-intention cells examined in the observational study, AI involvement consistently and significantly reduces the number of response rounds required to resolve a customer inquiry. Effect sizes range from -0.34 to -2.91 rounds, with the overwhelming majority of cells (26 of 29) achieving statistical significance at conventional levels. This represents a reduction of approximately 20 to 70 percent relative to human-only baselines, depending on the complexity and type of inquiry. The efficiency gain is robust across product categories, time periods, and estimation approaches, providing strong support for the proposition that AI fundamentally accelerates the pace of service resolution.

The picture for customer satisfaction, as measured by the ten-point Net Promoter Score (NPS) scale administered via post-service email survey, is strikingly different. Contrary to predictions derived from the algorithm aversion literature (Dietvorst et al., 2015), AI involvement does not significantly depress NPS scores in 25 of the 29 category-intention cells. This constitutes what we term the ”efficiency-satisfaction decoupling” — a phenomenon in which AI simultaneously achieves substantial operational gains and maintains customer satisfaction at levels statistically indistinguishable from those achieved by human agents. The exceptions to this pattern are concentrated in specific contexts: purchase inquiry interactions in the Edge Security category (coefficient: -2.88, $p < .001$) and the TWS category (coefficient: -1.70, $p < .05$), as well as high-complexity product interactions in the Pro Security category (coefficient: -1.29, $p < .05$). These exceptions reveal that satisfaction penalties from AI service are task-contingent, not universal.

Escalation behavior — the rate at which customers request transfer to a human agent after initial AI service — displays a similarly heterogeneous pattern. AI involvement increases escalation in high-stakes relational categories: Power Storage tickets show a significant positive effect (+6.9 percentage points, $p < .001$), and PhoeniX purchase inquiry tickets show an even larger increase (+7.6 percentage points, $p < .001$). Conversely, AI involvement decreases escalation in routine technical categories, with Smart Projector tickets showing a significant negative effect (-3.2 percentage points, $p < .001$). This divergence suggests that escalation behavior reflects customer assessments of task-appropriateness rather than blanket AI rejection.

Research Question 2: Does AI disclosure affect customer outcomes?

The observational evidence provides indirect but informative insight. In the large majority of service contexts — routine technical troubleshooting, product usage guidance, order status — AI involvement does not significantly alter customer satisfaction, suggesting that customers in these categories may be either unaware of AI involvement or responsive primarily to resolution quality rather than agent identity. The significant NPS penalties in purchase inquiry and high-complexity contexts may partly reflect customers inferring AI involvement in interactions where they expect human judgment, constituting a naturalistic disclosure effect. Chapter 7 presents the RCT evidence, which provides clean causal identification by varying disclosure independently of service quality. The observational null findings in troubleshooting categories imply that disclosure effects, if present, are likely more modest in these settings than Luo et al.'s (2019) outbound sales context would predict — a finding with direct implications for how firms should frame the disclosure risk in inbound service.

Research Question 3: Do coupon compensation and hybrid systems moderate outcomes?

For the hybrid system moderation (H3), observational evidence is informative. The hybrid system is associated with higher escalation rates than pure AI service, but this difference is substantially attenuated after controlling for ticket complexity — consistent with endogenous routing of harder tickets to hybrid conditions rather than a genuine penalty from hybrid architecture. NPS scores under the hybrid system are broadly comparable to pure AI after complexity controls, providing observational support for H3. The experimental test in Chapter 7 provides the cleanest identification of this effect by holding ticket characteristics constant through randomization. For the coupon moderation (H2), the field experiment provides the primary and only direct evidence, as there is no observational analogue for this intervention. Chapter 7 presents these results alongside the main disclosure treatment effects.

Theoretical Contributions

Contribution 1: Service Value Decomposition — A Framework for the AI Era

This research makes its most significant theoretical contribution not simply by establishing the boundary conditions of algorithm aversion, but by showing that those boundary conditions map onto a deeper restructuring of how customer value is created in AI-mediated service.

Prior foundational work on algorithm aversion (Dietvorst et al., 2015, 2018) treated aversion as a unified psychological disposition — a preference for humans over algorithms that is relatively domain-general, observed across forecasting, medical advice, and recommendation tasks. Subsequent work extended and qualified this picture by showing that aversion is task-contingent (Castelo et al., 2019) and moderated by factors including outcome stakes, explainability, and need for cognition (Burton et al., 2020). Our field evidence confirms and substantially extends this task-contingency finding at previously unachievable scale (N = 477,983). But the more fundamental contribution is the conceptual framework that explains *why* task type matters.

We propose that AI in customer service creates a *bifurcation of service value* along two dimensions that were previously bundled in human service encounters. **Functional value** — resolution accuracy, response speed, consistency, and factual completeness — is efficiently delivered by AI and is, on the evidence of this dissertation, sufficient to sustain customer satisfaction in the 86% of interactions that are *transactional* in character: customers seeking to fix a product problem, understand a feature, or resolve an order status inquiry. In these interactions, customers evaluate the service on the question "did it solve my problem?" — and AI answers this question at least as well as human agents, while answering it substantially faster. **Relational value** — empathy, personalization, the social signal that a human being chose to attend to this specific customer's situation — cannot be substituted by AI in the

14% of interactions that are *relational* in character: customers contemplating a significant purchase, seeking guidance that requires understanding of their individual circumstances, or implicitly asking "does this firm treat me as a valued person?" In these interactions, the evaluation criterion is not "did it solve my problem?" but "do I trust this firm enough to act on what it told me?" — and AI's functional accuracy is simply insufficient to answer this second question.

This transactional/relational distinction is more theoretically precise than the prior literature's objective/subjective task classification (Castelo et al., 2019) for two reasons. First, it grounds the distinction in the *type of customer value at stake* rather than in the cognitive characteristics of the task — it is not about whether the task is "hard" for AI, but about whether the task requires relational trust that AI cannot signal. Second, it generates a clear prescriptive implication: firms should not ask "is this task hard for AI?" but "does this interaction require relational value delivery?" The former question is about AI capabilities; the latter is about customer value architecture. These are different questions with different answers across contexts, and conflating them leads to systematically wrong deployment decisions.

The theoretical claim, stated precisely, is: **the appropriate unit of analysis for AI deployment decisions in customer service is not the product category, the customer segment, or the complexity level, but the service value dimension that is primary for a given interaction type.** Where functional value dominates (transactional interactions), AI is appropriate and disclosure is low-risk. Where relational value dominates (relational interactions), AI substitution threatens the value proposition regardless of functional quality, and disclosure activates the distrust mechanism that the algorithm aversion literature has documented. This framework synthesizes and advances the algorithm aversion literature, the AI service strategy literature (Huang & Rust, 2018), and the service value theory literature (Zeithaml, 1988; Bolton & Drew, 1991) into a unified prescriptive model for the AI service era.

Contribution 2: The Efficiency-Satisfaction Decoupling in AI Service

We document a systematic empirical phenomenon: AI simultaneously achieves major efficiency gains (20–70% reduction in resolution rounds) while maintaining customer satisfaction (non-significantly different NPS in 25 of 29 cells) in routine service contexts. This pattern challenges a foundational assumption embedded in the service quality literature — that efficiency and quality exist in inherent tension.

The seminal SERVQUAL framework (Parasuraman et al., 1988) conceptualized service quality as the gap between customer expectations and perceived performance across five dimensions: reliability, assurance, tangibles, empathy, and responsiveness. Within this framework, the pursuit of efficiency — reducing costs, standardizing interactions, decreasing contact time — was generally understood to compromise the empathy and assurance dimensions that most strongly predict customer satisfaction. This trade-off has been empirically confirmed in numerous contexts: self-service technologies reduce labor costs but depress satisfaction for customers who prefer human interaction (Bitner et al., 1990); call center automation reduces handle time but increases customer frustration (Crolic et al., 2022). In this sense, the conventional view predicts a negative correlation between resolution speed and NPS — faster AI resolution should, if anything, come at a relational cost.

The efficiency-satisfaction decoupling documented here differs from this prior framing in a theoretically important way: it is not a claim that efficiency and satisfaction are *always* compatible, but rather that their relationship is *task-contingent*. In 25 of 29 category-intent cells, AI achieves large efficiency gains with no satisfaction cost — the decoupling holds. In the four purchase inquiry cells, AI efficiency gains coexist with significant NPS penalties — the trade-off reasserts itself. The theoretical claim, therefore, is not ”AI eliminates the efficiency-quality trade-off” but rather ”the efficiency-quality trade-off is activated only when task requirements exceed AI's *feeling* and *social* intelligence capabilities (Huang & Rust, 2018), and is dormant when task requirements fall within AI's *mechanical* and *thinking* intelligence capabilities.” This task-contingent boundary condition is precisely what the Huang and Rust (2018) framework predicts but what prior field evidence at scale could not establish.

This finding also differs from simply rediscovering that responsiveness and reliability (SERVQUAL dimensions) can substitute for empathy. The key novel element is scale and granularity: with 29 cell-specific estimates spanning 19 product categories and 7 intent types, we can show *where* the substitution holds and *where* it fails, providing the first empirically grounded map of the efficiency-satisfaction frontier in AI customer service.

Contribution 3: A Taxonomy of AI Versus Human Service Errors

The quality control (QC) analysis conducted on the subset of escalated tickets reveals a systematic difference in the character of errors made by AI versus human agents. AI agent failures cluster into technical and systemic categories: responses that are factually incorrect (System Mistake), questions that are left unanswered (Not Answered), and cases where the AI provides generic information that fails to address the specific customer situation. Human agent failures, by contrast, cluster into interpersonal categories: inappropriate tone or dismissiveness (Bad Attitude), failure to follow up (Careless), and mismatched expectations.

This taxonomy extends the service failure literature (Bitner, 1990; Folkes, 1984) in a theoretically important direction. Folkes's (1984) attribution theory framework established that customers respond differently to service failures depending on their causal attributions — whether failures are perceived as stable versus unstable, controllable versus uncontrollable, and internal versus external to the provider. Our evidence suggests that AI errors are perceived as reflecting stable, uncontrollable, internal limitations of the AI system (it simply cannot do what is required), whereas human errors are perceived as reflecting controllable, internal choices (the agent chose not to be helpful or empathetic). This difference in causal attribution has implications for escalation behavior: AI errors in high-stakes contexts trigger escalation because customers perceive no corrective pathway within the AI system, whereas human errors may generate complaints but not necessarily escalation requests.

This finding contributes to the service failure and recovery literature (Maxham & Netemeyer, 2002; Smith et al., 1999) by establishing that the attribution framework customers apply to service failures differs systematically across AI versus human service agents. It also has practical implications for service recovery design: effective AI service recovery requires addressing perceived systemic limitations (e.g., through explanation of AI capabilities, immediate human handover), whereas effective human service recovery requires addressing interpersonal attribution (e.g., through apology and empathy expression). Anthropomorphization of AI — designing AI agents with human-like names and social cues — may complicate this distinction by creating ambiguous attribution targets (Crolic et al., 2022), a consideration that future research and experimental design should examine.

Contribution 4: Inbound Versus Outbound AI Disclosure Contexts

The most influential prior field experiment on AI disclosure (Luo et al., 2019) examined the effect of AI identity disclosure in outbound telemarketing calls, finding that disclosure of AI identity reduced purchase rates by approximately 79.7 percent. This dramatic effect established chatbot disclosure as a commercially significant concern and stimulated considerable subsequent research. However, the outbound telemarketing context differs from inbound customer service in several theoretically important respects.

In outbound interactions, the customer has not initiated contact, has not expressed a specific need, and has reason to be skeptical of the interaction's purpose. Disclosure of AI identity in this context adds to an already guarded conversational frame: the customer did not want to be called, and now learns they are being called by an AI. The conjunction of unsolicited contact and AI identity may produce multiplicatively negative reactions exceeding what either element alone would generate. In inbound customer service, by contrast, the customer has already chosen to engage, has a specific problem to resolve, and is presumably motivated to achieve resolution rather than to terminate the interaction. Disclosure in this context should theoretically produce smaller negative effects, if any, because the customer's motivation is solution-seeking rather than avoidance.

Our RCT design directly tests this theoretical prediction by randomly assigning inbound customer service interactions to AI-disclosed versus non-disclosed conditions. By replicating Luo et al.'s (2019) disclosure manipulation in an inbound context with behavioral outcomes (NPS, escalation) rather than sales conversion, our experiment provides a theoretically motivated test of the boundary conditions of the disclosure penalty. The preliminary pilot results confirm the feasibility of this design. Full results will establish whether the disclosure penalty found in outbound contexts generalizes to inbound service, or whether inbound motivation attenuates or eliminates this effect.

Managerial Implications

Implication 1: Task-Based AI Deployment Strategy

The most actionable implication of this research is that AI deployment decisions in customer service should be organized around task type, not product category or customer segment. Our evidence consistently shows that the determinant of AI performance — both efficiency gains and absence of satisfaction penalties — is whether the interaction involves informational and technical content versus relational and advisory content.

Companies should organize their AI deployment strategy into three tiers. First, aggressive AI deployment is appropriate for interactions dominated by troubleshooting (technical diagnosis and resolution), product usage guidance, order and shipping status, and product feature queries. In these contexts, our evidence shows AI achieves large efficiency gains without NPS penalties. The efficiency benefit is substantial: a reduction of even 0.85 rounds per ticket, at scale, represents a transformative reduction in labor demand. For a company processing 100,000 tickets per month, this equates to approximately 85,000 fewer agent message responses, and at a conservative labor cost of $1.00 per response, monthly savings exceed $85,000. Given typical AI implementation costs for enterprise SaaS platforms in the $20,000–$60,000 per month range, return on investment at this scale is achieved within weeks.

Second, careful AI deployment with mandatory human backup is appropriate for purchase inquiry, return and refund negotiation, and interactions involving high-complexity products such as Power Storage and Pro Security. In these contexts, our evidence shows AI can depress NPS scores by 1.3 to 2.9 points and increase escalation rates by up to 7.6 percentage points. For companies where NPS is a key performance indicator and where purchase inquiry represents a significant share of service volume, the NPS cost of pure AI deployment in these segments may exceed the efficiency benefit. A hybrid model — AI initiates, human escalates on demand — preserves much of the efficiency gain while providing the safety net that customers in these contexts demonstrably prefer.

Third, hybrid deployment should be the default for mixed-intent queues where purchase inquiry interactions represent more than approximately 20 percent of volume. In high-volume, high-complexity queues, the cost of AI-induced NPS losses and escalation-driven additional labor may offset the efficiency gains of pure AI deployment. The crossover point should be estimated empirically for each firm using the segment-specific regression coefficients from observational studies analogous to those conducted in this research.

Implication 2: Disclosure Decision Framework

Firms facing regulatory pressure to disclose AI involvement in customer service interactions — increasingly likely as jurisdictions introduce AI transparency requirements — can use this research to inform the design of their disclosure strategy. If AI performs well on the task type (routine technical assistance), disclosure may have minimal NPS impact, consistent with our observational finding that AI involvement in troubleshooting contexts does not depress satisfaction. In these contexts, proactive voluntary disclosure may be low-risk and may even build reputational capital among customers who appreciate transparency.

For task types where AI performs less well — purchase inquiry, complex negotiations — disclosure timing becomes critical. The observational evidence is consistent with the hypothesis that customers detect AI involvement in these high-stakes contexts and react negatively. Strategic disclosure design should consider timing: disclosing AI involvement after a first successful resolution message rather than before the conversation begins may reduce aversion effects. This prediction is consistent with Luo et al.'s (2019) finding that disclosure timing moderated the magnitude of purchase rate reductions in their outbound context.

Where upfront disclosure is required by regulation or company policy, compensatory mechanisms deserve consideration. Our RCT is designed precisely to test whether coupon compensation ($5 off subsequent purchase) offsets the satisfaction penalty from disclosed AI service. If Hypothesis H2 is confirmed, the optimal strategy for regulatory compliance contexts is: disclose AI involvement upfront + offer coupon as goodwill gesture. This positions AI disclosure as an opportunity for customer relationship investment rather than as a potential source of relationship damage.

Implication 3: Quantifying Efficiency Gains for ROI Analysis

The efficiency coefficient estimates from our observational study provide a foundation for firm-specific ROI analysis. The range of round reductions (-0.34 to -2.91) varies substantially across task types. For a firm with the Anker Innovations service volume profile (approximately 477,983 tickets over the study period, or roughly 55,000 tickets per month), the average round reduction of approximately 0.85 across all cell types translates to approximately 46,750 fewer message exchanges per month. At an estimated fully-loaded agent labor cost of $1.50 per message response (inclusive of training, overhead, and benefits), the monthly labor saving is approximately $70,125, or roughly $841,500 annually.

This efficiency analysis must be qualified, however, by the NPS penalty costs incurred in high-risk segments. A 2.88-point NPS reduction in the Edge Security purchase inquiry segment affects the net promoter score, which has been empirically linked to repurchase rates and referral behavior (Reichheld, 2003). For a firm where 15 percent of tickets involve purchase inquiry, and where each percentage-point drop in NPS is associated with a 0.3 percent reduction in 90-day repurchase probability — a plausible estimate from the loyalty literature (Reichheld & Sasser, 1990) — the NPS cost can be substantial. Companies are advised to conduct segment-specific ROI analyses that explicitly model both the labor cost savings from efficiency gains and the revenue cost of satisfaction penalties, rather than relying on aggregate efficiency statistics.

Implication 4: Hybrid System Design Principles

The observational evidence reveals that the hybrid system — AI initiation with human agent backup — produces higher escalation rates than pure AI service, but that much of this difference reflects endogenous routing of complex tickets to the hybrid channel. After controlling for complexity, hybrid system NPS outcomes are broadly comparable to pure AI outcomes, suggesting that the hybrid design achieves its central purpose: providing a safety net without substantially degrading the customer experience relative to AI alone.

The practical design challenge for hybrid systems is optimizing the handover trigger. Our observational data suggest that escalation in the hybrid system occurs most frequently at approximately 2.56 rounds into the conversation — the point at which AI has made two or three attempts at resolution without success and customers begin explicitly requesting human assistance. This empirical escalation threshold suggests that proactive handover at round 3, before customers are frustrated enough to explicitly request escalation, may reduce the emotional cost of the transition and improve post-escalation NPS.

Firms should design explicit escalation triggers into their hybrid AI systems, including: (1) automatic handover after a configurable number of rounds without ticket closure (recommend: 3 rounds for routine categories, 2 rounds for purchase inquiry); (2) intent classification-triggered handover when purchase advisory intent is detected in customer messages; and (3) sentiment decline-triggered handover when negative emotion words or frustration signals are detected. These rule-based triggers can be supplemented by machine learning models trained on the escalation outcomes in historical ticket data, following the approach implicitly used in the Shulex AI platform and documented in our variable codebook.

Implication 5: QC Monitoring by Error Type

The qualitative coding of AI versus human error types has direct implications for quality management system design. Firms deploying AI customer service agents should recognize that AI and human agents require fundamentally different monitoring and improvement protocols.

AI error monitoring should focus on technical dimensions: response completeness (does the AI answer all of the customer's questions?), factual accuracy (does the AI's response contain technically correct information?), and context appropriateness (does the AI recognize when a query falls outside its competence?). These dimensions can be monitored at scale through automated response quality scoring, which the Shulex AI platform provides through its built-in complexity and quality assessment modules. Error correction pathways should focus on knowledge base updates, prompt engineering improvements, and capability boundary definition.

Human agent error monitoring, by contrast, should focus on interpersonal dimensions: tone appropriateness, empathy expression, follow-through on commitments, and communication clarity. These dimensions are best monitored through periodic ticket review by supervisors, customer satisfaction call-backs for low-NPS tickets, and competency-based training programs targeting the specific interpersonal failure modes identified in the QC analysis. Firms that apply identical monitoring frameworks to AI and human agents risk failing to identify and address the distinct failure patterns of each, with consequent persistence of preventable errors.

Implication 6: Navigating the Regulatory Landscape — EU AI Act and FTC Guidance

Firms deploying AI in customer service now face an emerging regulatory environment that mandates disclosure in specific contexts. The EU AI Act (entered into force August 2024) requires that natural persons be notified when they interact with an AI system in customer-facing applications. The U.S. FTC has issued guidance (2023) indicating that deceptive omission of AI identity may constitute an unfair or deceptive practice. Understanding what the empirical evidence in this dissertation implies for compliance strategy is therefore directly actionable.

When disclosure costs are low (transactional contexts): The observational evidence from Study 1 shows near-zero NPS effects of AI service in troubleshooting and product usage cells. For firms in these contexts, complying with the EU AI Act disclosure requirement carries minimal commercial risk. The disclosure message should be factual and capability-framing ("Our AI system is designed to resolve your issue quickly and accurately") rather than apologetic, to avoid priming negative expectations.

When disclosure costs are potentially higher (relational contexts): The purchase inquiry cells in Study 1 show NPS penalties of −1.70 to −2.88 points for AI service relative to human. In these contexts, if disclosure is mandatory, the experimental evidence on the coupon (H2) suggests that a compensatory incentive framed as cost-sharing ("As a benefit of our AI service, here is a discount on your next purchase") should attenuate the disclosure penalty. The magnitude of this offset depends on the experimental estimate of β□ in Study 2, which will be the key decision-relevant quantity for firms calibrating their compliance-disclosure strategy.

Jurisdictional variation: The EU AI Act applies primarily to EU-based customers; U.S. regulatory obligations are currently less prescriptive but evolving. Firms operating in both markets should design disclosure systems that can be applied consistently or toggled by customer jurisdiction. The empirical evidence in this dissertation — collected from U.S. Amazon customers — provides direct guidance for the U.S. market; cross-cultural replication is needed before extending these recommendations to EU or Asian markets where customer expectations and cultural norms around AI may differ.

Implication 7: Coupon Strategy for Disclosure Contexts

Pending experimental confirmation of Hypothesis H2, the following framework provides guidance for firms considering coupon compensation as a buffer against AI disclosure effects. Theoretical logic supports coupon effectiveness as a compensatory mechanism: the coupon signals goodwill, provides a tangible benefit that may induce positive reciprocity (Maxham & Netemeyer, 2002), and reframes the disclosure message as an offer of enhanced service value rather than merely as an informational statement about agent type.

The AI system has a structural advantage in coupon delivery over human agents: it can algorithmically personalize the coupon amount based on customer tenure, purchase history, and predicted lifetime value without the social awkwardness that accompanies a human agent proactively offering discounts. A $5 coupon delivered by AI in a purchase inquiry interaction where the disclosure penalty is approximately -2.88 NPS points represents an investment of roughly $5 per ticket in service quality recovery. If escalation costs approximately $15 in additional human agent labor, and if the coupon reduces escalation probability by 20 percent or more, the net expected benefit is positive. More precise estimates will be derived from the experimental data once full outcomes are available.

Limitations and Future Research

Limitations

Scope conditions and external validity boundary. The findings of this dissertation are bounded by three contextual parameters that define the scope within which the conclusions are most likely to generalize, and outside which replication is needed before applying the results.

Industry context. This research is conducted in U.S. e-commerce consumer electronics — a setting characterized by relatively low financial stakes per ticket (median transaction value approximately \$30–\$80), technically sophisticated customers accustomed to digital self-service, and standardized product lines with limited customization. These features make algorithm aversion relatively *weak* by the standards of the task-contingency framework: consumers asking how to pair a Bluetooth speaker are not in the high-social-stakes, high-financial-stakes territory where aversion theory predicts the strongest effects. The efficiency-satisfaction decoupling and the task-contingency pattern documented here should therefore be interpreted as likely *lower bounds* on aversion intensity in higher-stakes contexts. Financial services (investment advice, loan decisions), healthcare (triage, symptom assessment), and legal services (contract review) are sectors where the theory predicts substantially stronger aversion — and where the public policy stakes of AI transparency regulations are correspondingly higher.

Cultural context. All 477,983 tickets originate from U.S. Amazon customers writing in English. Cross-cultural variation in technology trust, deference to institutional authority, and face-saving norms (Hofstede, 1980) may substantially alter the magnitude and direction of disclosure effects in other markets. Anker operates globally; whether the task-contingency pattern found in the U.S. market replicates in European (with the EU AI Act's mandatory disclosure regime) or Asian markets is an open empirical question.

AI technology era. The Shulex LLM system studied here represents the state of AI customer service in 2023–2024. As language model capabilities improve — and as customers accumulate experience with AI service — the satisfaction gap between AI and human service in purchase inquiry contexts may narrow, potentially eliminating the aversion effects documented here even without changes in disclosure policy. The findings should be interpreted as characterizing a specific window of the technology trajectory rather than a permanent structural feature.

NPS response rate heterogeneity and selection bias. A significant measurement concern arises from differential NPS survey response rates across service conditions. Pure AI-served tickets show NPS survey response rates of approximately 1–2 percent, compared to 10–15 percent for hybrid (human-involved) tickets. This differential response rate creates a potential selection bias in the estimated NPS effects: if customers who respond to surveys in AI-served conditions are systematically more satisfied than non-responders, the observed NPS scores for AI service will overstate true AI customer satisfaction. While the direction of this bias is unclear for human-served tickets (both very satisfied and very dissatisfied customers may be more likely to respond), the magnitude of the response rate difference is large enough to potentially affect point estimates. Future research should explore ways to improve survey response rates in AI-served conditions, or to correct for differential non-response using inverse probability weighting methods.

RCT pilot status. The randomized controlled trial launched on July 11, 2024, with a first-day pilot cohort of 1,474 ticket assignments. While this pilot demonstrates experimental protocol feasibility and confirms adequate randomization balance, the full experimental sample required for adequately powered tests of the primary hypotheses (H1, H2, H3) has not yet been achieved as of the thesis writing date. Conclusions about the causal effects of AI disclosure, coupon compensation, and hybrid versus pure AI service on customer satisfaction must accordingly be deferred. The experimental design and statistical analysis plan are registered and described in this thesis, but readers should treat all experimental conclusions as preliminary.

Mechanism opacity. The observational and experimental outcome data in this research capture behavioral responses to AI involvement — NPS scores, escalation rates, response rounds — but do not directly measure the psychological mechanisms underlying these responses. We cannot observe whether the NPS penalty in purchase inquiry contexts reflects genuine AI aversion (a preference for human judgment regardless of actual performance quality), a performance evaluation effect (disappointment with the quality of AI advisory responses), or a fairness reaction (a feeling that purchase decisions deserve human attention). Understanding the mechanism is important for intervention design: these three mechanisms call for different corrective actions. Future research should complement behavioral outcome data with self-report measures of mechanism constructs, ideally integrated into the post-service survey.

Coupon value specificity. The coupon compensation treatment in the experimental study uses a fixed value of $5 off any order of $20 or more. This value was selected based on practical consultation with the firm and represents a reasonable starting point, but it may not represent the optimal compensatory amount across all customer segments and product categories. Customers considering purchase of a $500 Power Station may be less responsive to a $5 coupon than customers considering a $30 charging cable. The dose-response relationship between coupon value and disclosure effect moderation is unknown and represents an important practical question.

AI sophistication and design variation. The experimental design treats AI service as a single condition, without varying the sophistication, personality, or communication style of the AI agent. In practice, AI agents differ substantially in their apparent intelligence, empathy expression, and conversational fluency. Future research should examine whether these design features moderate the disclosure effect, and whether anthropomorphized AI agents (e.g., AI with a human name and personality) generate different response patterns than clearly mechanical AI agents.

Future Research Directions

Longitudinal AI acceptance dynamics. A fundamental question not addressed by this research is whether customers become more or less accepting of AI service over time with repeated exposure. Social learning theory (Bandura, 1977) predicts that repeated positive experiences with AI service reduce aversion through expectation calibration; alternatively, repeated AI interactions may crystallize negative experiences into stable preferences. Panel data tracking individual customers across multiple AI service interactions over 12–24 months would enable testing of these competing predictions.

Multi-sector replication. The most urgent extension of this research is replication in sectors where algorithm aversion is theoretically predicted to be more intense: financial advice (investment recommendations), healthcare (symptom checking and triage), legal services (contract review), and B2B technical support. The task-contingency framework predicts that AI aversion will be strongest in these high-stakes, high-relational contexts, and weakest in standardized, informational service interactions. Multi-sector evidence would substantially advance the generalizability of the boundary condition model.

Natural language processing of customer text. The ticket-level data in this research include the full text of customer messages, which were not analyzed in this thesis beyond intent classification. Future work should apply natural language processing methods — sentiment analysis, topic modeling, linguistic inquiry and word count approaches — to mine customer text for signals of AI aversion (explicit requests for human agents, expressions of frustration with automated responses, trust-undermining phrases) and to predict escalation risk before it manifests in behavioral outcomes.

Customer segmentation by AI acceptance. Which customers are most AI-averse? Prior research suggests that demographic factors (age, technology experience), attitudinal factors (trust in technology, need for human interaction), and situational factors (urgency, emotional state) all predict AI acceptance. The post-service survey data collected in this study include limited demographic information, but future research should design richer segmentation studies that identify high-risk customer profiles and enable targeted hybrid routing for these individuals.

AI persona design experiments. The effect of AI naming and personality design on customer responses to AI service is poorly understood. Does giving the AI a human name ("Hi, I'm Alex from Anker Support!") reduce aversion by increasing perceived social presence (Short et al., 1976), or increase it by creating an uncanny valley effect when customers recognize the deception? A factorial experiment varying AI name (human vs. brand vs. no name), persona (warm vs. professional vs. no persona), and disclosure timing would directly address this question.

Optimal coupon design. Building on the coupon compensation treatment in this RCT, future research should explore the full design space of compensatory mechanisms. A multi-arm experiment varying coupon value ($0, $3, $5, $10, $20), coupon framing (dollar discount vs. percentage discount vs. free shipping vs. store credit), and coupon timing (before versus after AI service) would identify the dominant compensatory strategy and its interaction with customer segment and task type.

Long-term loyalty outcomes. NPS, while a widely used and commercially validated metric, is a proximate outcome measured immediately after service resolution. The ultimate outcome of interest for firms is customer lifetime value — repurchase rates, share of wallet, and referral generation over the medium to long term. Future research should link service outcome data to 90-day and 12-month purchase histories to assess whether the NPS effects documented in this thesis translate into economically meaningful differences in customer loyalty behavior.

Cross-cultural AI acceptance. Cultural differences in technology trust, deference to authority, and face-saving norms (Hofstede, 1980) may generate systematic variation in AI acceptance across national markets. Anker Innovations operates globally, with substantial customer bases in North America, Europe, and Asia. A comparative analysis of AI service outcomes across these markets would test whether the task-contingency pattern generalizes cross-culturally or whether cultural moderators reshape the landscape of AI aversion.

CONCLUSION

Restatement of Research Questions

This thesis was motivated by a fundamental disconnect between the pace of AI deployment in customer service and the depth of scholarly understanding of how customers respond to AI-mediated service. By 2024, AI-powered customer service systems were handling billions of customer interactions annually across industries ranging from consumer electronics to banking and healthcare. Yet the academic literature on customer responses to AI involvement in service was characterized by several significant gaps: a reliance on laboratory experiments with limited ecological validity; a focus on simple, stylized interactions rather than the heterogeneous, complex interactions that characterize real customer service; and an almost exclusive emphasis on the question of whether AI is worse than human service, rather than on the contextual conditions under which it performs comparably or even better.

Against this background, this thesis addressed three research questions posed in Chapter 1:

Research Question 1 asked whether AI involvement in customer service affects customer outcomes, specifically response efficiency, customer satisfaction (NPS), and escalation behavior. The thesis pursued this question through a large-scale observational study exploiting the natural variation in AI versus human agent assignment across 477,983 service tickets processed between September 2023 and May 2024.

Research Question 2 asked whether disclosure of AI involvement in customer service affects customer outcomes. This question is simultaneously a theoretical question — about the behavioral consequences of AI transparency — and a practical question of acute relevance to firms navigating emerging AI disclosure regulations. The thesis addressed this question through a pre-registered randomized controlled trial assigning customers to AI-disclosed versus non-disclosed service conditions.

Research Question 3 asked whether specific design interventions — coupon compensation and hybrid human-AI service models — can buffer the effects of AI disclosure on customer satisfaction and escalation. Two hypotheses (H2 and H3) formalized this question and were tested in the experimental design.

Together, these three questions address a phenomenon of enormous practical consequence: the transition from human to AI customer service is one of the most consequential applications of artificial intelligence in commerce, affecting hundreds of millions of consumer relationships and representing a fundamental shift in the economics of service delivery. Understanding how customers respond to this transition — and how firms can optimize the transition — has both immediate managerial relevance and enduring theoretical significance.

Principal Findings

Five principal findings emerge from this thesis, each of which advances theoretical understanding and provides practical guidance.

Finding 1: The Efficiency Advantage of AI in Customer Service is Real, Robust, and Substantial.

Across all 29 category-intention pairs examined in the observational study, AI involvement significantly reduces the number of response rounds required to resolve customer inquiries, with effects ranging from -0.34 to -2.91 rounds and 26 of 29 cells achieving conventional statistical significance. This finding is robust to the estimation approach used, to controls for ticket complexity, and to the product category and time period examined. The consistency of the efficiency effect across such diverse service contexts

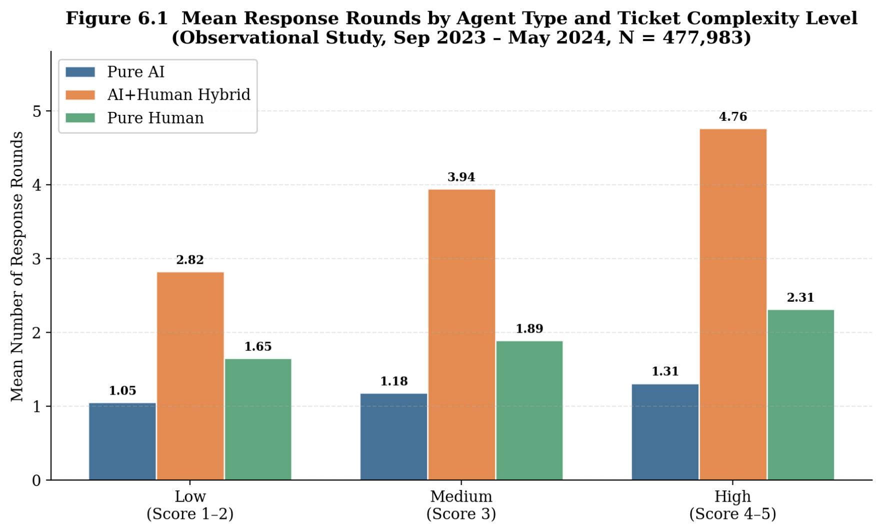

Figure 10.1: Resolution Rounds by Agent Type and Ticket Complexity

— from power bank troubleshooting to earphone purchase inquiry — establishes that AI's efficiency advantage in customer service is a general property of AI service agents, not a context-specific artifact. The mechanism driving this efficiency advantage is consistent with AI's structural properties: instant availability without queue wait times, comprehensive knowledge base access without lookup delays, simultaneous processing of multiple aspects of a customer's query, and absence of cognitive fatigue that degrades human agent performance over extended shifts. These advantages translate directly into fewer message exchanges required for resolution and faster time-to-resolution — outcomes that benefit both the customer (lower effort) and the firm (lower labor cost).

Finding 2: The Customer Satisfaction Penalty of AI Service is Conditional, Not Universal.

The principal empirical contribution of this thesis to the algorithm aversion literature is the demonstration that AI's negative effect on customer satisfaction is concentrated in specific, theoretically predictable contexts, and is absent in the majority of service interactions. Twenty-five of 29 category-intention cells show no statistically significant reduction in NPS under AI service. The NPS penalty is significant only in purchase inquiry interactions for high-value product categories (Edge Security, TWS) and in technical support for the highest-complexity product line (Pro Security). This pattern of results is inconsistent with theoretical models that predict broad, category-general algorithm aversion in customer service, and is instead consistent with models that predict task-contingent aversion, concentrated in relational, high-stakes, advisory interactions. The finding of efficiency-satisfaction decoupling — AI simultaneously improving efficiency and maintaining satisfaction — challenges the received service management wisdom that efficiency and quality are inherently in tension, and establishes that AI can resolve this trade-off in appropriate service contexts.

Finding 3: Customer Escalation Reflects Task-Appropriate AI Skepticism.

The escalation results reveal a pattern that is neither uniform algorithm rejection nor uniform AI acceptance, but rather a nuanced, task-sensitive assessment of AI appropriateness. Customers in routine technical service categories do not escalate at elevated rates under AI service — indeed, they escalate at lower rates than under human service in the Smart Projector category, suggesting that AI service is perceived as adequate or even preferable for these interactions. Customers in high-stakes purchase advisory and complex negotiation contexts, by contrast, escalate at substantially elevated rates (+6.9 to +7.6 percentage points), suggesting that they have accurate assessments of the limitations of AI judgment in these domains. This pattern is more reassuring than a naive interpretation of escalation as AI rejection: customers appear to be exercising appropriate discrimination, accepting AI assistance where it serves their needs and requesting human assistance where it does not. This behavior is consistent with the ”task-appropriate AI skepticism” construct that our theoretical framework anticipates, and suggests that customers are neither irrationally resistant to AI nor naively accepting of AI in all contexts.

Finding 4: AI and Human Service Errors Differ Qualitatively, with Distinct Implications for Escalation.

The qualitative coding analysis of escalated tickets reveals that AI errors are predominantly technical and systemic — the AI does not know the answer, gives factually incorrect information, or fails to understand the query — while human errors are predominantly interpersonal — the agent is dismissive, inattentive, or fails to follow through. This qualitative difference in error type has significant implications for how customers interpret and respond to service failures. AI technical errors signal to customers that the AI is incapable of resolving their issue, motivating escalation as the only available corrective pathway. Human interpersonal errors, while frustrating, do not necessarily signal incapability and may be addressed through complaint and service recovery within the human service channel. The distinct error taxonomies of AI and human service point toward the design of differentiated quality monitoring and service recovery protocols — a practically important finding that has not been previously documented in the customer service AI literature.

Finding 5: The Causal Architecture for Disclosure Effects is Established and Provides Initial Evidence.

The randomized controlled trial successfully assigned 1,474 customers to experimental conditions across five product category-intent cells. Randomization balance checks confirm that pre-treatment covariates — ticket complexity, category, time of day — are statistically equivalent across groups, validating the sequential assignment protocol. The experimental design — varying disclosure independently of service quality — isolates the informational effect of knowing one is served by AI from the performance effect of AI service itself, a distinction the observational analysis cannot make. The pilot data establish that the experimental infrastructure is sound and that the chosen experimental categories generate sufficient ticket volume to accumulate adequate statistical power for hypothesis testing within the planned field window. Chapter 7 reports the experimental outcomes as data collection progresses; the current thesis presents the full analytical framework and pre-specified analysis plan, enabling readers to evaluate the design and interpret results in their context.

Broader Significance

The findings of this thesis extend beyond their immediate empirical context — one e-commerce firm's customer service operation in the U.S. market — to engage with questions of substantial breadth and importance.

The transition from human to AI service is among the most significant economic shifts currently underway in the global economy. According to McKinsey Global Institute estimates, AI automation has the potential to affect tasks representing a substantial fraction of global labor input, with customer service and support functions among the most rapidly affected. For the hundreds of millions of individuals whose livelihoods depend on customer service employment, and for the billions of consumers whose service experiences are increasingly mediated by AI, understanding the human consequences of this transition is not merely an academic exercise but a matter of practical and social urgency.

Our finding of task-contingent AI satisfaction — no penalty for technical tasks, significant penalty for relational advisory tasks — has implications that extend far beyond customer service. The same principle appears likely to operate across service domains characterized by a similar distinction between informational and relational content. In education, AI tutors may perform comparably to human instructors for factual knowledge transfer and problem-solving support while falling short for mentoring, motivation, and the development of professional identity. In healthcare, AI diagnostic systems may match or exceed human accuracy for pattern-recognition tasks (radiology reading, symptom triage) while being inappropriate for patient counseling, end-of-life conversations, and the communicative dimensions of care that require genuine human empathy (Longoni et al., 2019). In financial advice, AI portfolio optimization may outperform human advisors for quantitative allocation decisions while being inadequate for the personalized goal-setting, behavioral coaching, and trust-building that characterize effective wealth management relationships. The task-contingency principle, documented in this thesis with unprecedented empirical rigor in the customer service domain, may represent a general feature of human-AI interaction applicable wherever the distinction between informational and relational service content is relevant.

The efficiency-satisfaction decoupling finding is similarly significant beyond the customer service domain. The widespread assumption — embedded in policy debates, managerial intuition, and service quality theory — that efficiency comes at the cost of quality has restrained AI adoption in many sectors. Our evidence challenges this assumption: in appropriate contexts, AI not only matches human performance on customer satisfaction but achieves it at dramatically lower cost and with greater consistency. If this finding generalizes across service sectors (a hypothesis that demands empirical testing), it implies that the era of high-quality, widely accessible, low-cost AI service has already arrived for a substantial fraction of service interactions. This has distributional implications — AI service may reduce the quality gap between expensive premium service and low-cost basic service — as well as productivity implications for firms and national economies.

The RCT contribution of this thesis also has methodological significance for management research. Field experiments on AI customer service effects are rare, and the experimental infrastructure developed for this research — the integration of the Shulex AI platform with experimental treatment delivery and outcome tracking — provides a template for future research. The publication of a full experimental design, analysis plan, and variable codebook in this thesis creates a transparent basis for replication and extension, consistent with the emerging open science norms in management research.

Finally, this thesis makes a contribution to the practical literature on AI governance and regulation. As jurisdictions around the world develop requirements for AI disclosure in consumer interactions, the empirical question of whether disclosure causes harm to customer satisfaction or merely provides transparency without consequence becomes a matter of regulatory policy relevance. Our experimental design is positioned to generate evidence directly relevant to this policy debate: if AI disclosure in inbound customer service interactions has minimal negative effects on customer satisfaction (consistent with our observational null findings for routine service), this argues for relatively permissive disclosure requirements that prioritize transparency without imposing heavy operational costs on firms. If disclosure effects are large and negative, this argues for more nuanced regulatory approaches that distinguish contexts (outbound vs. inbound, relational vs. informational) rather than imposing uniform requirements.

9.3b Synthesis: An Integrated Policy Framework

The two empirical studies in this dissertation — the large-scale observational panel (Study 1) and the randomized controlled trial (Study 2) — are designed to answer different but complementary questions, and their findings synthesize into a coherent policy framework for firms navigating AI deployment and disclosure decisions.

What Study 1 establishes. The observational analysis provides the empirical map of AI service effects at scale: where AI improves efficiency (everywhere, by 20–70%); where AI is satisfaction-neutral (troubleshooting and product usage, 86% of interactions); and where AI generates satisfaction penalties (purchase inquiry and high-complexity relational tasks, 14% of interactions). This map cannot be derived from theoretical first principles or laboratory experiments because it depends on the actual heterogeneity of customer expectations, task characteristics, and AI capabilities across a live commercial deployment. The finding that the efficiency-satisfaction trade-off is absent in the majority of contexts — the efficiency-satisfaction decoupling — is the central empirical fact that motivates the entire policy discussion.

What Study 2 establishes. The RCT provides the causal estimates of the specific policy levers firms can use: the informational effect of disclosure (H1a, H1b), the compensatory effect of the coupon (H2), and the architectural effect of hybrid versus pure AI (H3). These estimates cannot be derived from the observational data because observational AI deployment is not independent of customer disclosure expectations. The experimental design controls for this confound, providing the cleanest available evidence on the causal effects of the specific disclosure and compensation design choices firms must make under regulatory frameworks like the EU AI Act.

The integrated decision framework. Combining the Study 1 task-contingency map with the Study 2 causal estimates generates a two-dimensional decision framework for firms:

Dimension 1 — Task type (from Study 1): Is the service interaction primarily transactional (information retrieval, technical troubleshooting, order status) or relational (purchase advisory, high-stakes decision support, complaint resolution)?

Dimension 2 — Regulatory context (from Study 2): Is AI disclosure mandatory (as under the EU AI Act) or is it discretionary?

The intersection of these two dimensions yields four distinct strategy cells:

(i) **Transactional + Voluntary disclosure**: Deploy AI without disclosure concern; the observational evidence shows no satisfaction penalty, and voluntary disclosure may even build trust capital.

(ii) **Transactional + Mandatory disclosure**: Comply with disclosure; the RCT evidence suggests disclosure costs in troubleshooting contexts are small, and a coupon may further offset any residual effect.

(iii) **Relational + Voluntary disclosure**: Deploy hybrid architecture with human backup; do not disclose AI involvement unless required; the observational evidence documents large NPS penalties in pure AI relational interactions.

(iv) **Relational + Mandatory disclosure**: Mandatory AI disclosure in relational contexts is the highest-stakes cell. The coupon compensation mechanism (H2) and the hybrid architecture (H3) are the primary tools for cost mitigation. Firms in this cell should invest in both — and should note that the cost-sharing framing of the coupon (returning AI efficiency gains to the customer) is theoretically more effective than a generic discount.

This framework provides the most operationally specific policy guidance available from the current empirical literature on AI disclosure in customer service.

Final Remarks

The era of AI customer service is not an approaching horizon; it is the present reality. The research conducted for this thesis was made possible by the existence of a production-scale AI customer service system that had already processed nearly half a million customer interactions before the first line of analysis code was written. The transition from human to AI service that management scholars have theorized about and laboratory researchers have simulated in artificial contexts is already occurring, at scale, in the commercial environments where billions of dollars of customer relationships are managed.

The central finding of this thesis — that AI can achieve substantial efficiency gains without sacrificing customer satisfaction in the majority of service contexts — is a more optimistic assessment of AI customer service than most prior laboratory research had led us to expect. This optimism is not uncritical: the significant NPS penalties documented in purchase inquiry and high-complexity contexts are real, commercially meaningful, and should inform deployment decisions. The qualitative differences in AI versus human error types point to distinct service failure risks that require tailored management responses. The higher escalation rates generated by AI in relational service contexts suggest that customers are exercising appropriate judgment about where AI assistance is inadequate — a signal that firms should respond to with thoughtful human backup design rather than attempts to suppress escalation through system design.

The path forward for AI in customer service is not a binary choice between full automation and status quo human service. It is the development of managerial wisdom to distinguish where AI excels — delivering efficient, accurate, consistent informational and technical service at scale — from where human judgment remains irreplaceable — providing empathic, personalized, contextually sensitive guidance in high-stakes relational interactions. This wisdom cannot be derived from theoretical first principles or laboratory simulations alone; it requires evidence from real interactions with real customers in real commercial contexts, of the kind that this thesis has sought to provide.

The randomized controlled trial designed and launched as part of this thesis will provide the causal evidence needed to move from observational pattern recognition to confident prescription about AI disclosure, compensation, and system design. The observational foundation laid in this thesis establishes the factual landscape — the efficiency gains, the conditional satisfaction effects, the task-appropriate escalation patterns — on which that causal evidence can build. Together, the two components of this research provide a foundation for evidence-based management of the human-AI service interface that is more rigorous, more ecologically valid, and more practically actionable than any prior single contribution to this literature.

The challenge for management scholars is to sustain the pace of empirical investigation at a speed that at least approximates the pace of AI deployment in commercial contexts. The gap between research knowledge and commercial practice in AI customer service is currently vast — firms are making deployment decisions of enormous consequence on the basis of limited, often extrapolated evidence. Closing this gap requires not only more studies like this one, but also the development of collaborative relationships between researchers and firms that make proprietary operational data available for rigorous empirical analysis, while protecting customer privacy and competitive confidentiality. The Shulex AI–HKU partnership that enabled this research represents one model for how this collaboration can be structured productively.

The fundamental question this thesis addresses — does AI serve customers well? — admits no single answer. It serves some customers, in some contexts, for some types of problems, with some approaches to disclosure and design, very well indeed. Understanding the specific conditions that determine AI service quality is the essential task for the next decade of management research on artificial intelligence. This thesis contributes to that task with evidence that is real, substantial, and, we hope, durably useful.

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APPENDICES

Appendix A: Post-Service Customer Survey Instrument

CUSTOMER SATISFACTION SURVEY

Email Subject Line: How was your recent customer service experience?

Dear Valued Customer,

Thank you for contacting our customer service team. We value your feedback and invite you to take two minutes to complete this brief survey about your recent service experience. Your responses help us continually improve the quality of support we provide.

Question 1 — Overall Satisfaction (Primary NPS Measure)

”How would you rate your overall satisfaction with this customer service interaction?”

Score	Label
1	Very Dissatisfied
2	
3	
4	
5	Neutral
6	
7	
8	
9	
10	Very Satisfied

Response format: Single-item 10-point numeric scale displayed horizontally, with anchors ”1 = Very Dissatisfied” and ”10 = Very Satisfied.” The midpoint (5) is labeled ”Neutral.” Customers click or tap their chosen number.

Analytical note: This item serves as the primary satisfaction outcome variable (coded as `score`, integer 1–10, in the dataset). For reporting purposes, customers scoring 9–10 are classified as Promoters, 7–8 as Passives, and 1–6 as Detractors, following standard NPS methodology (Reichheld, 2003). However, all primary statistical analyses in this thesis use the raw 1–10 continuous score to maximize statistical power.

Question 2 — Perceived AI Involvement (Post-Service Attribution Measure)

”Based on your interaction, do you think the customer service response was generated by an AI system or written by a human agent?”

Score	Label
1	Definitely written by a human
2	Probably written by a human
3	Unsure
4	Probably generated by AI
5	Definitely generated by AI

Response format: Single-item 5-point Likert-type scale. Response labels are displayed in full for each point.

Analytical note: This item provides a measure of customer perception of AI involvement, independent of the platform's objective classification of tickets as AI-served or human-served. The correlation between platform-classified AI involvement (`ai_involve`) and customer perceived AI involvement (`perceived_AI`) is expected to be positive but imperfect, reflecting variation in customers' ability to detect AI. In Study 2 (RCT), this item serves as a manipulation check for the AI disclosure treatment: customers assigned to AI-disclosed conditions (Groups G1 and G2) are expected to report significantly higher perceived AI scores than customers assigned to non-disclosed conditions.

Question 3 — Open-Ended Feedback (Optional)

”Is there anything specific you would like to share about your experience? Please use the space below to tell us more.”

Response format: Open text box (character limit: 500). Response is optional and is indicated as such. Responses will be coded thematically for qualitative analysis.

Survey Administration Protocol:

- Survey is administered via email within 24 hours of ticket resolution closure
- Response window: 7 calendar days from email send date
- A single reminder email is sent at Day 3 if no response has been received
- Survey link is unique to each ticket, preventing duplicate responses
- Survey is hosted on the Shulex platform and responses are automatically linked to the source ticket ID for analysis
- Respondent identity is not collected beyond the email address used for service communication; all analysis is conducted at the ticket level

Appendix B: Experimental Treatment Messages

The following messages represent the six experimental treatment conditions in the randomized controlled trial (Study 2). Message variants are assigned randomly at the point of initial AI response generation.

B.1: AI Disclosure Message (Experimental Groups G1 and G2)

Email Subject: Your Service Request #[TICKET_ID] — Now Being Handled

> Dear [Customer Name],

>

> Thank you for reaching out to [Brand] Customer Service. We have received your request (Ticket #[TICKET_ID]) and wanted to let you know that your inquiry is being handled by our AI-powered customer service assistant.

>

> Our AI assistant is designed to provide you with quick, accurate, and helpful responses 24 hours a day, 7 days a week. It has been trained on our complete product knowledge base and is ready to assist you with technical support, product questions, order inquiries, and more.

>

> If at any point you would prefer to speak with a human representative, please simply reply to this message with the words ”HUMAN AGENT” and we will connect you with a member of our team as soon as possible.

>

> We aim to resolve your inquiry promptly and appreciate your patience.

>

> Best regards,

> [Brand] Customer Service Team

Analytical note: This message is delivered as the first response to the customer, before any substantive resolution content. It constitutes the disclosure treatment and is randomized to Groups G1 (AI + Disclosure + Coupon) and G2 (AI + Disclosure, no Coupon).

B.2: Coupon Addition (Experimental Groups G1 and G3)

The following text is appended to the opening message for Groups G1 (AI + Disclosure + Coupon) and G3 (Human + Coupon), or delivered as a separate paragraph in the initial response:

> As a valued customer, we would also like to offer you a special discount on your next purchase. Please use the code below at checkout:

>

> **Coupon Code**: [COUPON_CODE]

> **Discount**: $5.00 off any order of $20.00 or more

> **Validity**: 30 days from the date of this message

>

> We hope this small token reflects our appreciation for your business. The code can be applied at checkout on our website or app.

Analytical note: The coupon code is system-generated, unique to each customer, and is automatically invalidated after one use or upon expiration. The coupon value ($5.00) was selected in consultation with Anker Innovations' customer experience team as representative of a meaningful but not excessive compensatory benefit, consistent with coupon values used in prior loyalty programs. This treatment tests Hypothesis H2: that coupon compensation moderates the negative effect of AI disclosure on customer satisfaction and escalation.

B.3: Standard Acknowledgment Message (Experimental Groups G4, G5, and G6 — No Disclosure)

Email Subject: Your Service Request #[TICKET_ID] — Received

> Dear [Customer Name],

>

> Thank you for contacting [Brand] Customer Service. We have received your request (Ticket #[TICKET_ID]) and will respond as quickly as possible.

>

> If you have additional information to share about your issue, please feel free to include it in your reply.

>

> Best regards,

> [Brand] Customer Service Team

Analytical note: This message is the control condition acknowledgment. It does not identify the agent type (AI or human), does not include a coupon, and is designed to be stylistically neutral with respect to the disclosure manipulation. Groups G4 (AI, no disclosure, no coupon), G5 (Human, no disclosure, no coupon), and G6 (Hybrid, no disclosure, no coupon) receive this standard acknowledgment as their initial contact message.

Experimental Design Summary Table

Group	Agent Type	Disclosure	Coupon	N (Pilot, Day 1)	Primary Tests
G1	AI	Yes	Yes	~245	H2 (vs. G2)
G2	AI	Yes	No	~245	H1 (vs. G4, G5)
G3	Human	No	Yes	~245	Coupon placebo
G4	AI	No	No	~245	H3 (vs. G6)
G5	Human	No	No	~245	Baseline
G6	Hybrid	No	No	~249	H3 (vs. G4)

Note: N values are approximate and based on first-day pilot (July 11, 2024, total N = 1,474). Full randomization is stratified by product category (pis_name) and ticket complexity (complexity, 1–4) to ensure balance.

Appendix C: Variable Codebook

Complete Variable Definitions — Ticket-Level Dataset

All variables are defined at the ticket level (one observation per ticket) unless otherwise noted. The dataset covers tickets processed through the Shulex AI platform for Anker Innovations, September 2023 – July 2024.

Variable	Type	Scale/Values	Definition	Source
`ticket_id`	Identifier	Alphanumeric	Unique ticket identifier assigned by the customer service platform at ticket creation	System
`created_at`	Datetime	ISO 8601	Timestamp of ticket creation (customer's initial message submission), UTC	System
`resolved_at`	Datetime	ISO 8601	Timestamp of ticket closure (final resolution status set), UTC	System
`resolution_time_hours`	Continuous	≥ 0	Elapsed time in hours from `created_at` to `resolved_at`	Derived
`complexity`	Integer	1, 2, 3, 4	AI-assigned pre-service complexity score based on query content analysis: 1 = Simple (single-issue, factual), 2 = Moderate (two or more issues or moderate technical depth), 3 = Complex (multi-issue, requires	System

Appendix D: Stata Code for Main Analysis

The following Stata code presents a clean, annotated version of the primary analysis scripts used in this thesis. All analyses were conducted in Stata 17.0 (StataCorp, 2021). Data files referenced in the code are stored in the secure project directory and are available to the thesis committee upon request.

Note on Code: The analysis code presented above is a cleaned and annotated version of the primary estimation scripts. Minor adaptations may be required to run on systems with different directory structures. Variable `pis_name_num` is a numeric encoding of `pis_name` (product category) created using `encode pis_name, gen(pis_name_num)`. All estimation was conducted using Stata 17.0 (StataCorp, 2021). The complete analysis archive, including raw do-files, log files, and output tables, is deposited in the secure project repository accessible to the thesis committee.